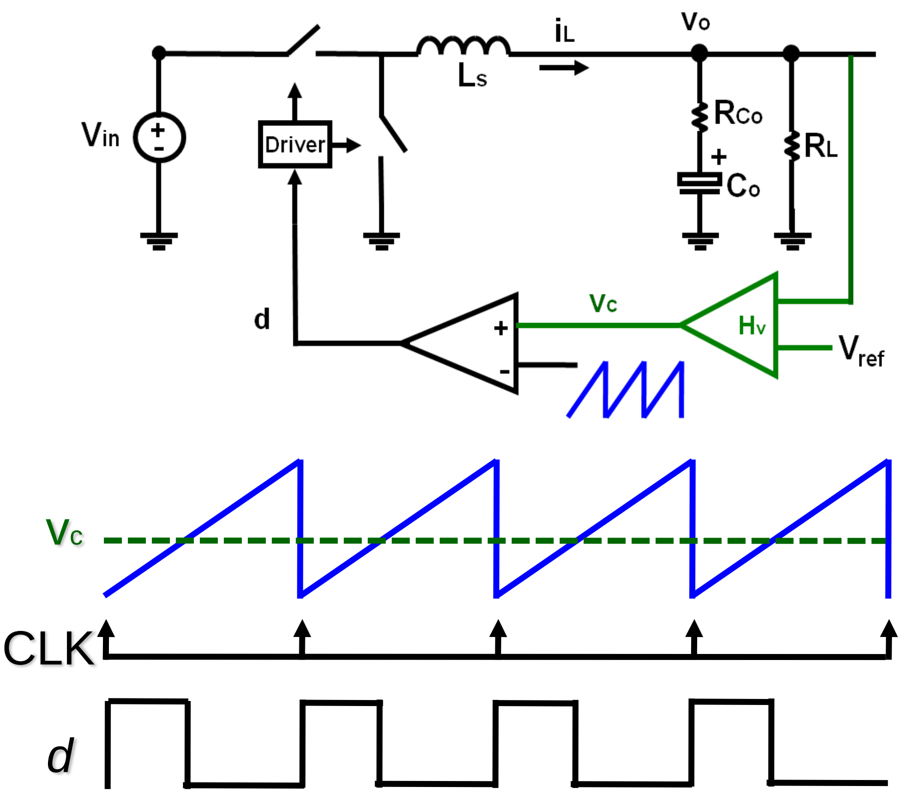
Chapter 5

New Current-Mode Control Modeling based on Describing Function Method

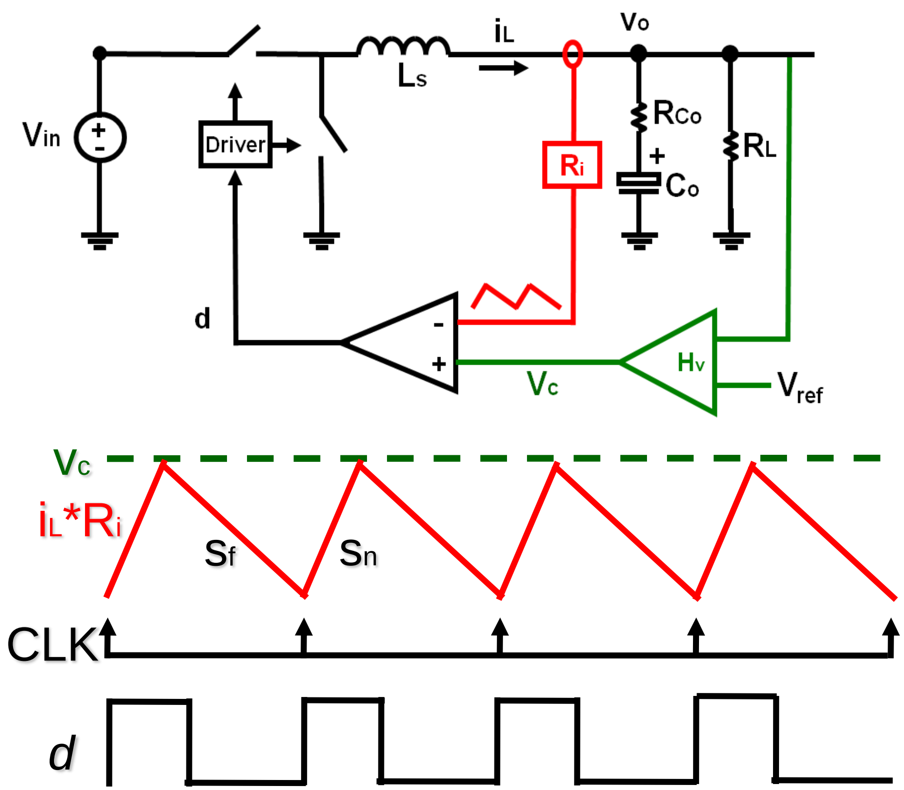
Voltage-Mode vs. Current-Mode

- *Voltage-Mode Control*



External ramp is used

- *Peak Current-Mode Control*

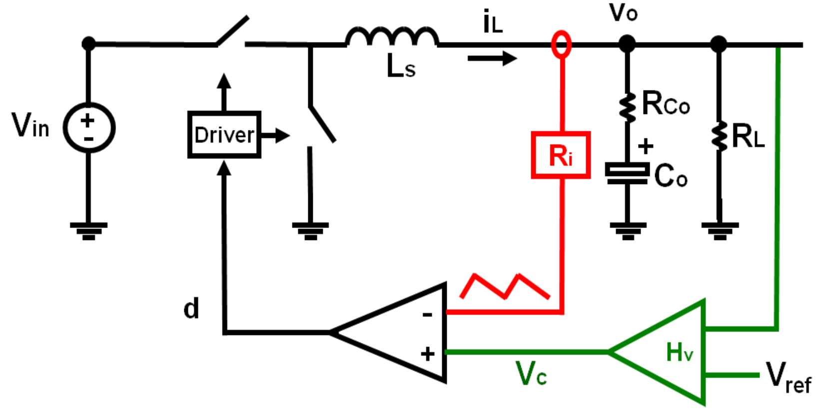


Inductor current ramp is used

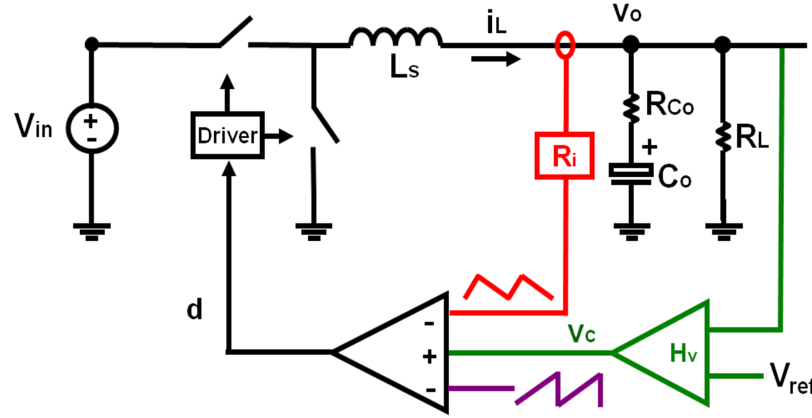
In current-mode control,
state variable i_L is used in the pulse-width modulator

Subharmonic Oscillations
in Current-Mode Control

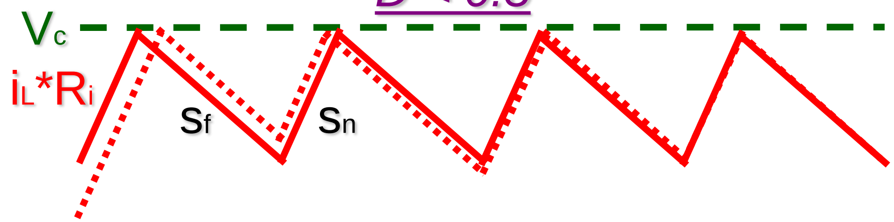
➢ *Peak Current Mode Control*



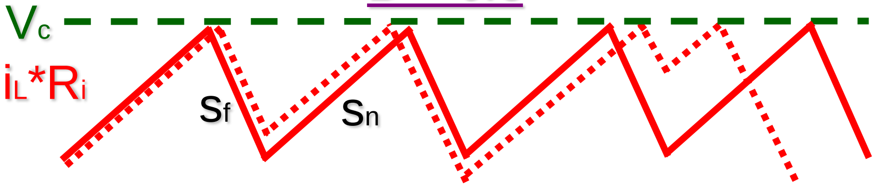
➢ *Adding an External Ramp*



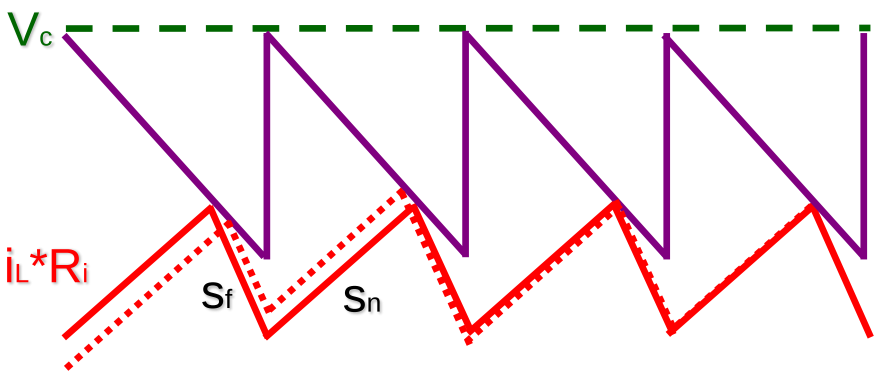
D < 0.5



D > 0.5



Subharmonic Oscillations



External ramp is used to eliminate subharmonic oscillations

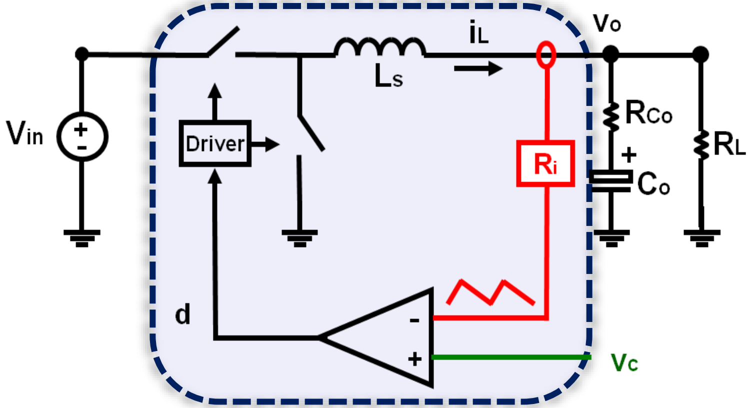
An accurate model is indispensable to system design

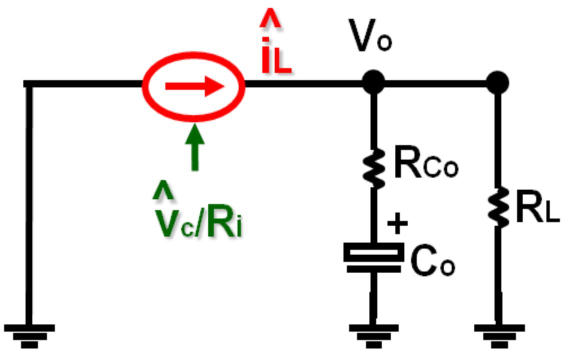
Review of Previous Models for Peak Current-Mode Control

- "Current Source" Model
- Average Model
- Discrete-Time Model
- Sample-Data Model
- Modified Average Model
- Other Models

The Concept of "Current Source"*

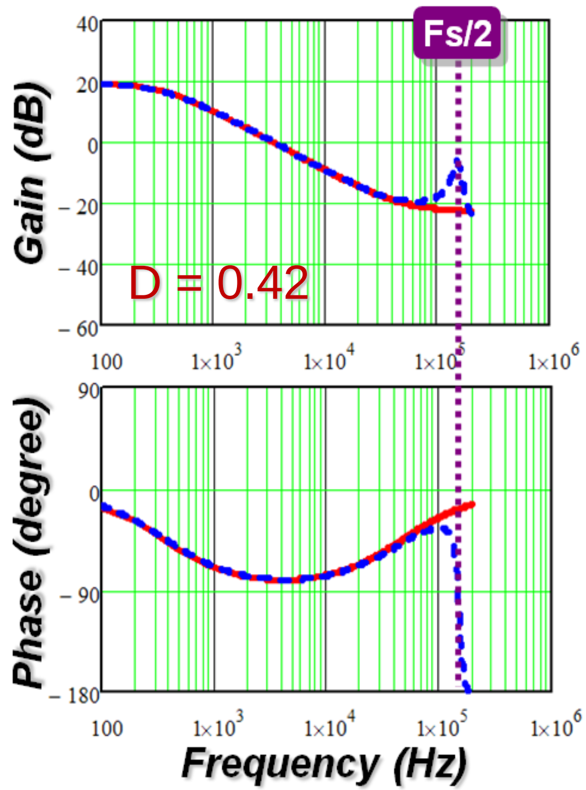
➢ *The current loop is treated as one entity to model*





A 1st order system

Control-to-output Transfer Function



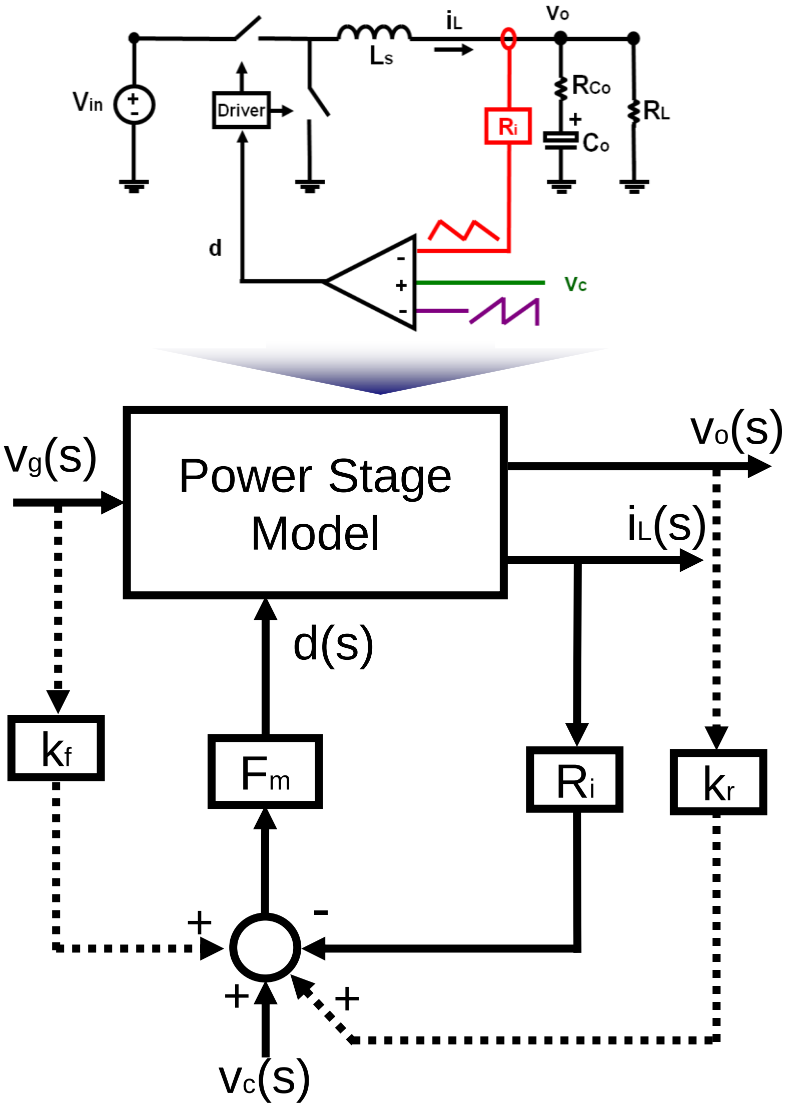
◆ *"Current Source" Model*

◆ *Simplis Simulation ≈ Measurement*

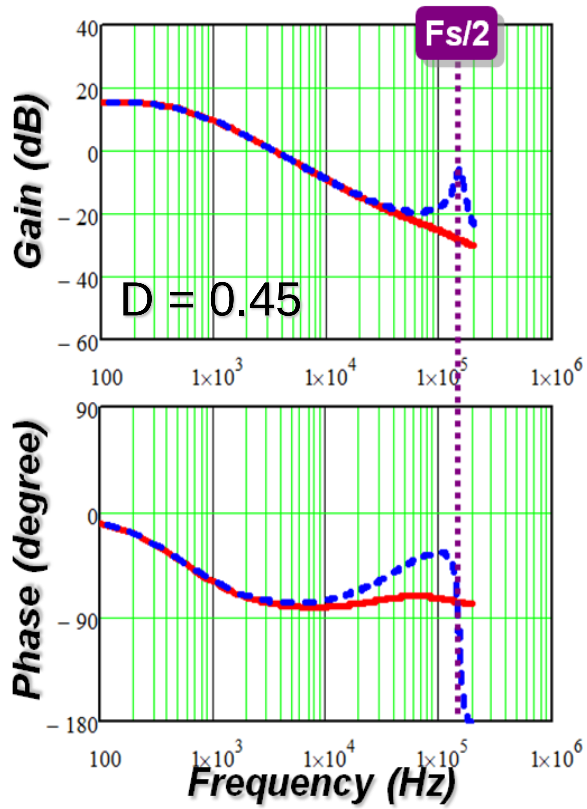
This concept is too simple to predict subharmonic oscillations

* S. P. Hsu, R. D. Middlebrook, etc. "Modeling and analysis of switching dc-to-dc converters…," PESC'79

Average Model



Control-to-output Transfer Function



Average Model
Simplis Simulation ≈ Measurement

The average model can't predict the response @ high frequency

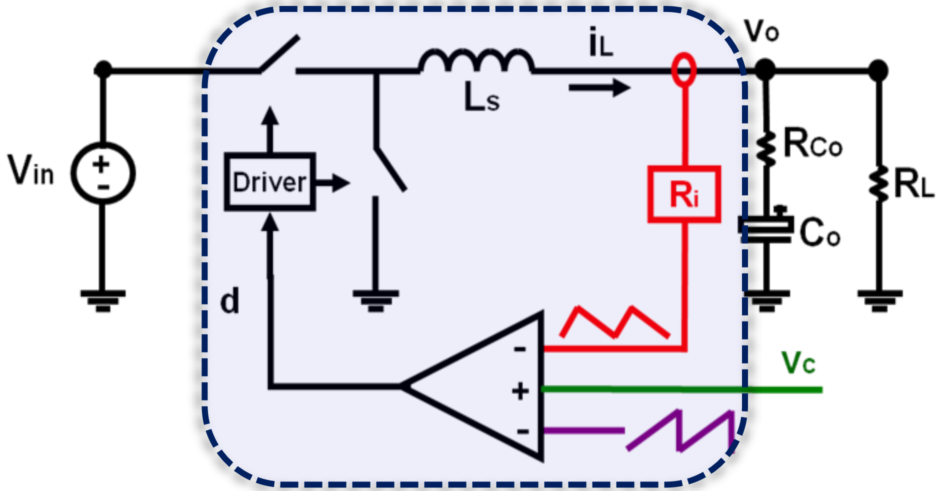
R. D. Middlebrook, etc. "Topics in Multiple-Loop Regulators and Current Mode Programming," PESC'85

Sample-Data Model

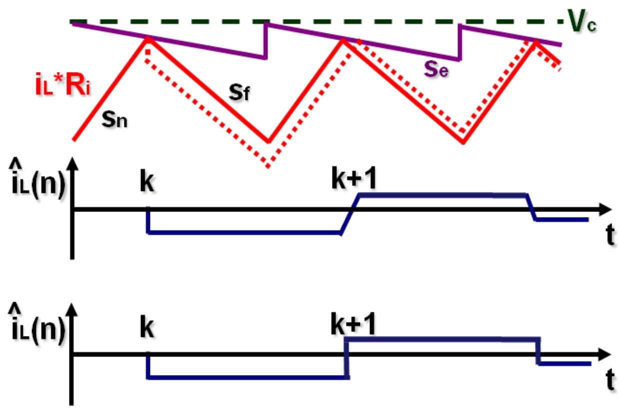
for Peak Current-Mode Control

by Arthur Brown and Middlebrook

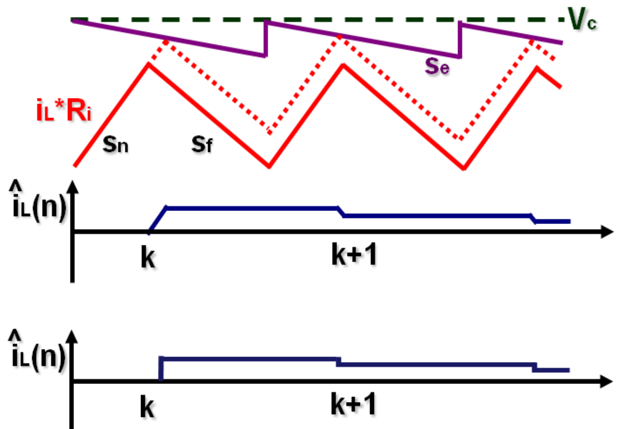
- *The current loop is treated as one entity to model*



1. Natural Response



2. Forced Response



$$H(z)=\frac{\hat{i}_L(z)}{\hat{v}_c(z)}=\frac{1+\alpha}{R_i}\frac{z}{z+\alpha}$$

where, $\alpha=(s_f-s_e)/(s_n+s_e)$

Assumption: inductor current slopes are constant

Current loop instability can be shown in this model

Modified Average Model

R. Ridley's Frequency-Domain Analysis

CPES

- *Discrete time*

$$H(z)=\frac{\hat{i}_L(z)}{\hat{v}_c(z)}=\frac{1+\alpha}{R_i}\frac{z}{z+\alpha}$$

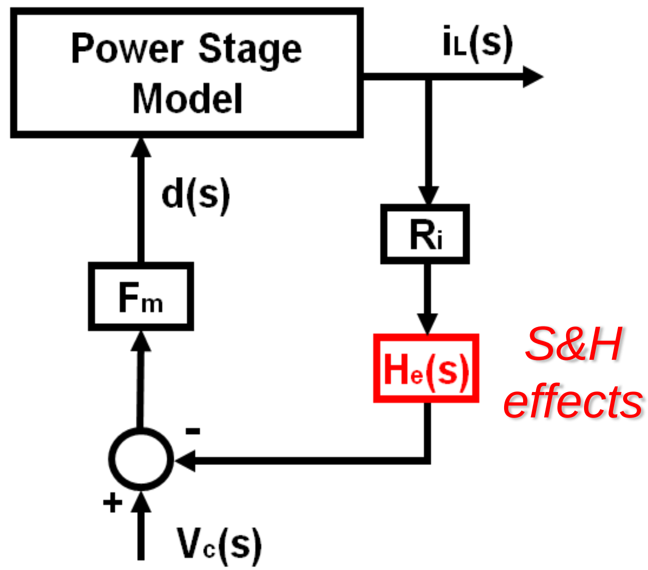
to

- *Continuous*

$$\frac{i_L(s)}{v_c(s)}=\frac{1}{R_i}\frac{1+\alpha}{sT_{sw}}\frac{e^{sT_{sw}}-1}{e^{sT_{sw}}+\alpha}$$

=

- *Modified Average model*



$$\frac{i_L(s)}{v_c(s)}=\frac{F_mF_i(s)}{1+F_mF_i(s)R_iH_e(s)}$$

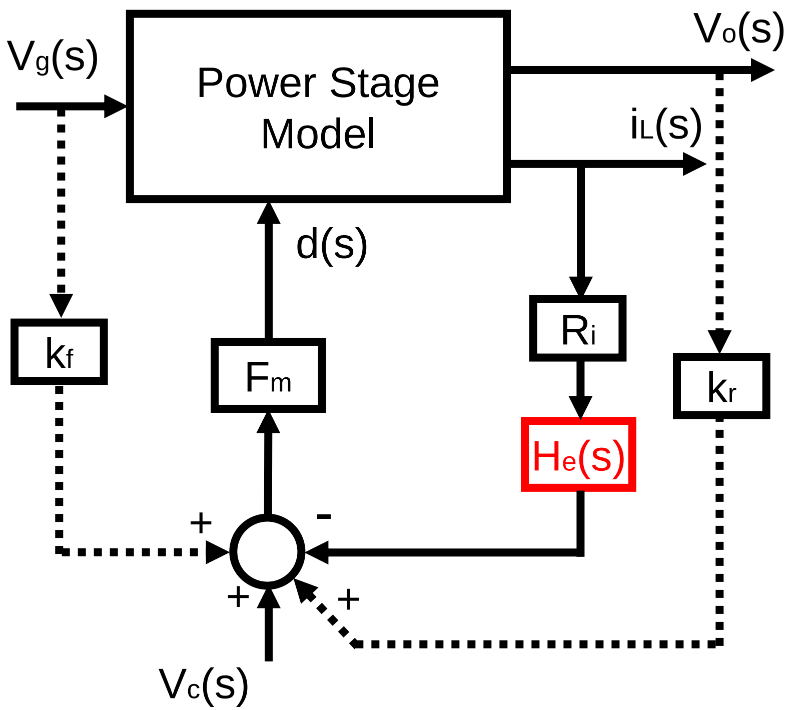
$$H_e(s)=\frac{sT_{sw}}{e^{sT_{sw}}-1}$$

Hypothesis method is used in R. Ridley's model

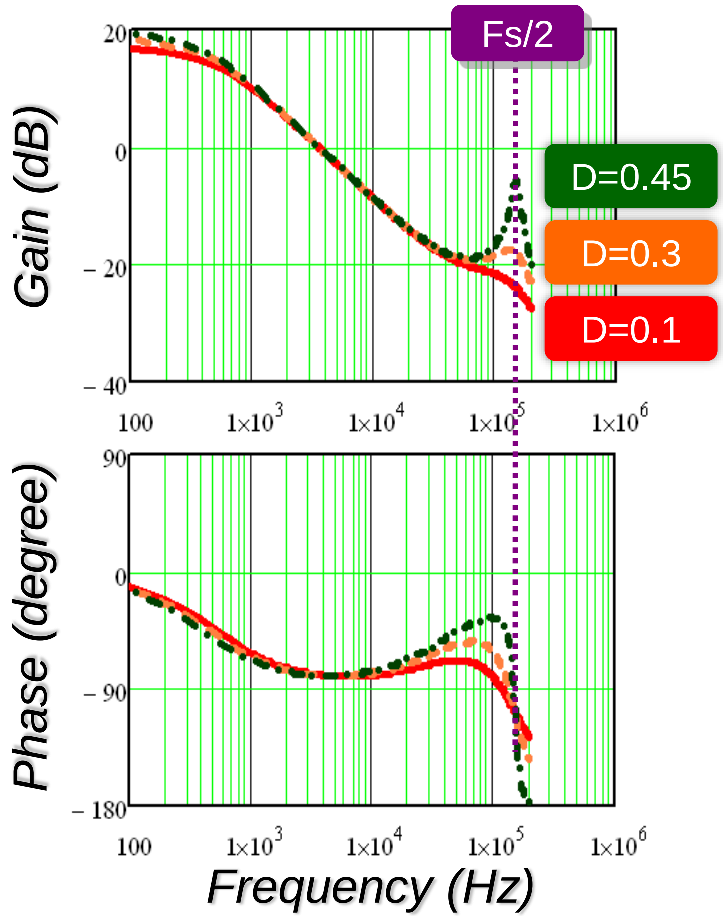
R. B. Ridley, "A new, continuous-time model for current-mode control," IEEE Trans. on Power Electronics, 1991.

R. Ridley's Model

for Peak Current-Mode Control

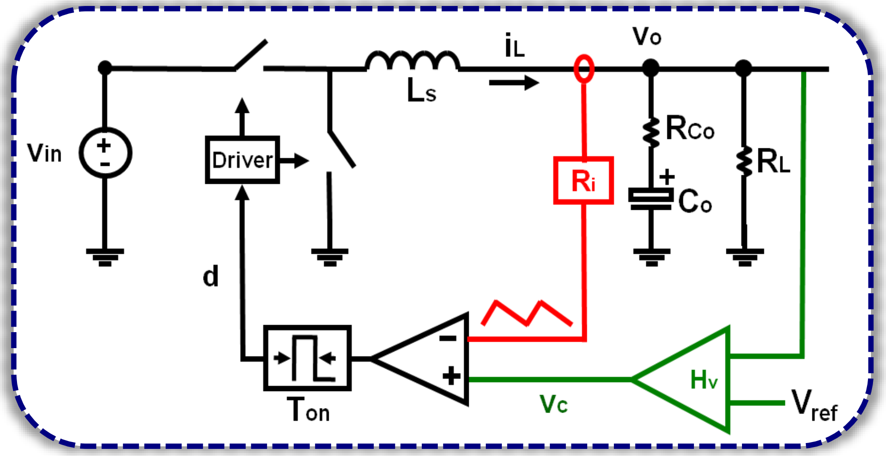


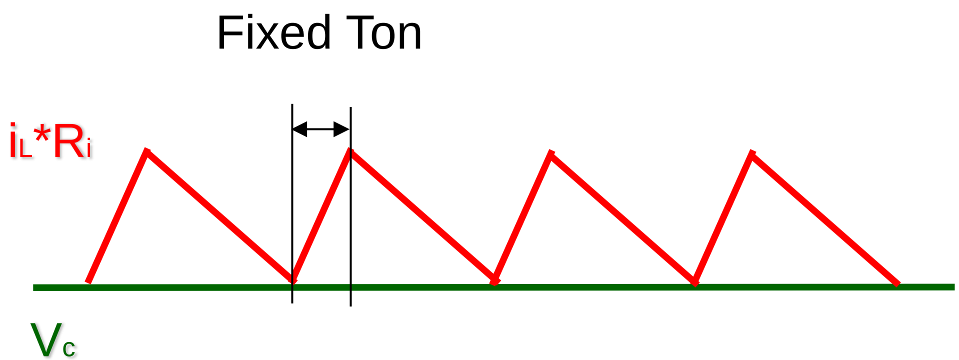
Control to vo Transfer Function

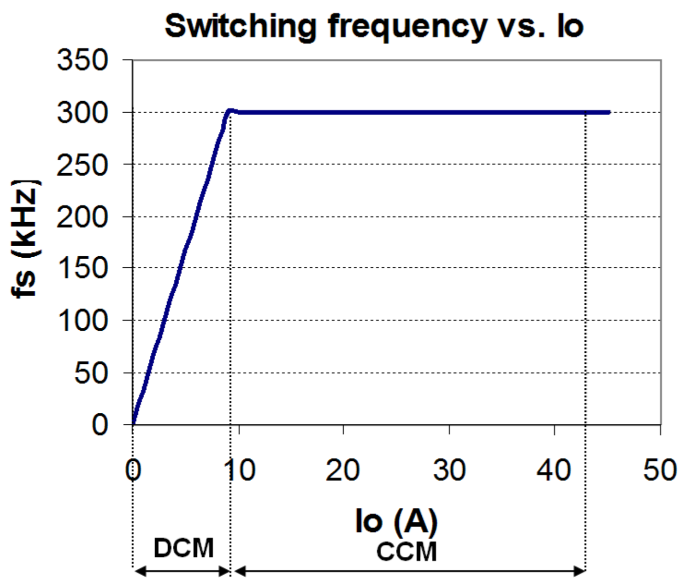


- R. Ridley's model is good for constant frequency case:
  1. Peak current model control
  2. Valley current mode control

Constant On-time Control

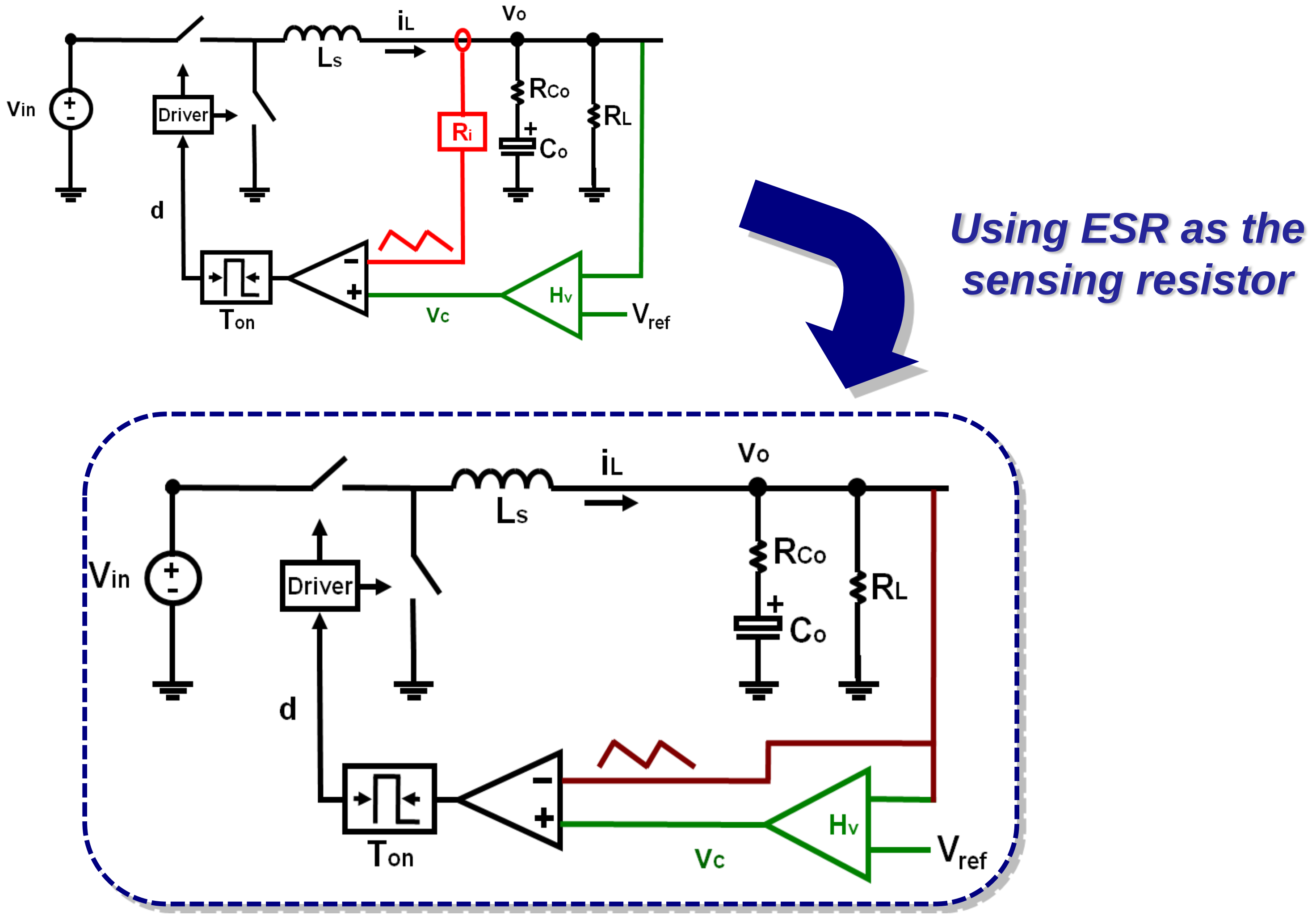






Constant on-time control is widely used to improve light-load efficiency

V^2 Type Implementation

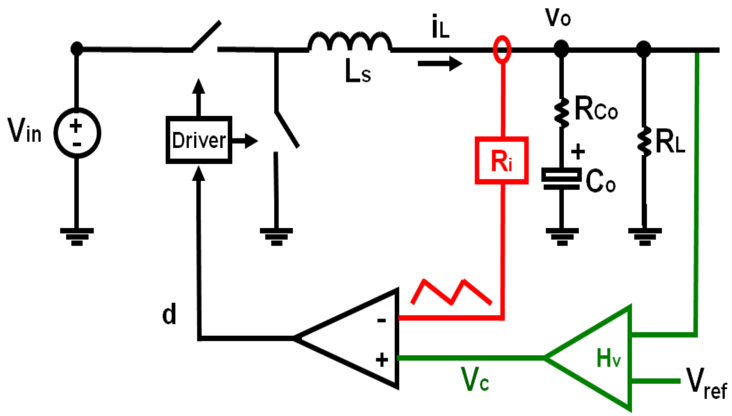


Benefits: 1. Simple implementation; 2. Good efficiency @ light load

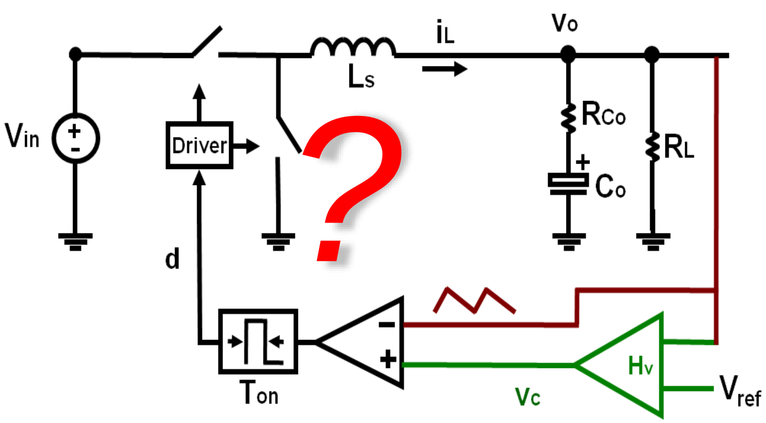
3. Fast transient response

Can R. Ridley's Model be Extended to Variable Freq.?

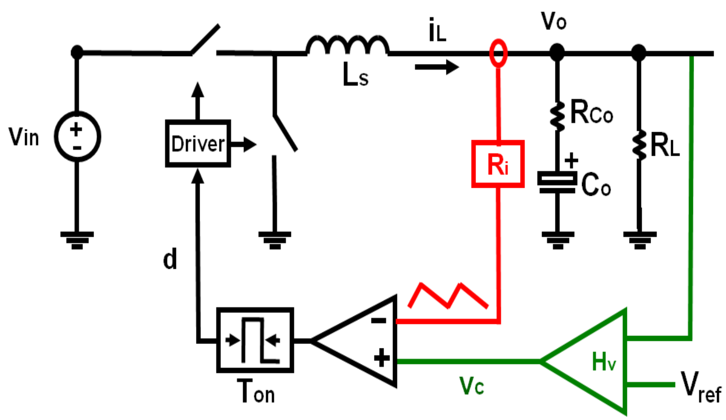
- *Peak Current-Mode Control*



- *V^2 Constant On-time Control*



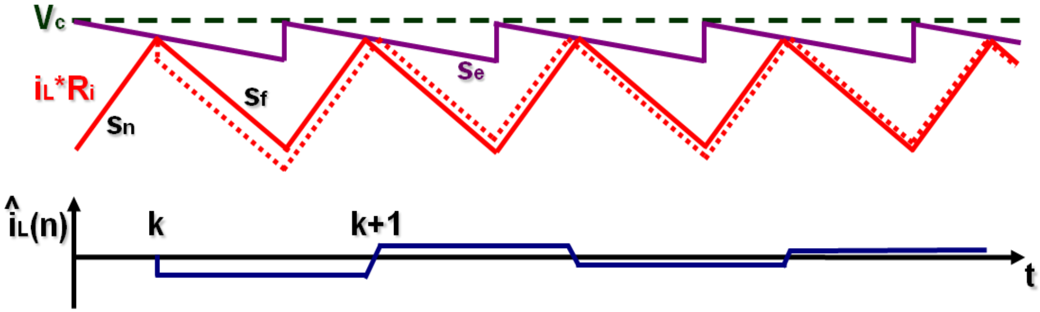
- *Constant On-time Control*



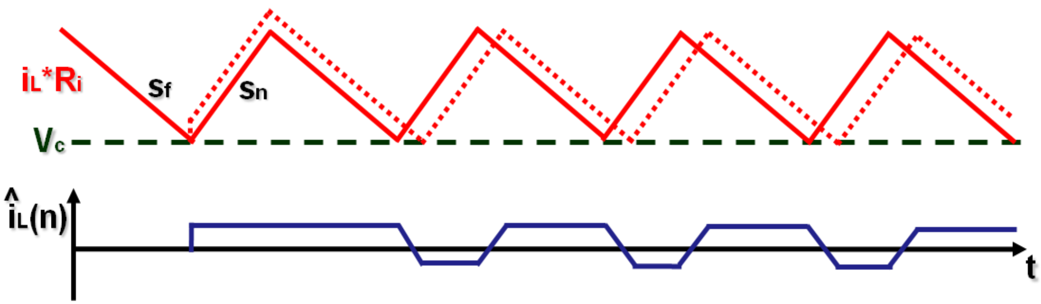
Extension of Ridley's Model

to Constant On-time Control*

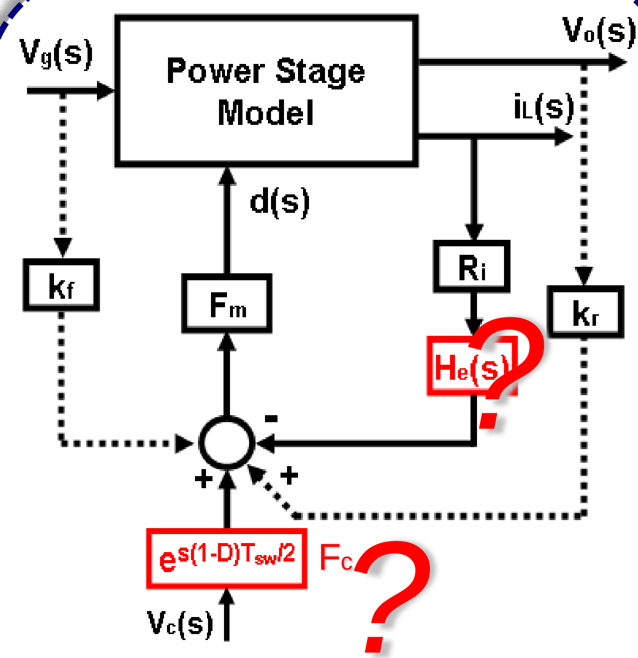
- *Peak Current-Mode control*



- *Constant On-time Control*



Variable frequency,
No dynamic exhibited,
No sample and hold effect.

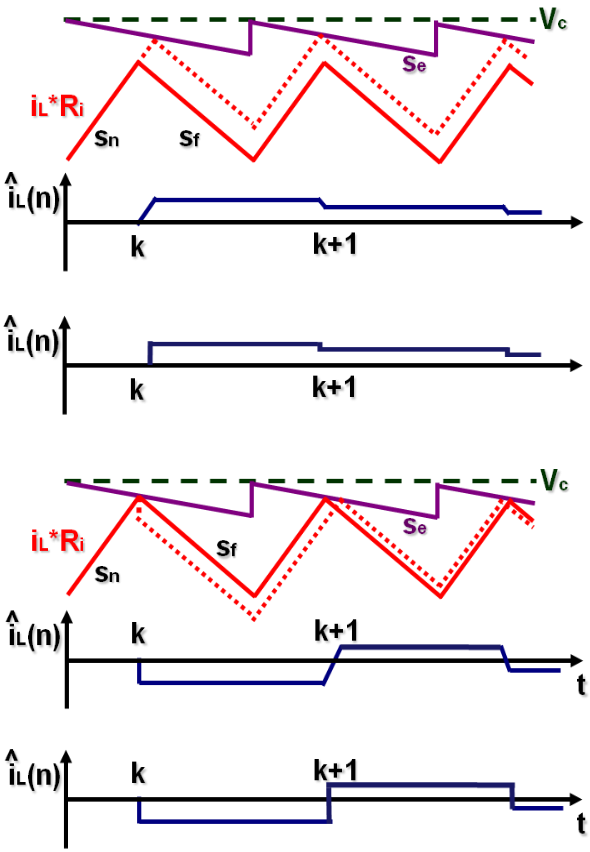


R. Ridley's model for constant on-time control

R. Ridley, "A new continuous-time model for current-mode control with constant frequency, constant on-time, and constant off-time, in CCM and DCM", 1990

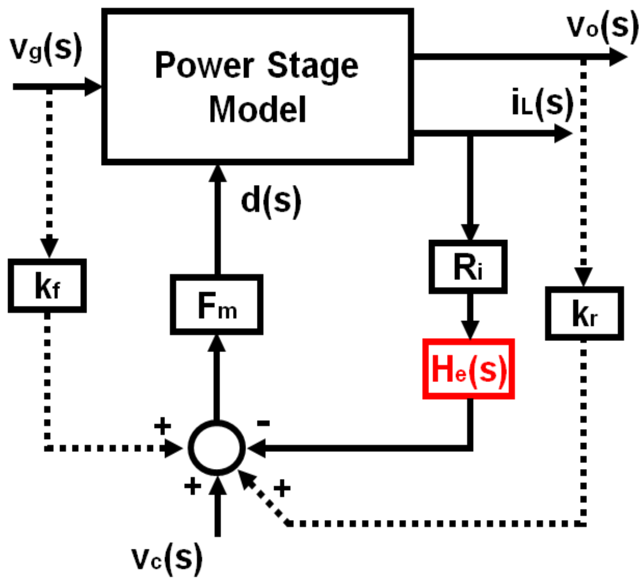
Fundamental Limitation
of R. Ridley's Model

➤ *Discrete-time analysis* (Sample-Data Model)



$$H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1+\alpha}{R_i}\frac{z}{z+\alpha}$$

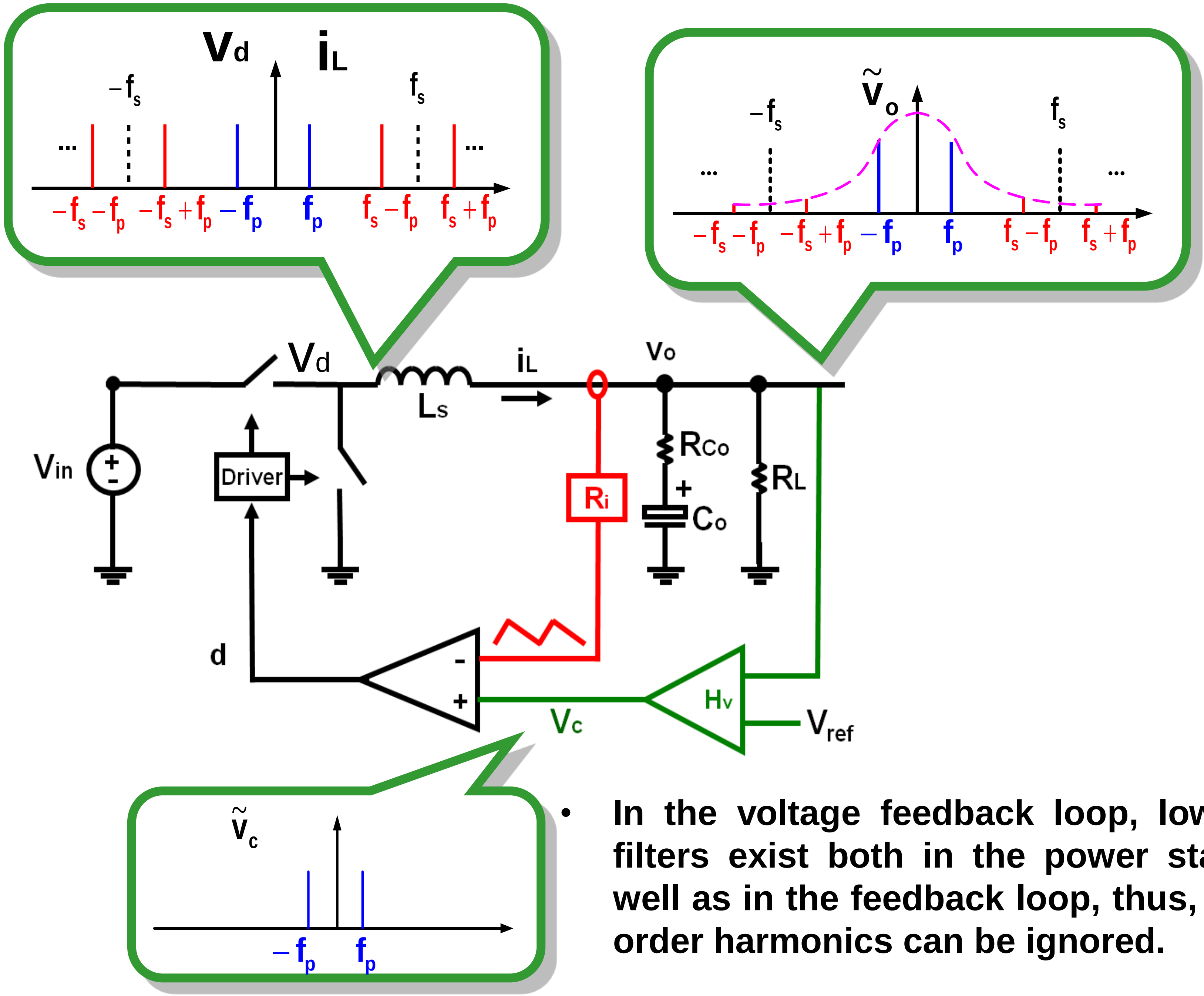
➤ *R. Ridley's model*



1. R. Ridley's model is based on discrete-time analysis (Sample-Data Model);
2. Discrete-time analysis is not applicable to Variable-frequency modulation;

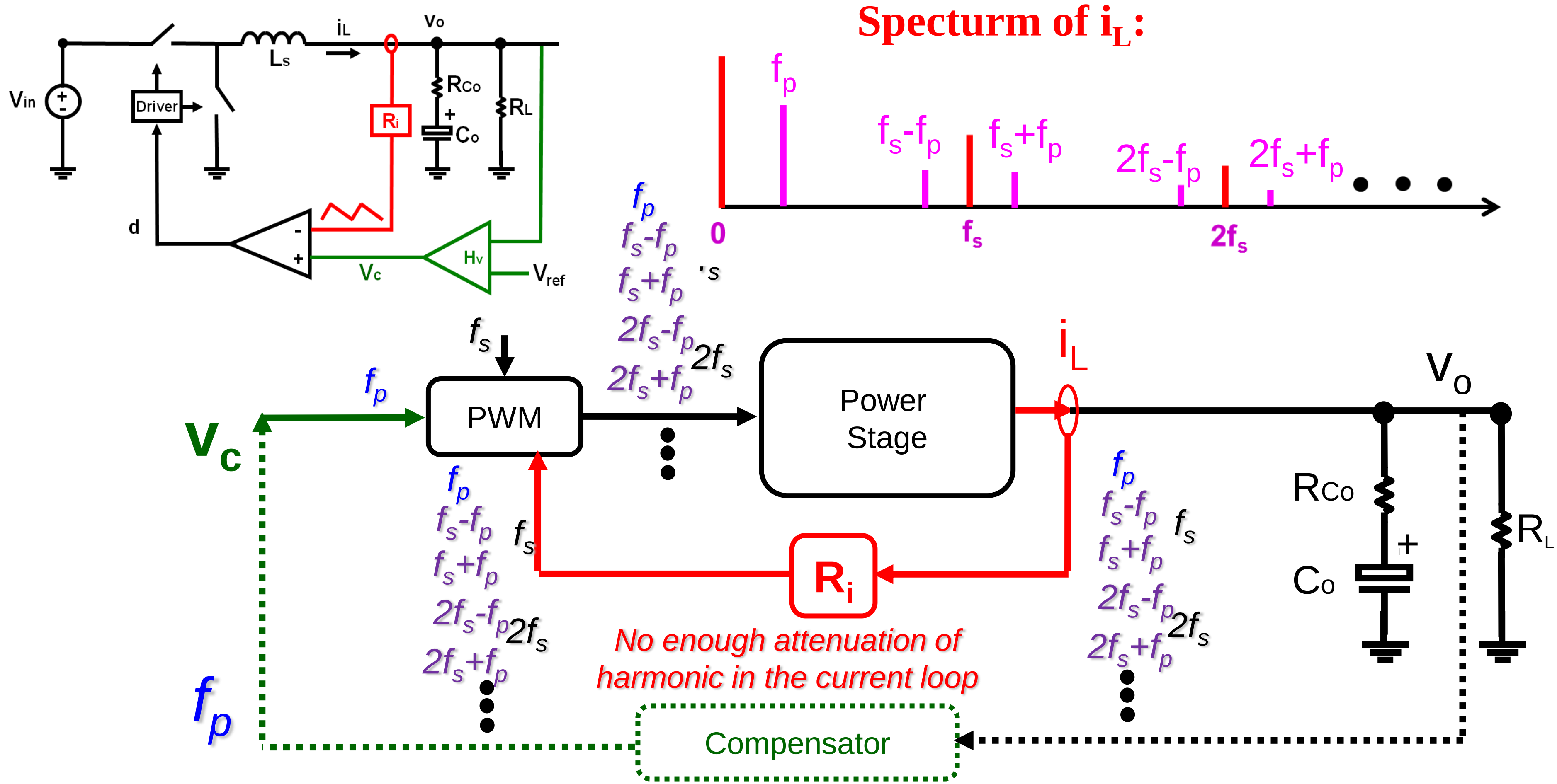
New Modeling Approach
for
Current-Mode Control

Frequency Spectrum in Voltage feedback



- In the voltage feedback loop, low pass filters exist both in the power stage as well as in the feedback loop, thus, higher order harmonics can be ignored.

Complexities Introduced by Current feedback

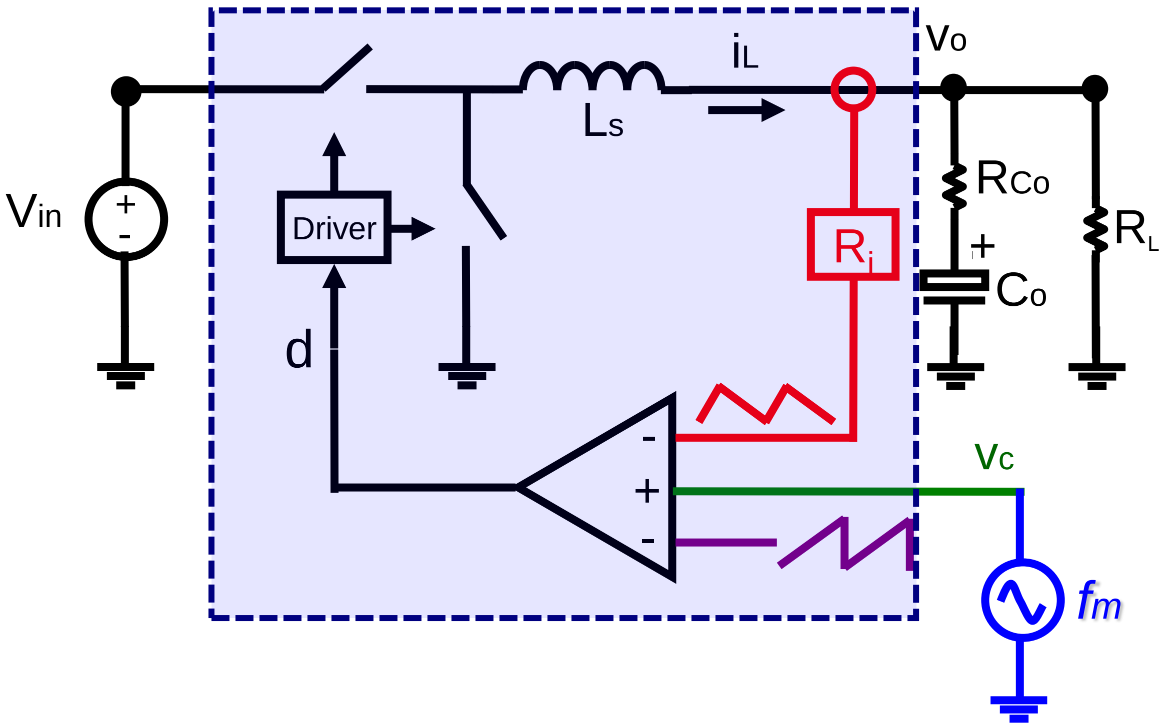


- No low-pass filter exists in the current feedback loop, thus higher order harmonics should be included in the modeling process
- Current-loop has to be considered as one single entity

New Modeling Strategy*

for Peak Current-Mode Control

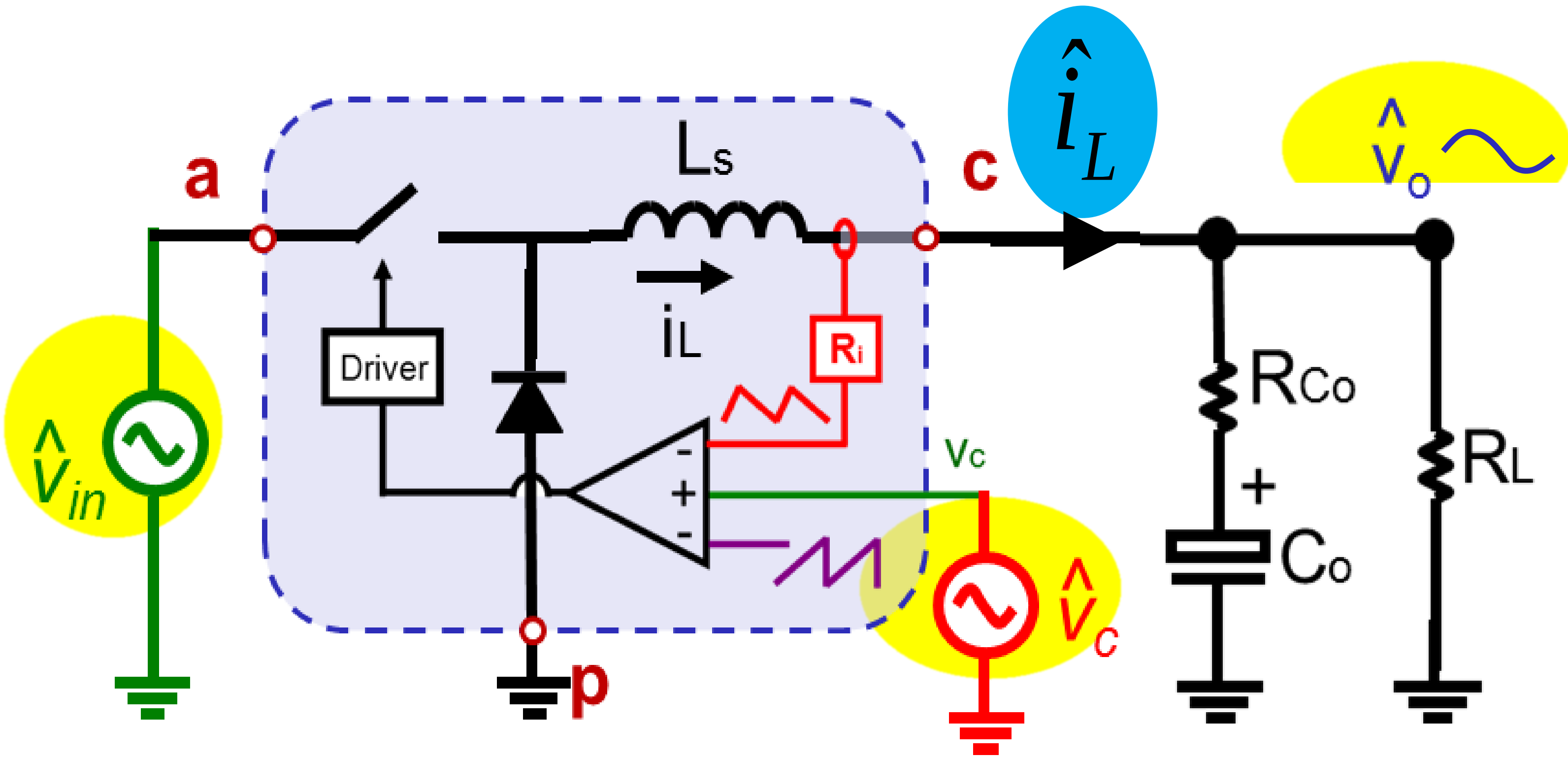
by Jian Li



- *Treat the inductor, modulator and switches as a single entity*
- *Describing function method is used*

* Jian Li; Lee, F.C.; , "New Modeling Approach and Equivalent Circuit Representation for Current-Mode Control," *Power Electronics, IEEE Transactions on* , vol.25, no.5, pp.1218-1230, May 2010

Modeling in Time Domain

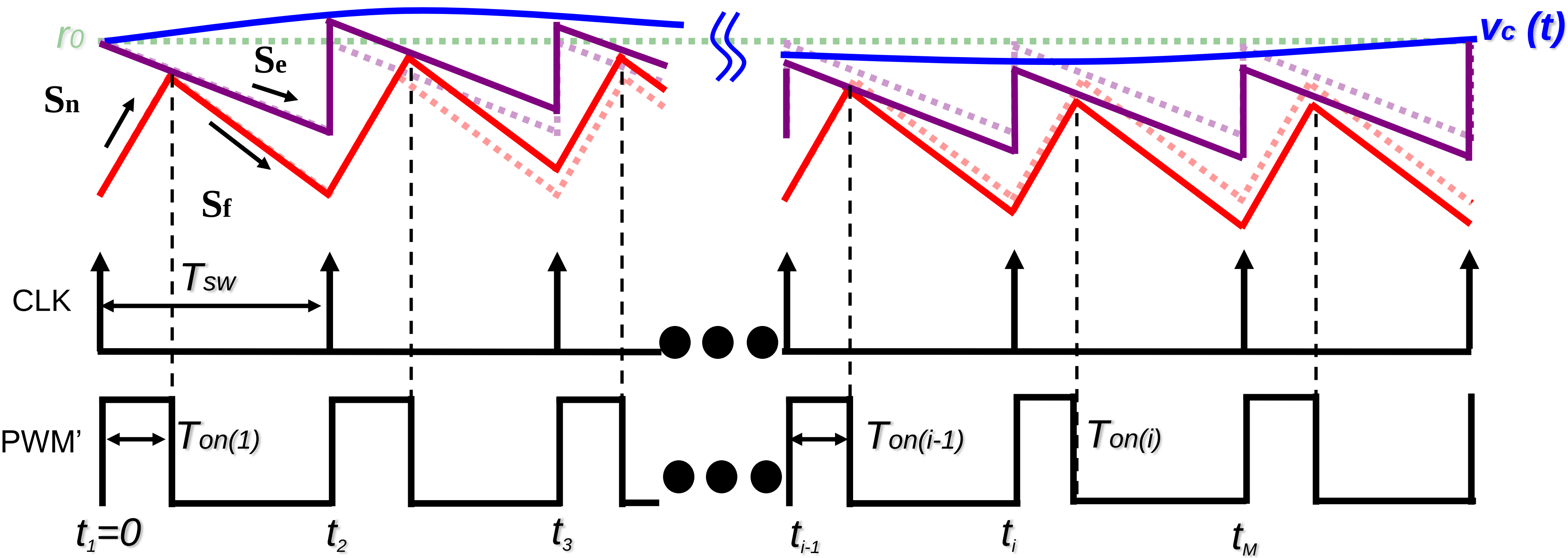


$$\hat{i}_L(s) \approx K_{vc}(s)\hat{v}_c(s) + K_{vo}(s)\hat{v}_o(s) + K_{vin}(s)\hat{v}_{in}(s)$$

K_{vc}, K_{vo} and K_{vin} are derived separately

* Jian Li; Lee, F.C.; , "New Modeling Approach and Equivalent Circuit Representation for Current-Mode Control," *Power Electronics, IEEE Transactions on* , vol.25, no.5, pp.1218-1230, May 2010

Small Signal Perturbation on v_c



- *Sinusoidal perturbation (small signal):*

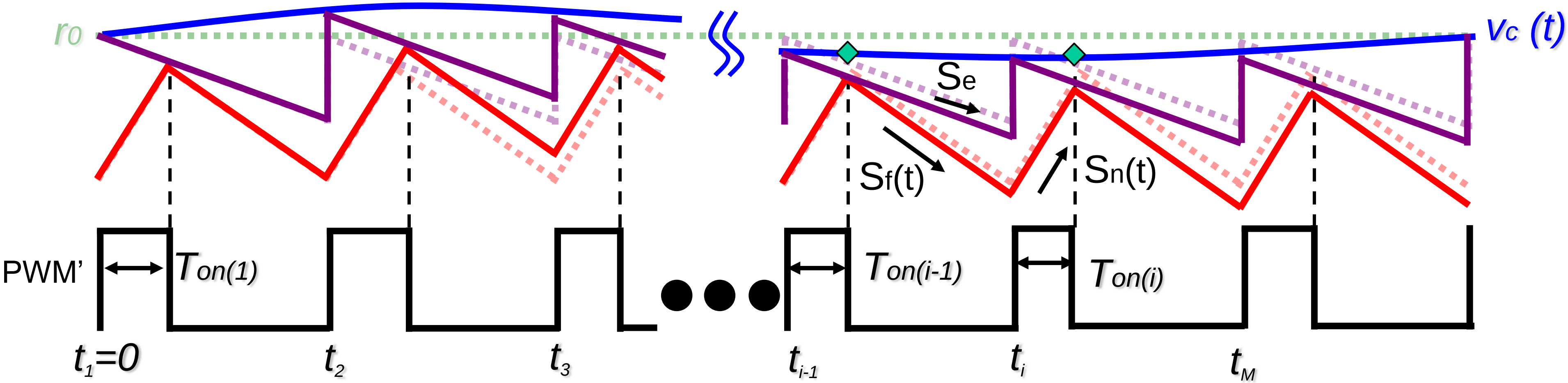
$$v_c(t) = r_0 + \hat{r}\sin(2\pi f_m \cdot t + \theta)$$

- On-time calculation: $$T_{on(i)} = T_{on} + \Delta T_{on(i)}$$

where

$$(s_e + s_n)\Delta T_{on(i)} = -\hat{v}_c(t_{i-1} + T_{on(i-1)}) + \hat{v}_c(t_i + T_{on(i)}) + (s_e - s_f)\Delta T_{on(i-1)}$$

Calculation of Inductor Current



➤ **Calculation of Duty cycle waveform**

$$d(t)\Big|_{0\le t\le NT_m} = \sum_{i=1}^{M}[u(t-t_i)-u(t-t_i-T_{on(i)})]$$

Where, u(t)=1, when t>0.

➤ **Calculation of inductor current:**

$$i_L(t) = \int_0^t \left\{ \frac{V_{in}-V_o}{L_s} d(t) - \frac{V_o}{L_s}[1-d(t)] \right\} \cdot dt + I_{L0}$$

$$= \int_0^t \left\{ \frac{V_{in}}{L_s} d(t) - \frac{1}{L_s} V_o \right\} \cdot dt + I_{L0}$$

Where, I_{L0} is the initial value of the inductor current.

Fourier Analysis of Inductor Current

➢ Calculation of the Fourier coefficient at fm for inductor current $i_L(t)$ and $v_c(t)$:

For $\hat{i}_{L(f_m)}$ ➡ $c_{m(i_L)} = j\dfrac{2f_m}{N}\displaystyle\int_0^{NT_m} i_L(t)\cdot e^{-j2\pi f_m t}dt$

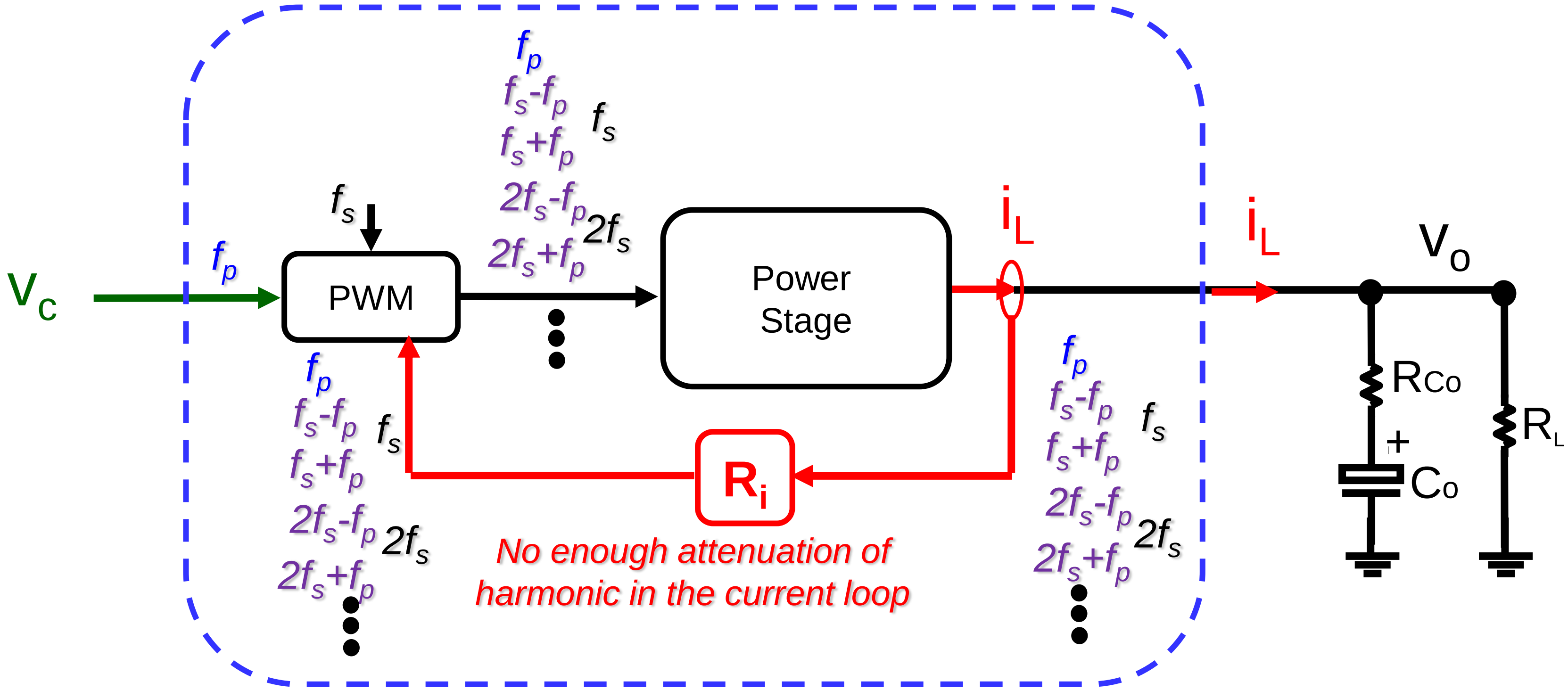
$$c_{m(iL)} = \hat{r}\cdot e^{-j\theta}\cdot\frac{f_s(1-e^{-j2\pi f_m T_{sw}})}{(s_n+s_e)+(s_f-s_e)e^{-j2\pi f_m T_{sw}}}\frac{V_{in}}{L_s\cdot j2\pi f_m}$$

For $\hat{v}_{c(f_m)}$ ➡ $c_{m(v_c)} = \hat{r}e^{-j\theta}$

➢ Calculation of describing function:

$$\frac{\hat{i}_{L(f_m)}}{\hat{v}_{c(f_m)}} = \frac{c_{m(i_L)}}{\hat{r}e^{-j\theta}} = \frac{f_s(1-e^{-j2\pi f_m T_{sw}})}{(s_n+s_e)+(s_f-s_e)e^{-j2\pi f_m T_{sw}}}\frac{V_{in}}{L_s\cdot j2\pi f_m}$$

Control-to-i_L Transfer Function

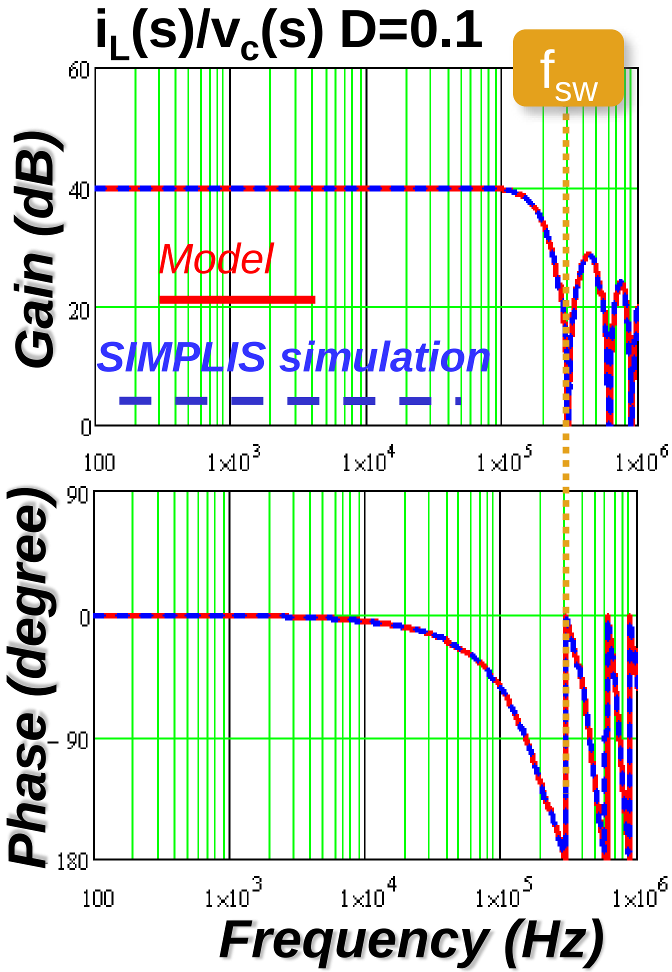


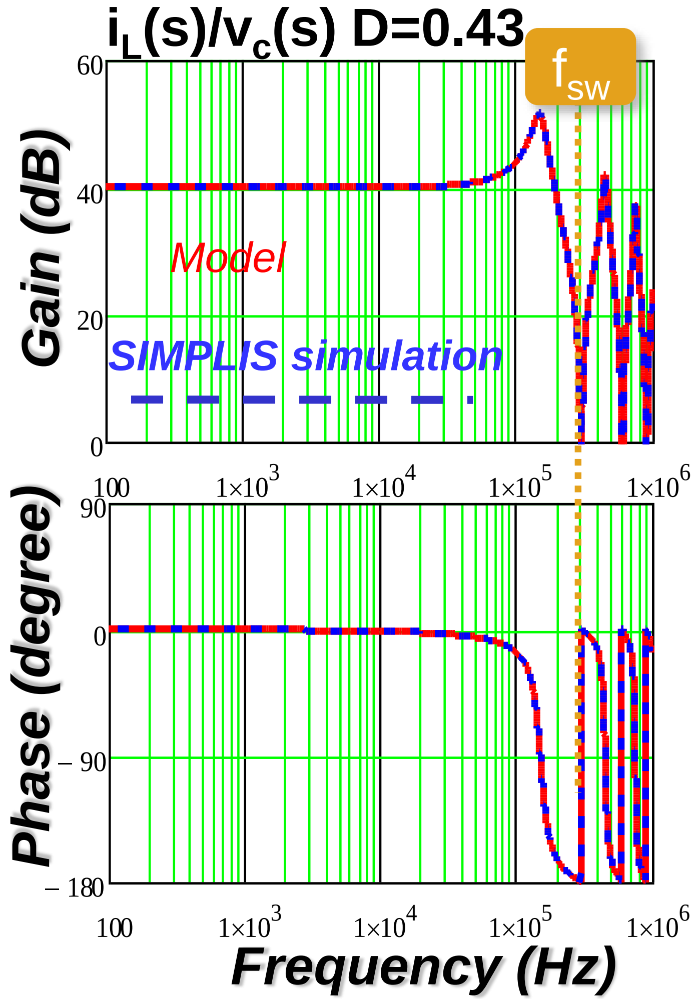
$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{f_s(1-e^{-sT_{sw}})}{(s_n+s_e)+(s_f-s_e)e^{-sT_{sw}}}\frac{V_{in}}{L_s s}$$

$$s = j2\pi f_m$$

Control-to-i_L Transfer Function

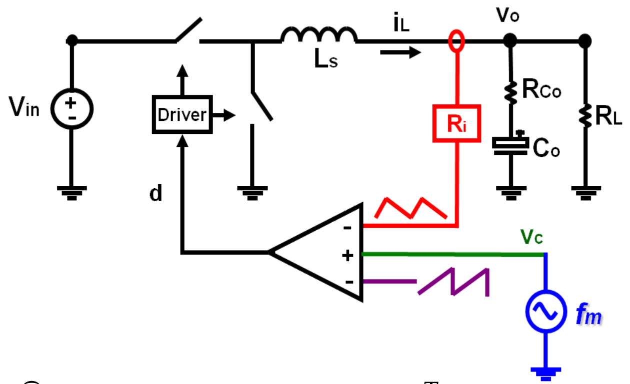
$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{f_s}{(s_n + s_e) + (s_f - s_e)e^{-sT_{sw}}} \frac{V_{in}}{L_s s} \left(1 - e^{-sT_{sw}}\right)$$





Control-to-i_L Transfer Function

➢ *New modeling result*



$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{f_s(1-e^{-sT_{sw}})}{(s_n+s_e)+(s_f-s_e)e^{-sT_{sw}}}\frac{V_{in}}{L_s s}$$

$$\boxed{\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{1}{R_i}\frac{1+\alpha}{sT_{sw}}\frac{e^{sT_{sw}}-1}{e^{sT_{sw}}+\alpha}}$$

where, $\alpha = (s_f - s_e)/(s_n + s_e)$

$=$

➢ *Sample-Data Model result*

$$\boxed{H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1+\alpha}{R_i}\frac{z}{z+\alpha}}$$

to

➢ *Continuous*

$$\boxed{\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{1}{R_i}\frac{1+\alpha}{sT_{sw}}\frac{e^{sT_{sw}}-1}{e^{sT_{sw}}+\alpha}}$$

The result is the same as previous Sample-Data analysis

Approximation

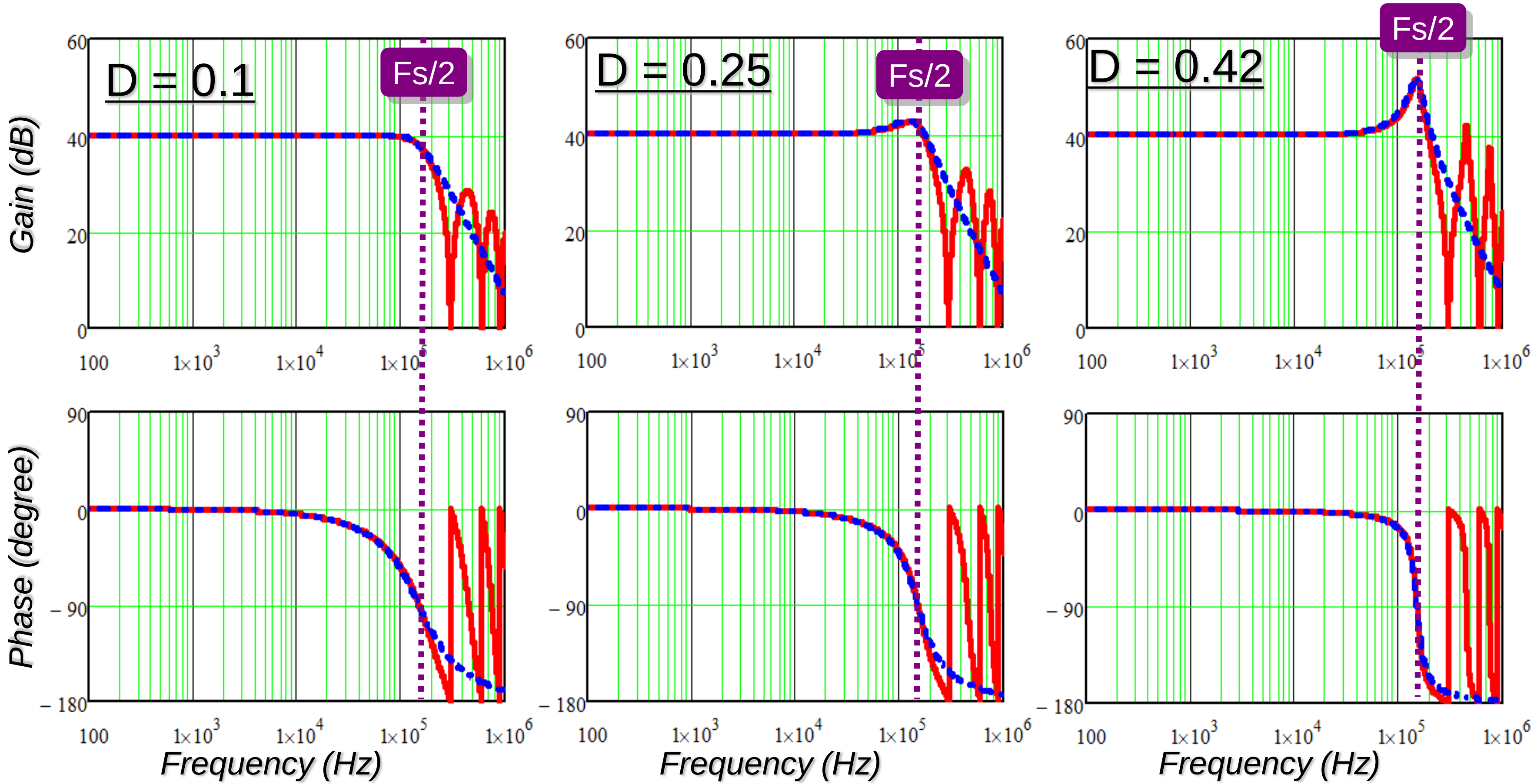
$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{1}{R_i}\frac{1+\alpha}{sT_{sw}}\frac{e^{sT_{sw}}-1}{e^{sT_{sw}}+\alpha}$$

$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} \approx \frac{1}{R_i}\frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}$$

where,

$$\omega_2 = \frac{\pi}{T_{sw}} \quad Q_2 = \frac{1}{\pi(\frac{s_n+s_e}{s_n+s_f}-0.5)}$$

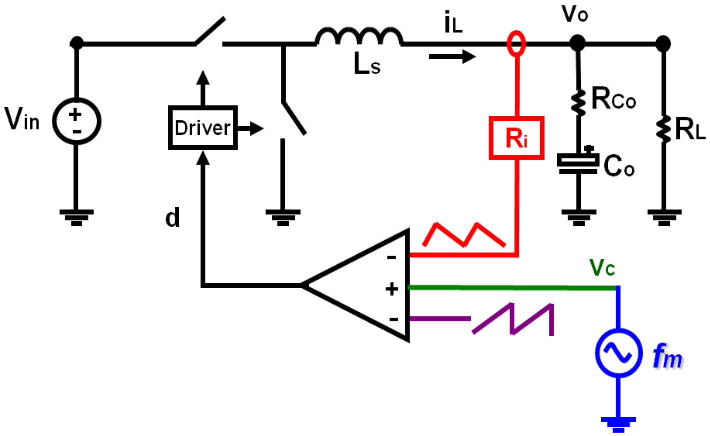
Pade approximation



◆ *Exact form* ◆ *Approximation*

Fourier Analysis of Output Voltage

➢ *Calculation of the Fourier coefficient at fm for output voltage $v_o(t)$:*

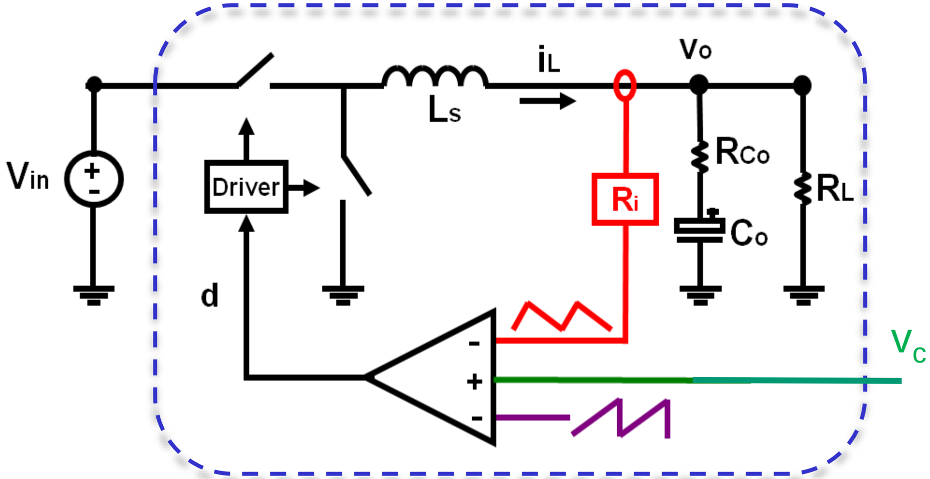


$$c_{m(vo)} = c_{m(iL)} \frac{R_L(R_{Co}C_o \cdot j2\pi f_m + 1)}{(R_L + R_{Co})C_o \cdot j2\pi f_m + 1}$$

➢ *Calculation of describing function:*

$$\frac{\hat{v}_{o(f_m)}}{\hat{v}_{c(f_m)}} = \frac{f_{sw} \cdot [1 - e^{-j(2\pi f_m \cdot T_{sw})}]}{(s_n + s_e) + (s_f - s_e)e^{-j \cdot 2\pi f_m \cdot T_{sw}}} \frac{V_{in}}{j2\pi f_m \cdot L_s} \frac{R_L(R_{Co}C_o \cdot j2\pi f_m + 1)}{(R_L + R_{Co})C_o \cdot j2\pi f_m + 1}$$

Simplified Control-to-v_o Transfer Function



$$\frac{\hat{v}_o(s)}{\hat{v}_c(s)} \approx K_p' \frac{R_{Co}C_o s + 1}{s/\omega_p' + 1} \frac{1}{1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2}}$$

where, $K_p' = \frac{R_L}{R_i}$ $\omega_p' = \frac{1}{(R_L + R_{Co})C_o}$ $\omega_2 = \frac{\pi}{T_{sw}}$ $Q_2 = \frac{1}{\pi(M_c D' - 0.5)}$

Peak Current Mode Control

w/o External Ramp

- *Peak Current Mode Control*

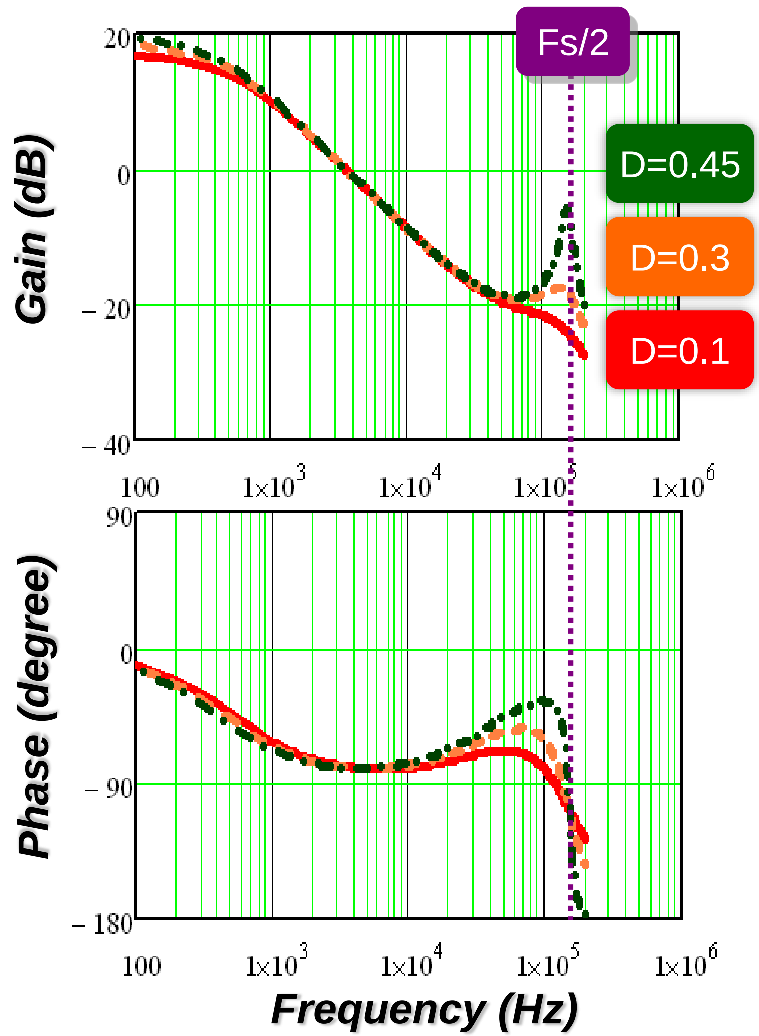
$$\frac{v_o(s)}{v_c(s)} \approx K_p' \frac{R_{Co}C_o s+1}{s/\omega_p'+1} \frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}$$

where

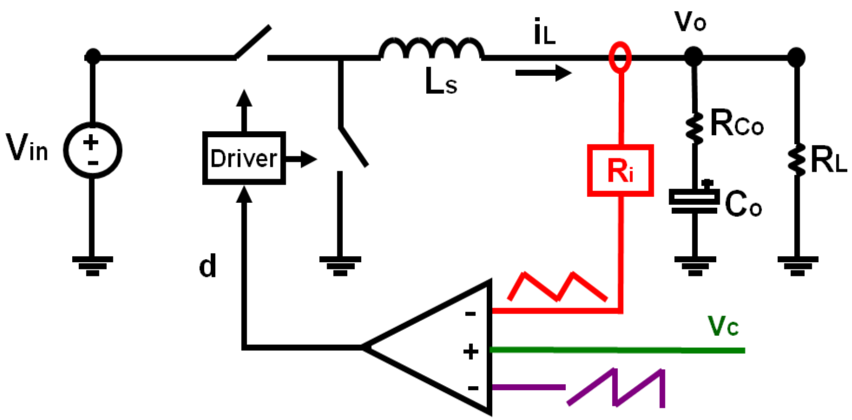
$$K_p' = \frac{R_L}{R_i} \qquad \omega_p' = \frac{1}{(R_L+R_{Co})C_o}$$

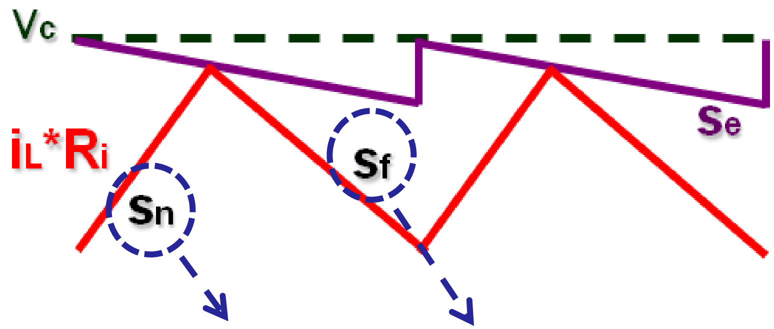
$$\omega_2 = \frac{\pi}{T_{sw}} \qquad Q_2 = \frac{1}{\pi(M_c D' - 0.5)}$$

Control-to-output Transfer Function



Inductor Current Slope Variation

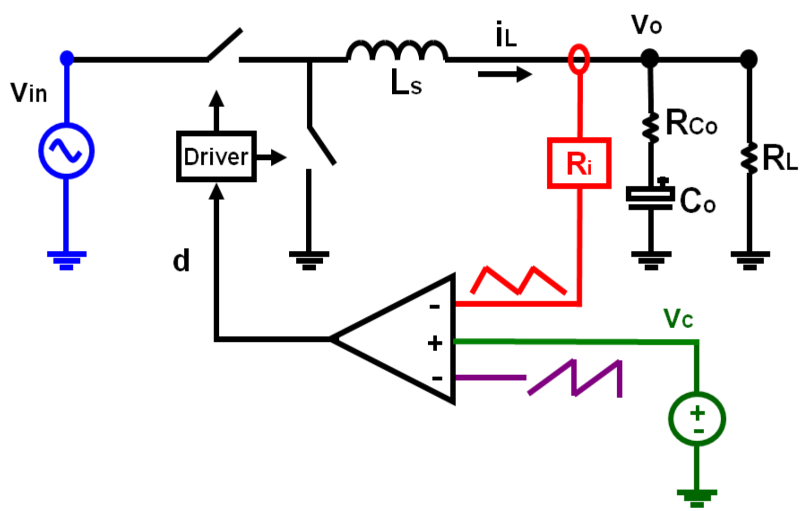




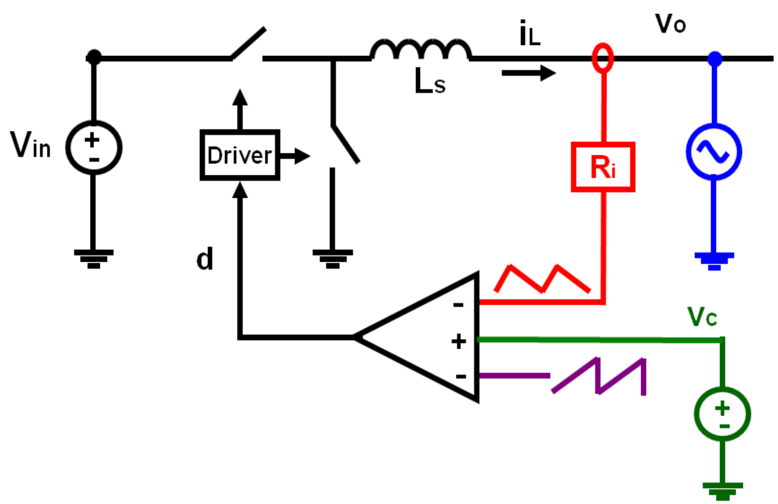
$$s_n = \frac{v_{in} - v_o}{L_s} R_i \qquad s_f = \frac{v_o}{L_s} R_i$$

- *Additional feed-forward and feedback gain should be considered*

- *The influence of the input voltage*

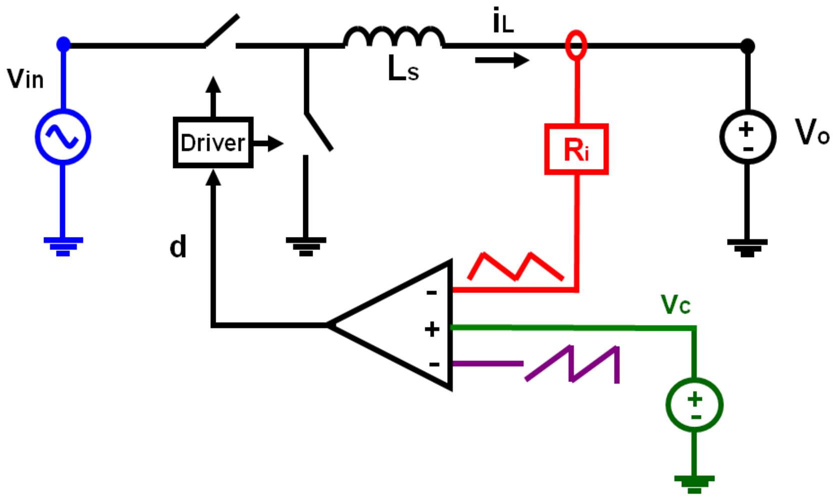


- *The influence of the output voltage*

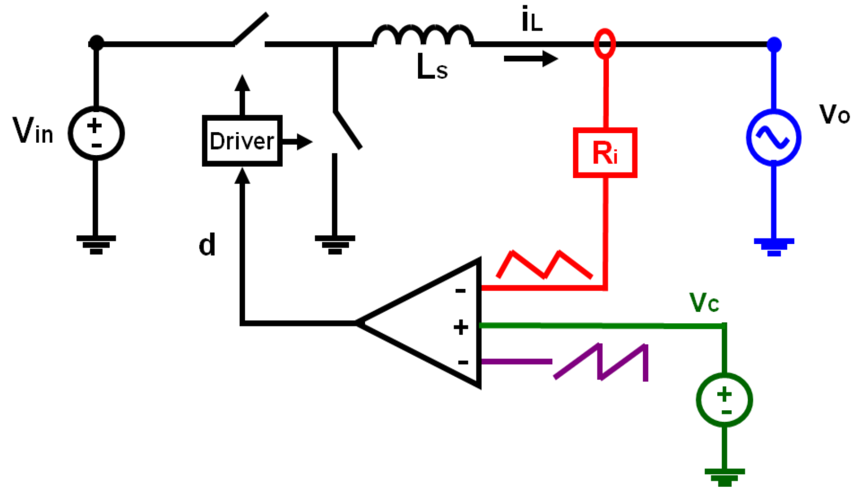


Similar Derivation Process

- *The influence of the input voltage*



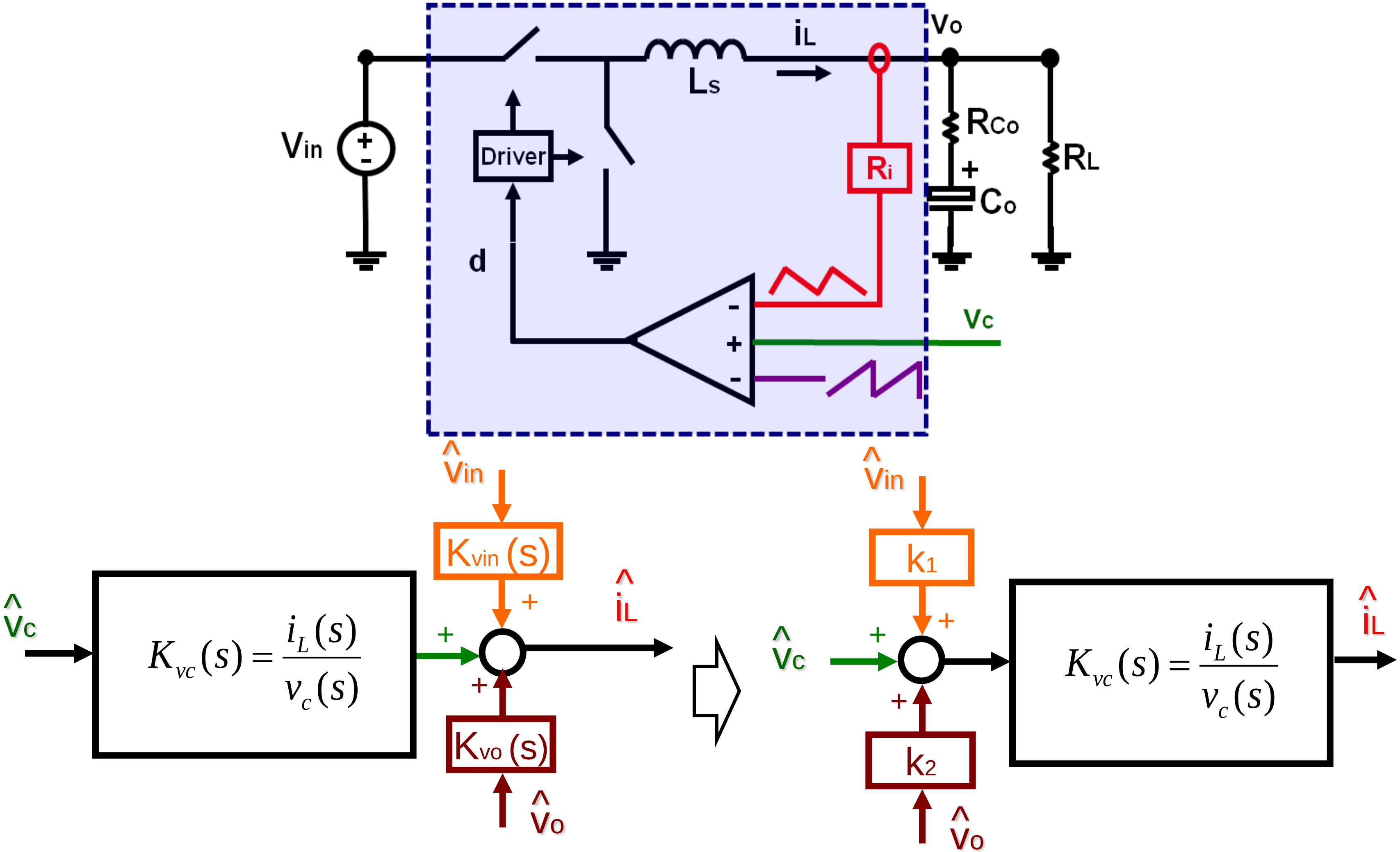
- *The influence of the output voltage*



$$K_{vin}(s)=\frac{i_L(s)}{v_{in}(s)}=\frac{1}{L_s s}[-\frac{f_s}{(s_n+s_e)+(s_f-s_e)e^{-sT_{sw}}}\frac{(1-e^{-sT_{on}})}{s\cdot L_s/R_i}\cdot V_{in}+D]$$

$$K_{vo}(s)=\frac{i_L(s)}{v_o(s)}=\frac{1}{L_s s}\cdot[\frac{f_s(1-e^{-sT_{sw}})}{(s_n+s_e)+(s_f-s_e)e^{-sT_{sw}}}\cdot\frac{1}{s\cdot L_s/R_i}\cdot V_{in}-1]$$

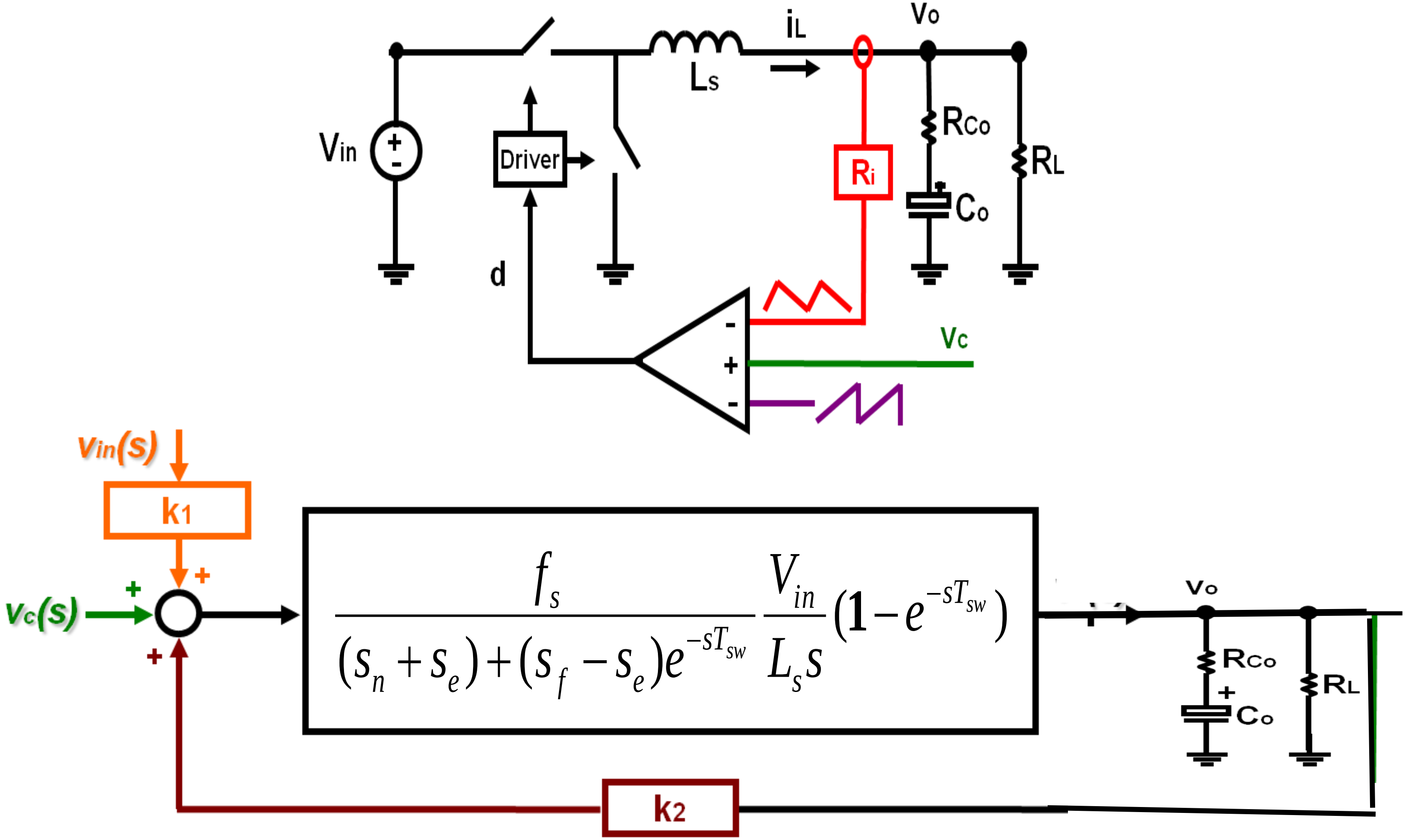
Complete Model



$$k_1(s)=\frac{K_{vin}(s)}{K_{vc}(s)}\approx\frac{DR_i}{L_s}\left[\frac{1}{Q_2\omega_2}-\frac{T_{off}}{2}\right]\doteq k_1$$

$$k_2(s)=\frac{K_{vo}(s)}{K_{vc}(s)}\approx-\frac{R_i}{L_sQ_2\omega_2}\doteq k_2$$

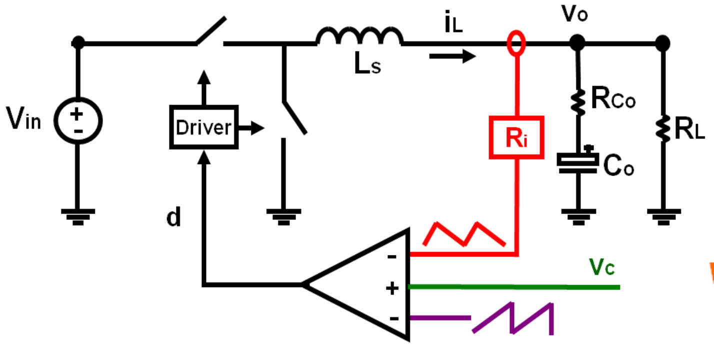
Complete Model



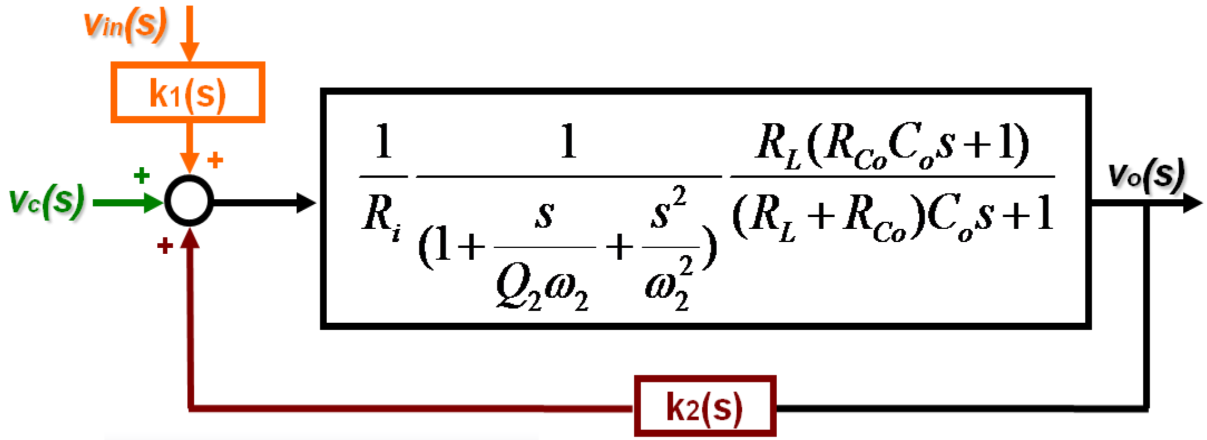
$$k_1(s)=\frac{K_{vin}(s)}{K_{vc}(s)}\approx\frac{DR_i}{L_s}\left[\frac{1}{Q_2\omega_2}-\frac{T_{off}}{2}\right]\doteq k_1$$

$$k_2(s)=\frac{K_{vo}(s)}{K_{vc}(s)}\approx-\frac{R_i}{L_s Q_2\omega_2}\doteq k_2$$

Complete Model with Approximation



$$k_1(s) \approx \frac{DR_i}{L_s}\left[\frac{1}{Q_2\omega_2} - \frac{T_{off}}{2}\right] \qquad k_2(s) \approx -\frac{R_i}{L_s Q_2 \omega_2}$$



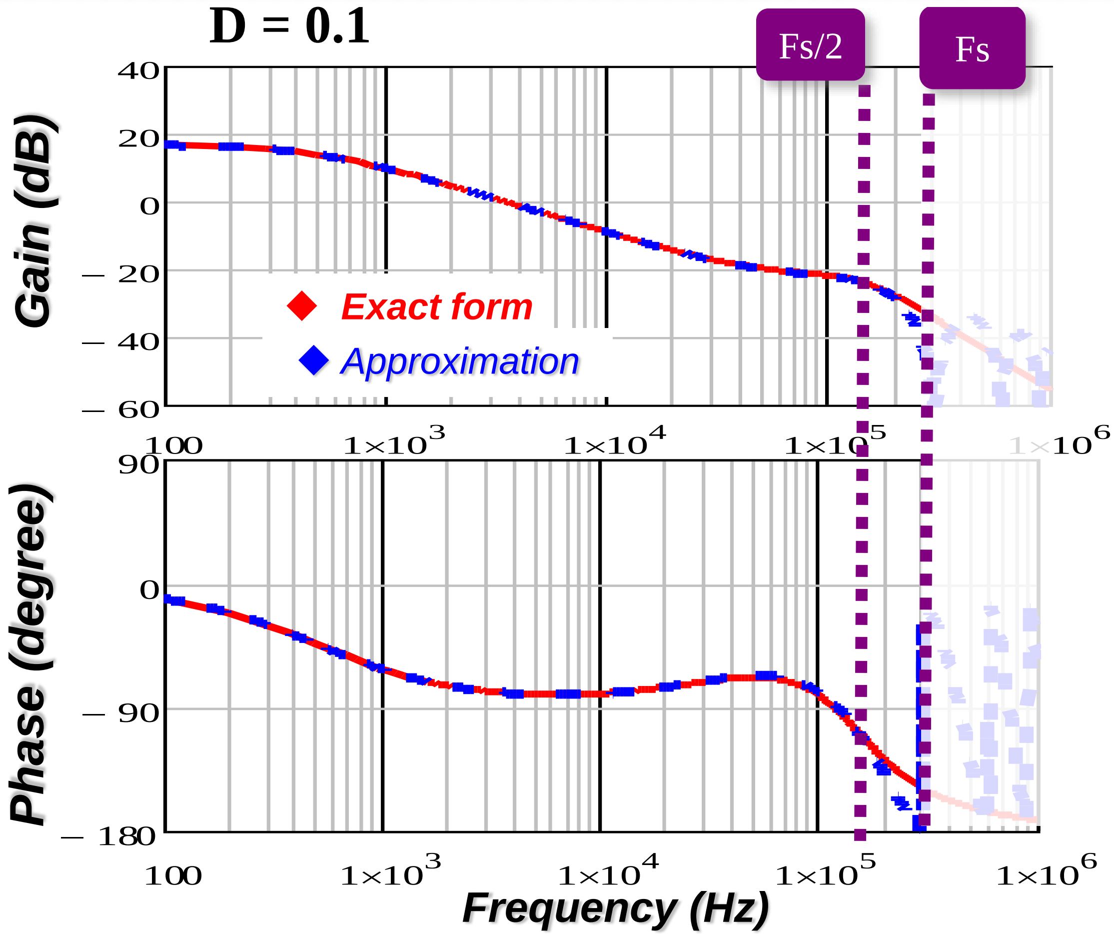
$$\frac{v_o(s)}{v_c(s)} \approx K_p \frac{R_{Co}C_o s+1}{s/\omega_p+1} \frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}$$

where, $K_p = \frac{R_L}{R_i(1-k_2/R_i \cdot R_L)}$ $\quad \omega_p = \frac{1-R_L k_2/R_i}{(R_L+R_{Co})C_o - R_L R_{Co} C_o k_2/R_i}$ $\quad \omega_2 = \frac{\pi}{T_{sw}}$ $\quad Q_2 = \frac{1}{\pi(\frac{S_n+S_e}{S_n+S_f}-0.5)}$

Simplified Model

$$\frac{v_o(s)}{v_c(s)} \approx K_p \frac{R_{Co}C_o s+1}{s/\omega_p+1}\frac{1}{1+\dfrac{s}{Q_2\omega_2}+\dfrac{s^2}{\omega_2^2}}$$

where, $\omega_2 = \dfrac{\pi}{T_{sw}}$ $\quad Q_2 = \dfrac{1}{\pi(\dfrac{s_n+s_e}{s_n+s_f}-0.5)}$

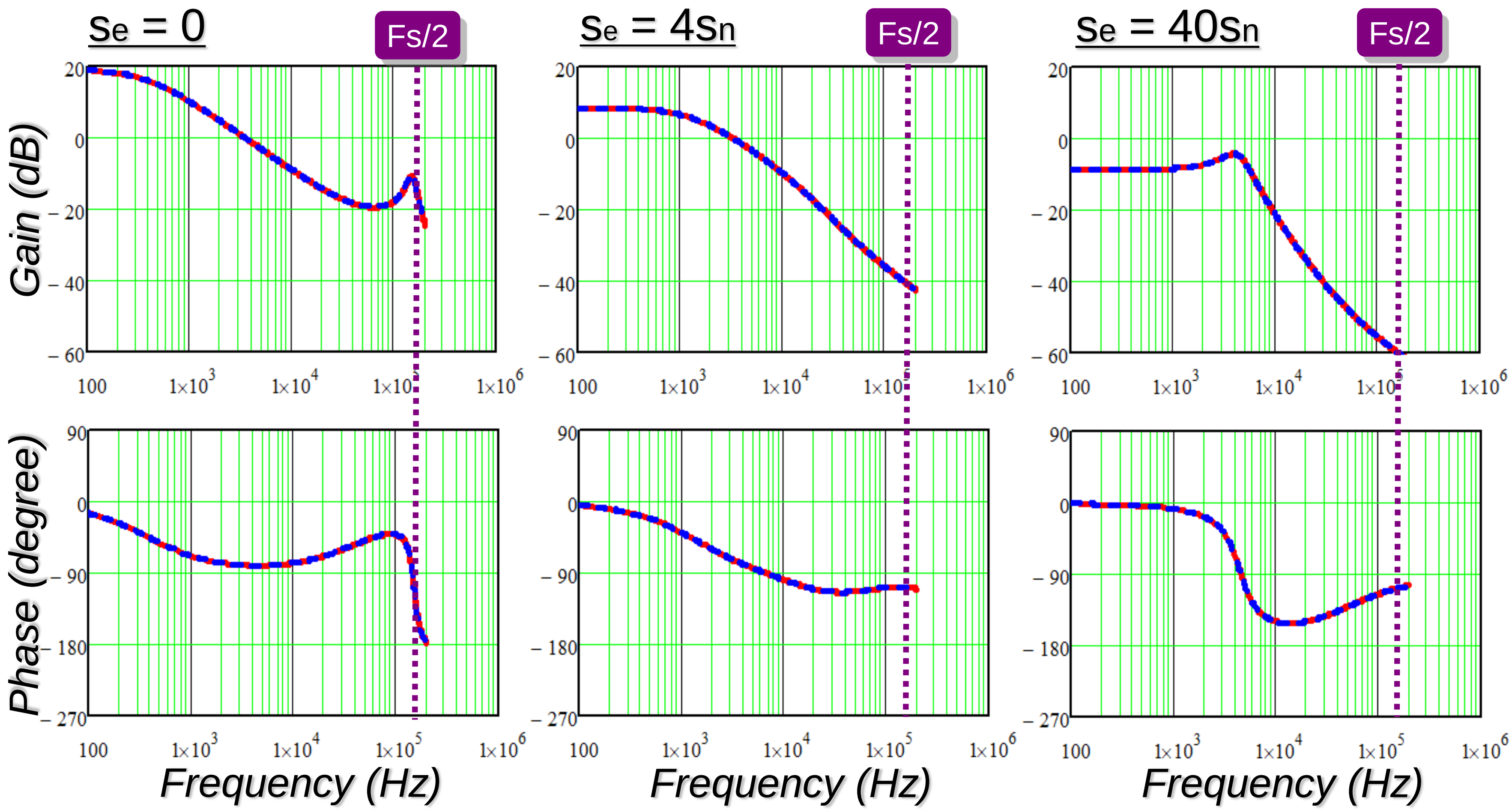


Model accurate at 1/2 f_s, and useful up to f_s

Simulation Verification
for Peak Current-Mode Control

CPES

Control-to-output Transfer Function (D = 0.42)

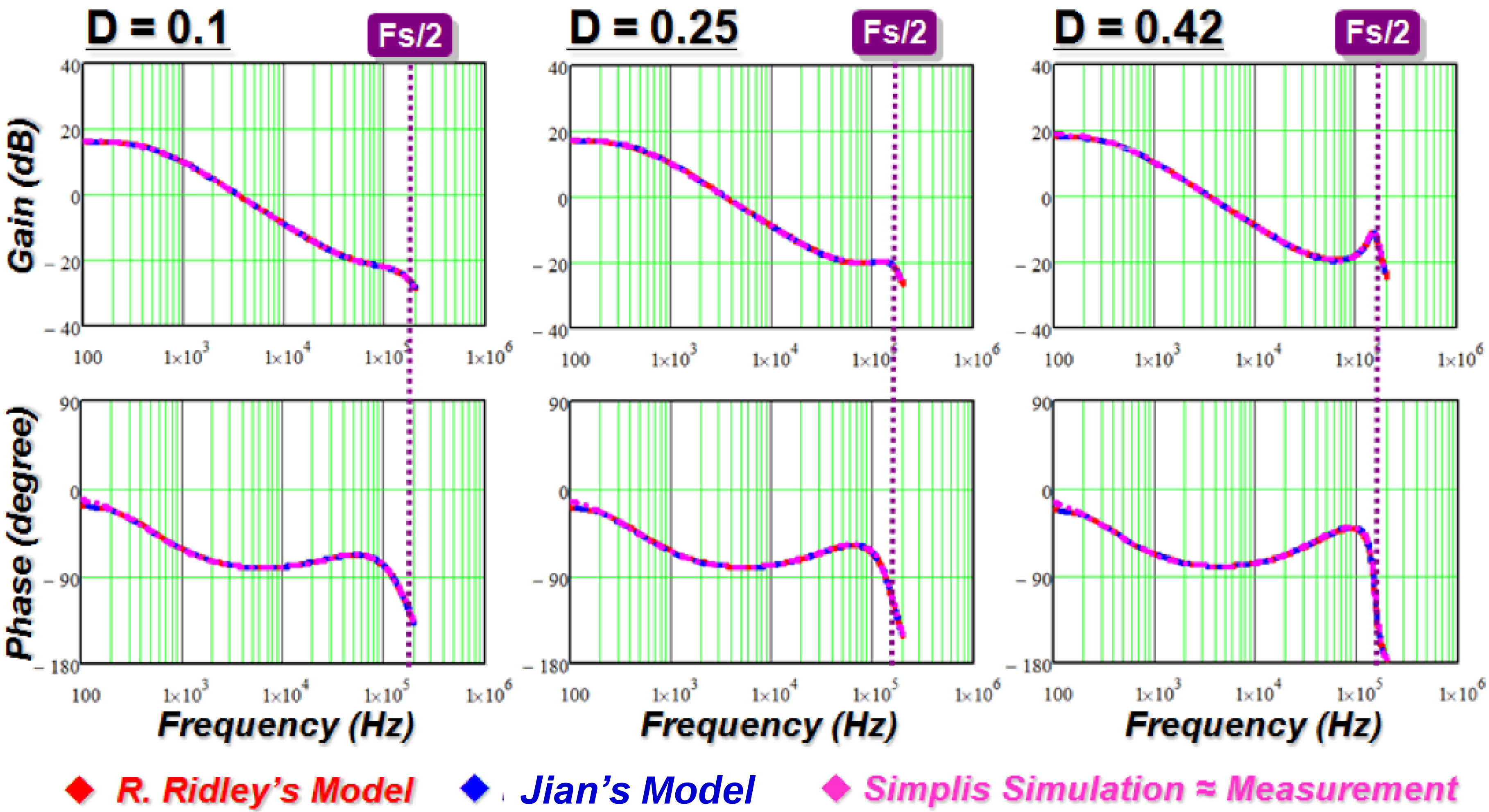


◆ *Jian's Model* ◆ *Simplis Simulation ≈ Measurement*

Model Comparison

Control-to-output transfer function of Peak Current Mode Control

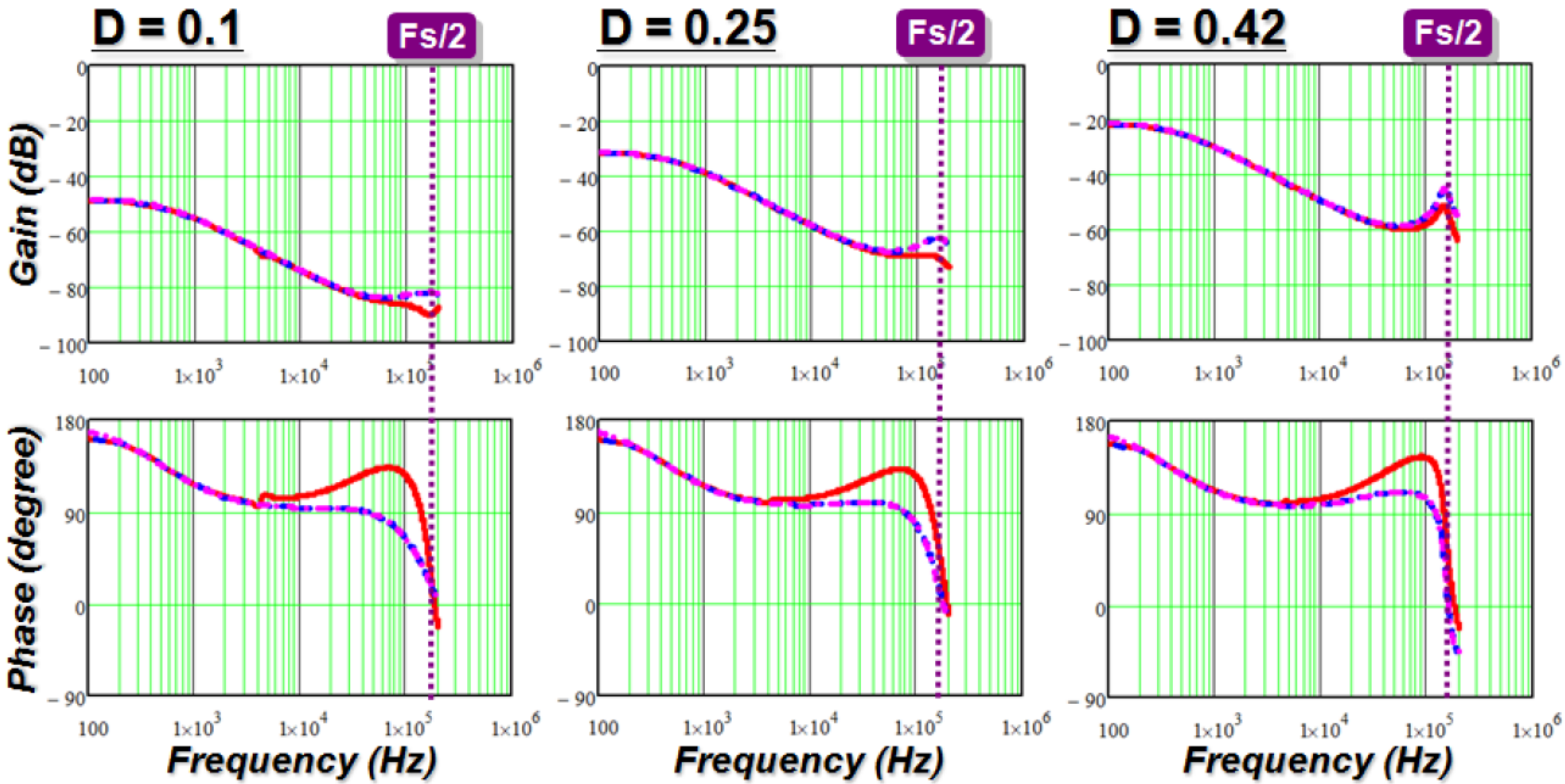
Se = 0



Model Comparison

v_{in}-to-v_o transfer function of Peak Current Mode Control

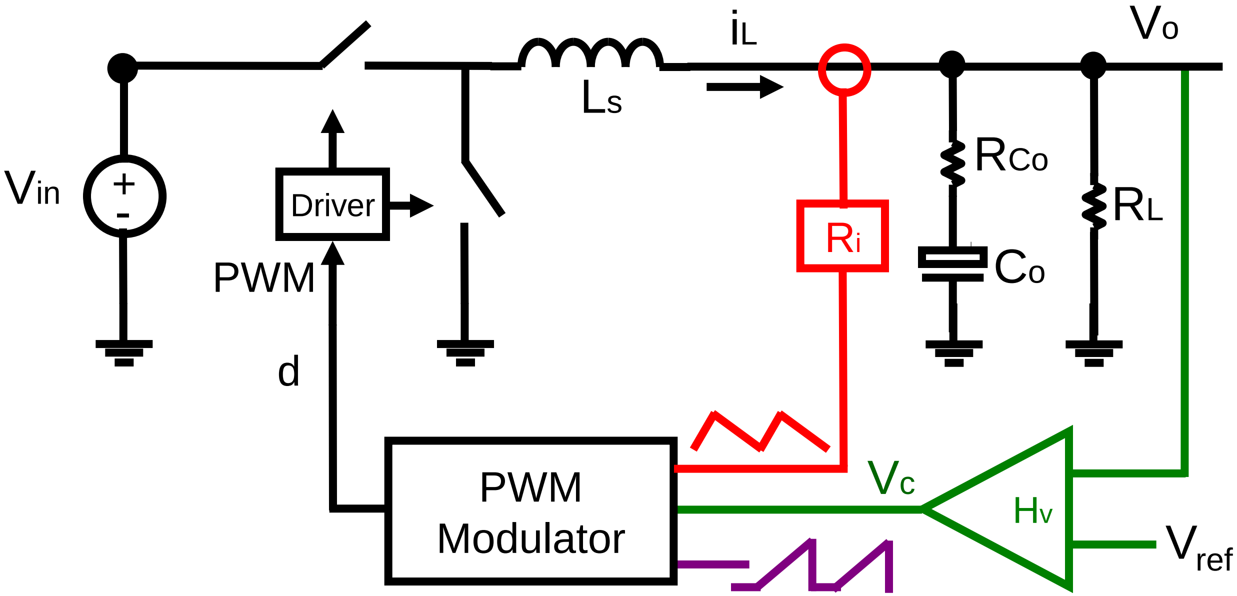
Se = 0



◆ R. Ridley's Model ◆ Jian's Model ◆ Simplis Simulation ≈ Measurement

Extension to

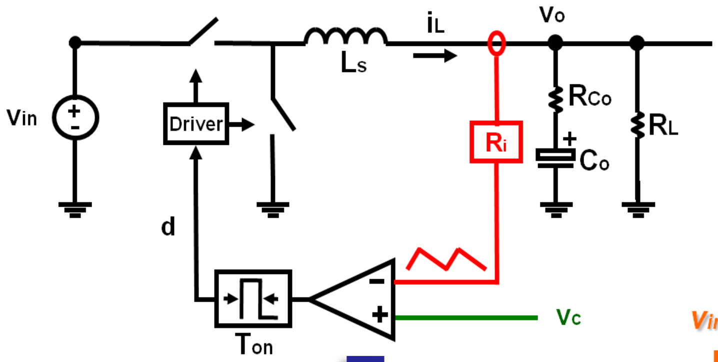
Other Current Mode Controls



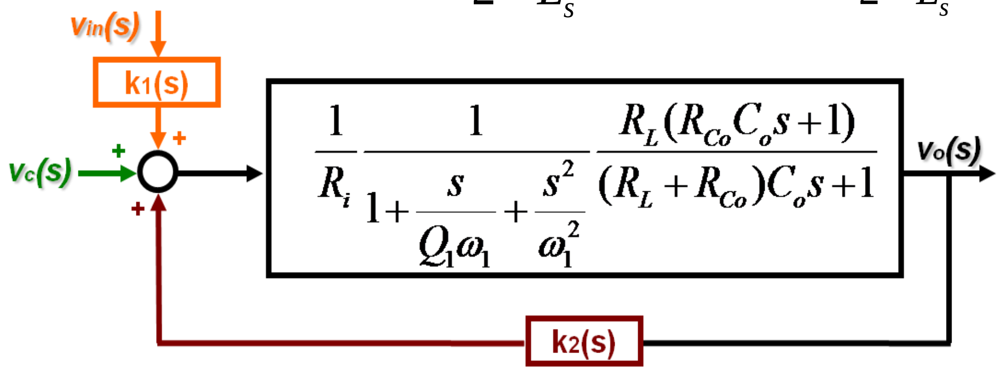
- ***Valley Current-Mode Control***
- ***Constant On-time Control***
- ***Constant Off-time Control***
- ***Charge Control***

Model Extension

to Constant on-time control



$$k_1(s) = \frac{1}{2}\frac{T_{on}R_i}{L_s} \qquad k_2(s) = -\frac{1}{2}\frac{T_{on}R_i}{L_s} = -\frac{R_i}{L_s Q_1 \omega_1}$$



$$\frac{1}{R_i}\frac{1}{1+\frac{s}{Q_1\omega_1}+\frac{s^2}{\omega_1^2}}\frac{R_L(R_{Co}C_o s+1)}{(R_L+R_{Co})C_o s+1}$$

$$\frac{v_o(s)}{v_c(s)} \approx K_c \frac{R_{Co}C_o s+1}{s/\omega_a+1}\frac{1}{1+\frac{s}{Q_1\omega_1}+\frac{s^2}{\omega_1^2}}$$

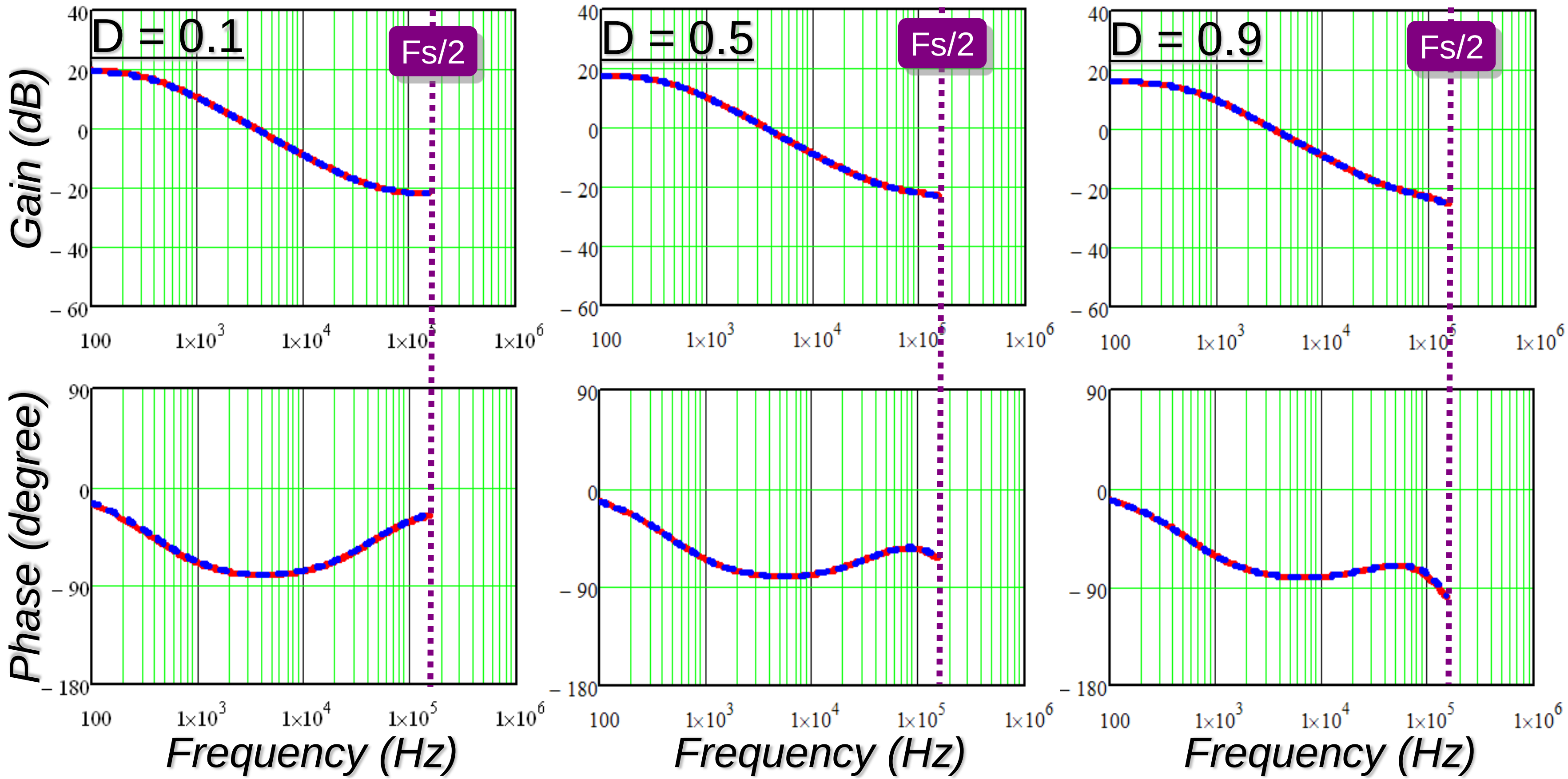
where, $K_c = \dfrac{R_L}{R_i(1-k_2/R_i \cdot R_L)}$ $\quad \omega_a = \dfrac{1-R_L k_2/R_i}{(R_L+R_{Co})C_o - R_L R_{Co} C_o k_2/R_i}$ $\quad \omega_1 = \dfrac{\pi}{T_{on}}$ $\quad Q_1 = \dfrac{2}{\pi}$

Simulation Verification
for Constant On-time Control

CPES

Control-to-output Transfer Function

$$\frac{v_o(s)}{v_c(s)} \approx K_c \frac{R_{Co}C_o s+1}{s/\omega_a+1} \frac{1}{1+\frac{s}{Q_1\omega_1}+\frac{s^2}{\omega_1^2}} \qquad \omega_1 = \frac{\pi}{T_{on}} \qquad Q_1 = \frac{2}{\pi}$$



◆ Proposed Model ◆ Simplis Simulation ≈ Measurement

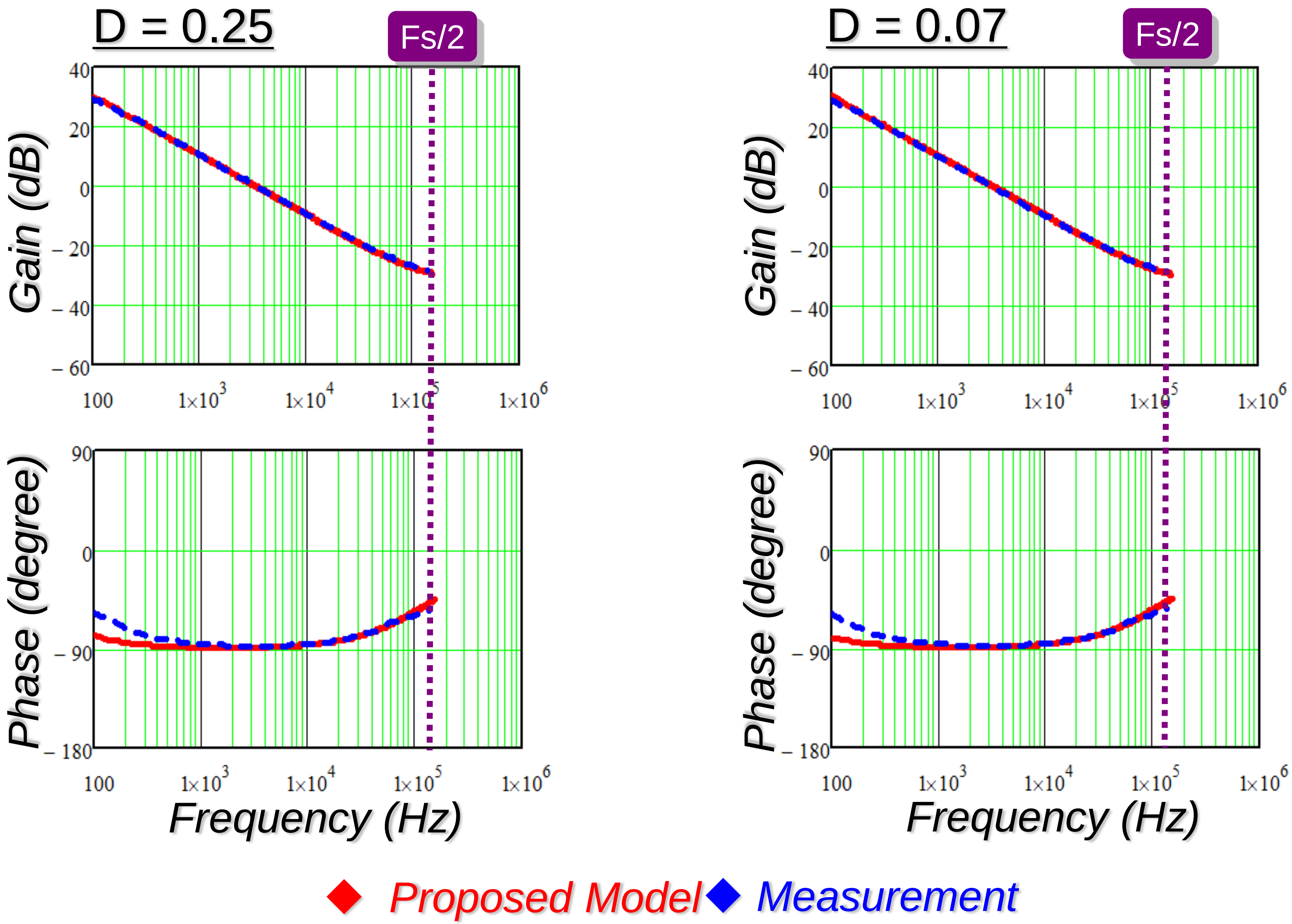
Experiment Verification

for Constant On-time Control

- Based on the demo-board from LINEAR TECHNOLOGY

*LTC3812 Const. On-time Controller V_o = 3.3V, T_{on} = 0.26μs, L_s = 3.1μF, C_o = 2×150μF, and R_{Co} = 9/2mΩ.

Control-to-output Transfer Function

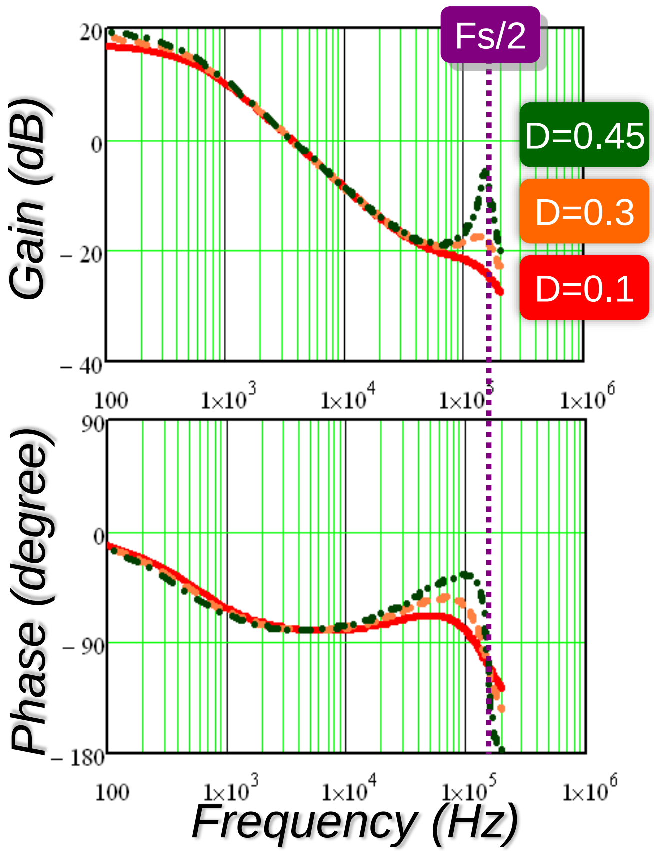


Transfer Function Comparison

➢ *Peak Current Mode-Control*

$$\frac{v_o(s)}{v_c(s)} \approx K_p \frac{R_{Co}C_o s+1}{s/\omega_p+1} \cdot \frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}$$

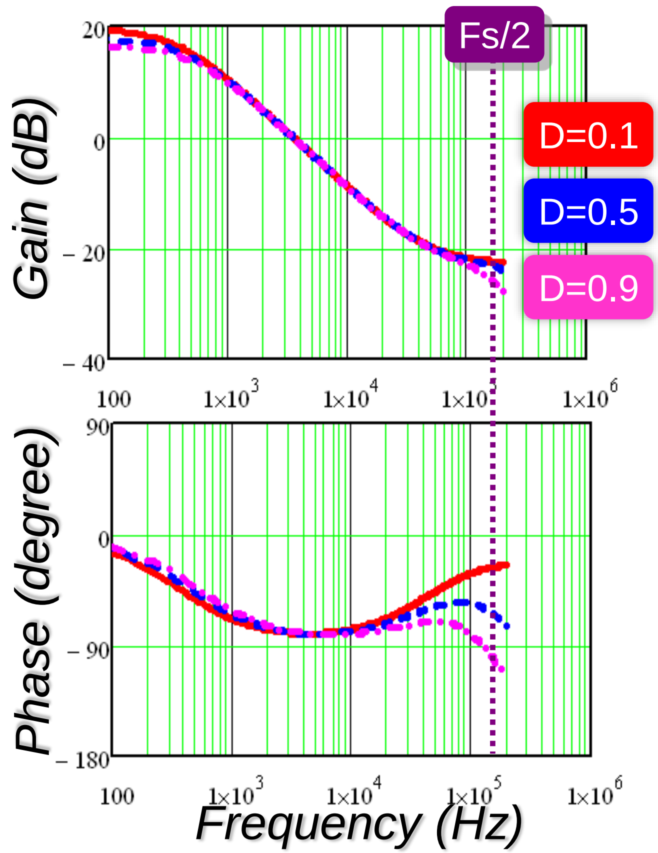
$$\omega_2=\frac{\pi}{T_{sw}} \qquad Q_p=\frac{1}{\pi[(s_n+s_e)/(s_n+s_f)-0.5]}$$



➢ *Constant On-time Control*

$$\frac{v_o(s)}{v_c(s)} \approx K_c \frac{R_{Co}C_o s+1}{s/\omega_a+1} \cdot \frac{1}{1+\frac{s}{Q_1\omega_1}+\frac{s^2}{\omega_1^2}}$$

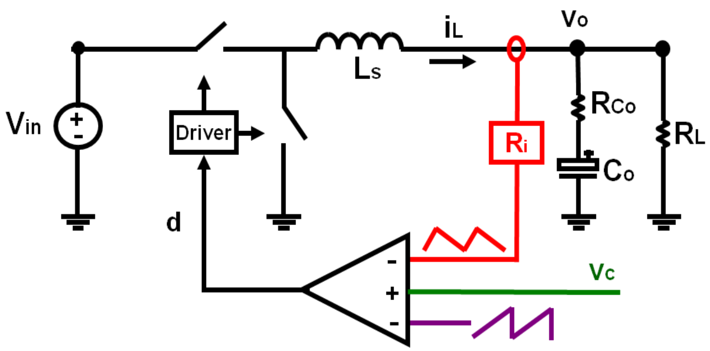
$$\omega_1=\frac{\pi}{T_{on}} \qquad Q_1=\frac{2}{\pi}$$



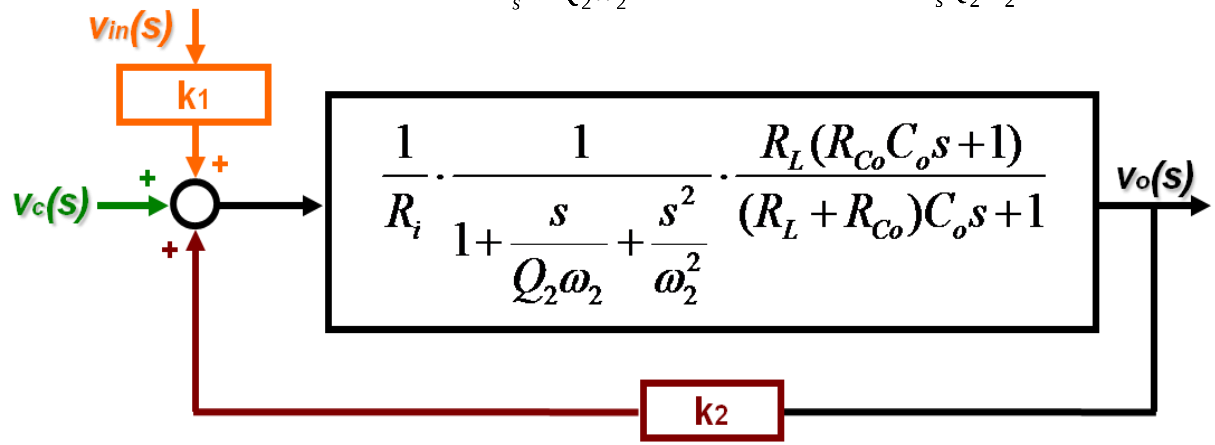
Equivalent Circuit Representation of Current-Mode Control

Complete Model

for Peak Current-Mode control



$$k_1(s) \approx \frac{DR_i}{L_s}\left[\frac{1}{Q_2\omega_2} - \frac{T_{off}}{2}\right] \qquad k_2(s) \approx -\frac{R_i}{L_s Q_2 \omega_2}$$



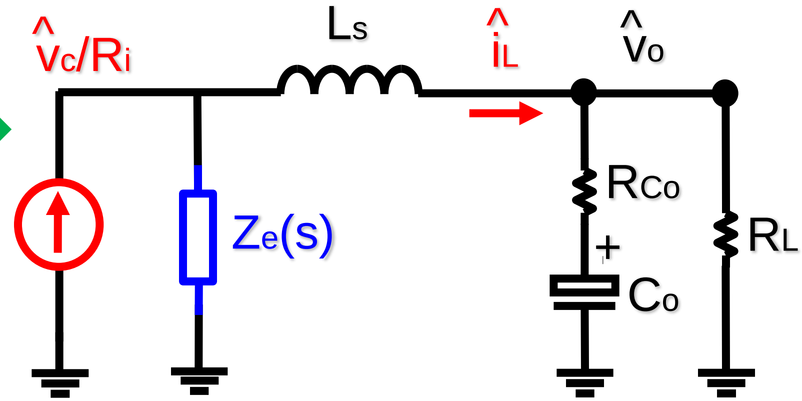
$$\frac{1}{R_i}\cdot\frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}\cdot\frac{R_L(R_{Co}C_os+1)}{(R_L+R_{Co})C_os+1}$$

$$\frac{v_o(s)}{v_c(s)} = \frac{\frac{1}{R_i}\frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}\frac{R_L(R_{Co}C_os+1)}{(R_L+R_{Co})C_os+1}}{1-k_2\frac{1}{R_i}\frac{1}{1+\frac{s}{Q_2\omega_2}+\frac{s^2}{\omega_2^2}}\frac{R_L(R_{Co}C_os+1)}{(R_L+R_{Co})C_os+1}}$$

Equivalent Circuit Model

for Peak Current-Mode Control

$$v_o(s) = \boxed{\frac{v_c(s)}{R_i}} \cdot \frac{\boxed{Z_e(s)} \cdot \dfrac{R_L(R_{Co}C_o s + 1)}{(R_L + R_{Co})C_o s + 1}}{\boxed{Z_e(s)} + L_s s + \dfrac{R_L(R_{Co}C_o s + 1)}{(R_L + R_{Co})C_o s + 1}}$$

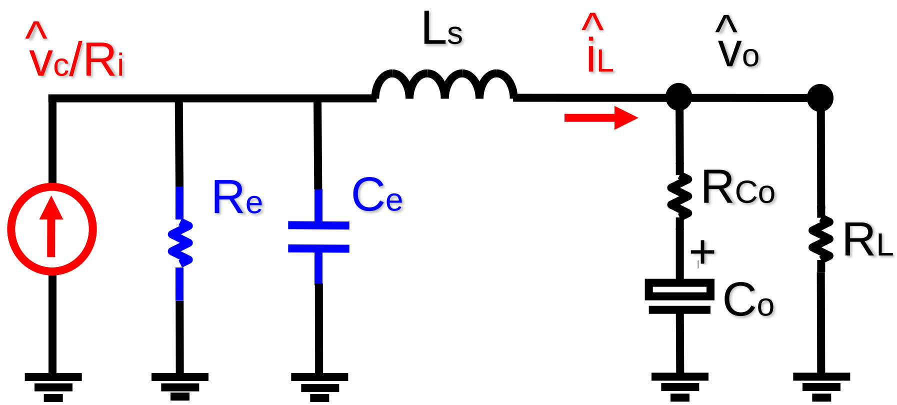


$$Z_e(s) = \frac{1}{\dfrac{1}{L_s Q_2 \omega_2} + \dfrac{s}{L_s \omega_2^2}}$$

where

$$Q_2 = \frac{1}{\pi\left(\dfrac{s_n + s_e}{s_n + s_f} - 0.5\right)}$$

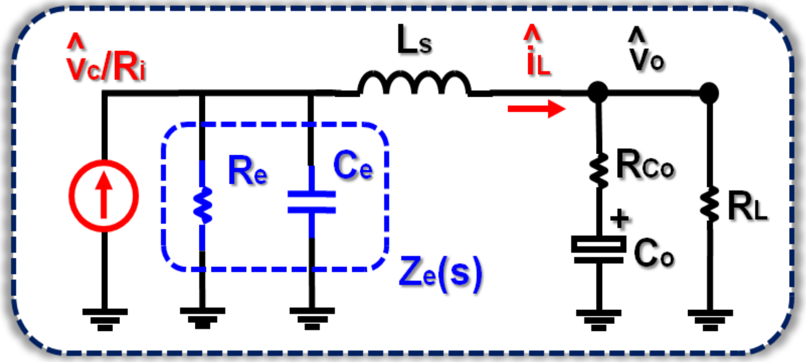
$$\omega_2 = \frac{\pi}{T_{sw}}$$



$$C_e = \frac{1}{L_s \omega_2^2} = \frac{T_{sw}^2}{L_s \pi^2} \qquad R_e = L_s Q_2 \omega_2 = L_s / \left[T_{sw}\left(\frac{s_n + s_e}{s_n + s_f} - 0.5\right)\right]$$

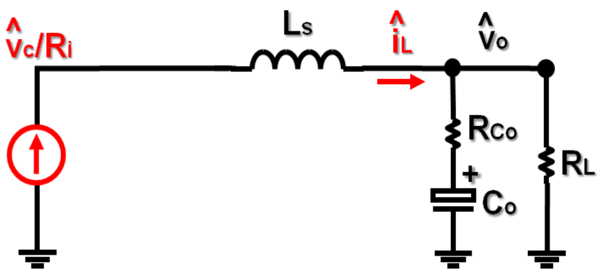
Equivalent Circuit Model

for Peak Current-Mode Control

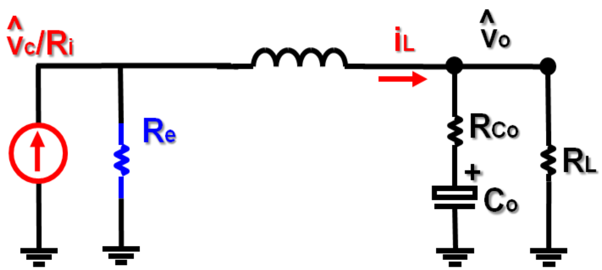


➢ R_e: *the current loop effect*

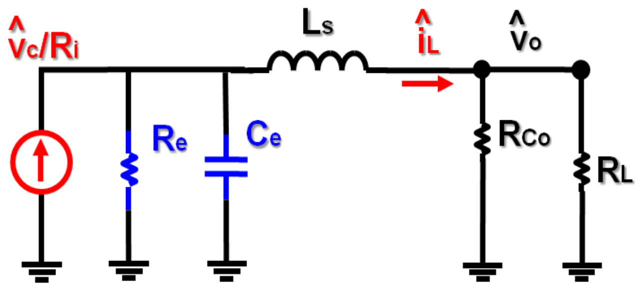
1. R_e *is large: current-loop effect is strong*



2. R_e *is small: current-loop effect is weak*



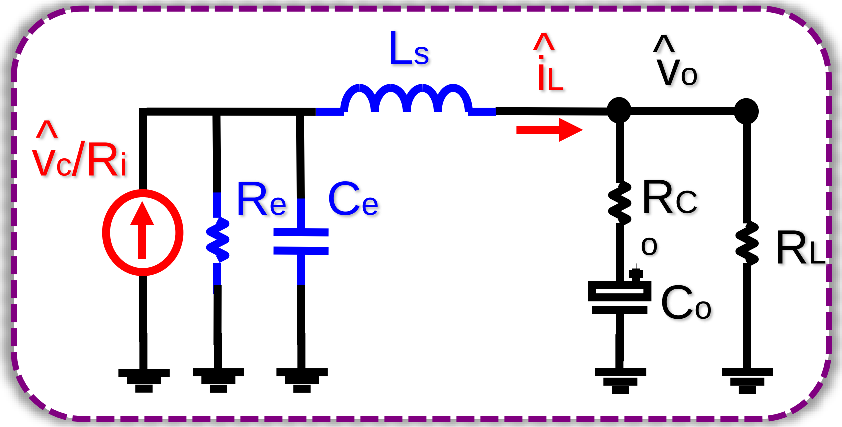
➢ C_e: *the high frequency response*



1. ***The resonant frequency between*** L_s ***and*** C_e ***is fs/2***
2. ***R_e may be negative which results instability issue***

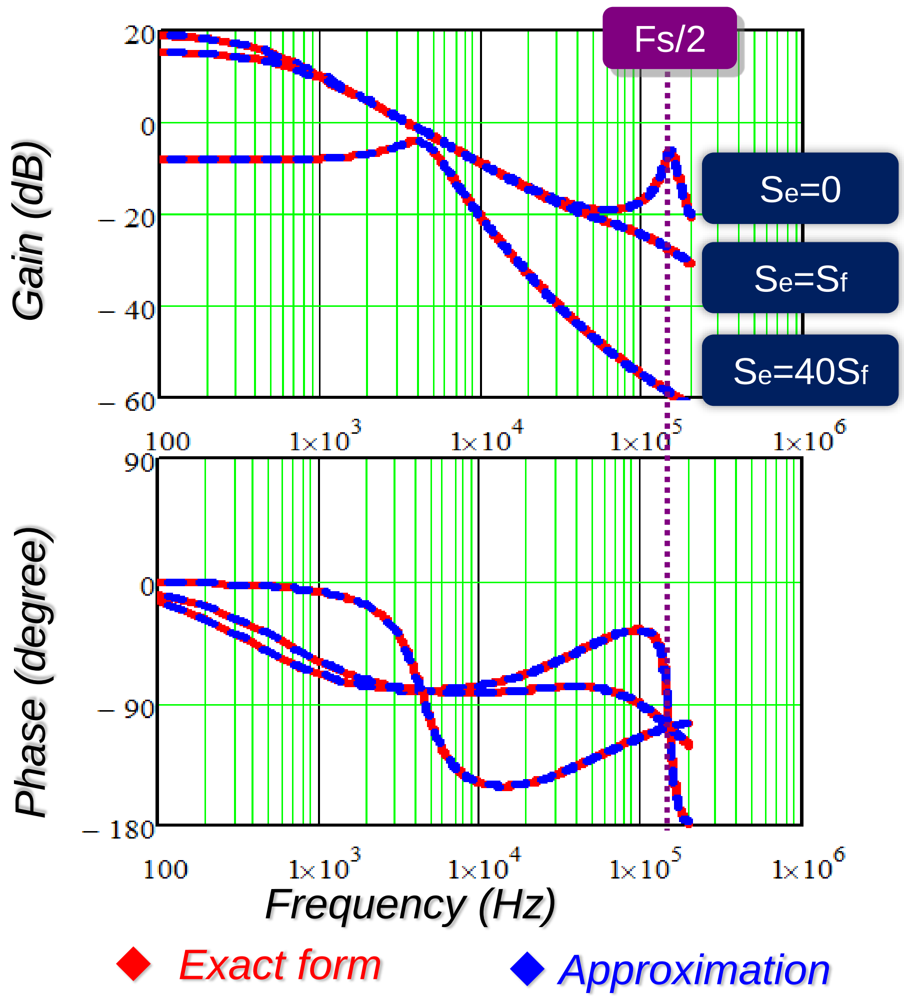
Equivalent Circuit Model

for Peak Current-Mode Control



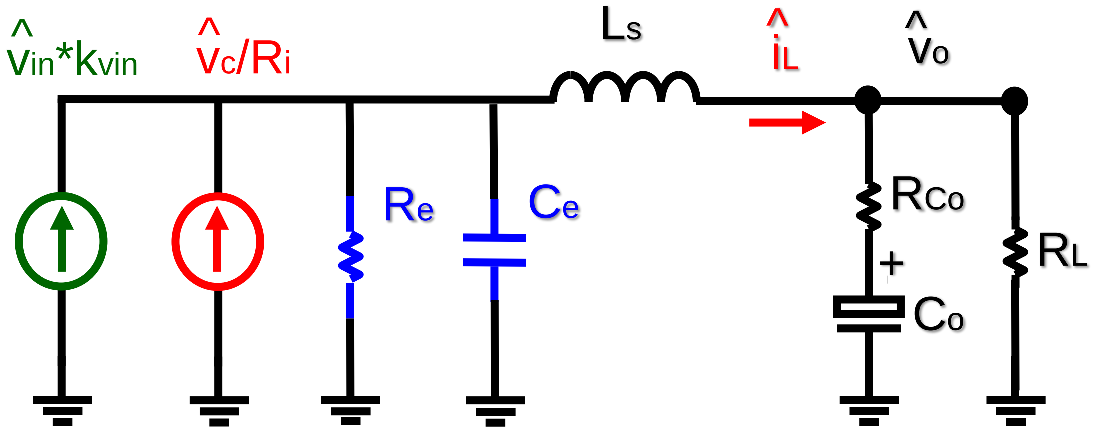
$$R_e = L_s / [T_{sw} (\frac{S_n + S_e}{S_n + S_f} - 0.5)]$$

$$C_e = T_{sw}^2 / (L_s \pi^2)$$



Complete Equivalent Circuit Model

for Peak Current-Mode Control



$$C_e = \frac{1}{L_s\omega_2^2} = \frac{T_{sw}^2}{L_s\pi^2}$$

$$R_e = L_s Q_2 \omega_2 = L_s / [T_{sw}(\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

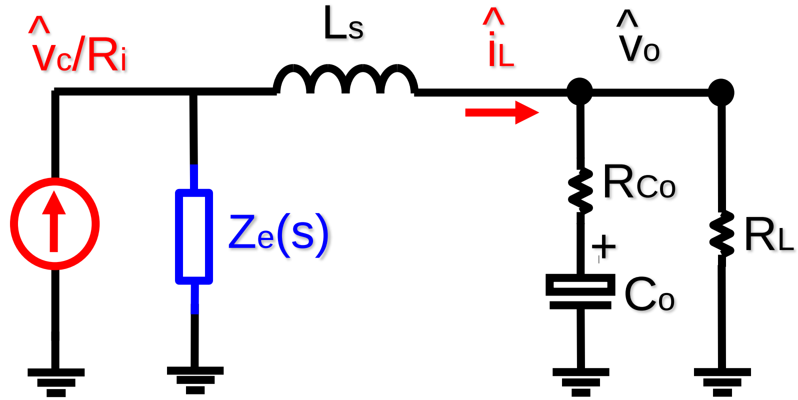
$$k_{vin} = \frac{D}{L_s}[\frac{1}{Q_2\omega_2} - \frac{T_{off}}{2}]$$

where $Q_2 = \dfrac{1}{\pi(\frac{s_n + s_e}{s_n + s_f} - 0.5)}$ $\quad \omega_2 = \dfrac{\pi}{T_{sw}}$

Equivalent Circuit Model

for Constant On-time Control

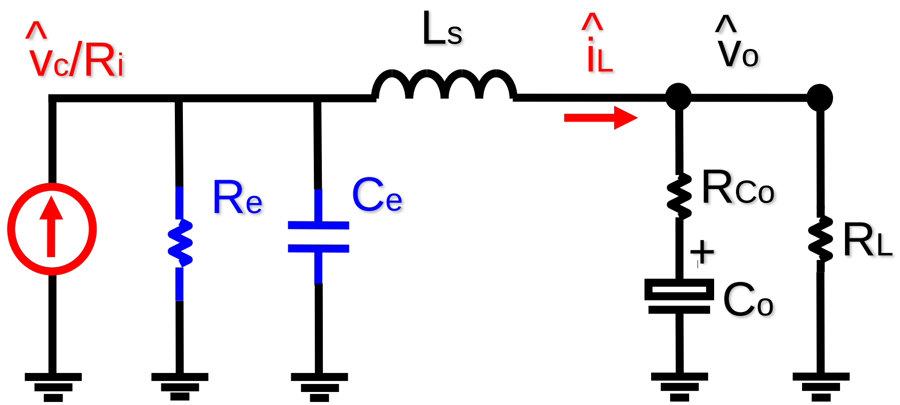
$$v_o(s)=\frac{v_c(s)}{R_i}\cdot\frac{Z_e(s)\cdot\dfrac{R_L(R_{Co}C_o s+1)}{(R_L+R_{Co})C_o s+1}}{Z_e(s)+L_s s+\dfrac{R_L(R_{Co}C_o s+1)}{(R_L+R_{Co})C_o s+1}}$$



$$Z_e(s)=1/(\frac{1}{\omega_1 Q_1 L_s}+\frac{s}{\omega_1^{\,2} L_s})$$

where

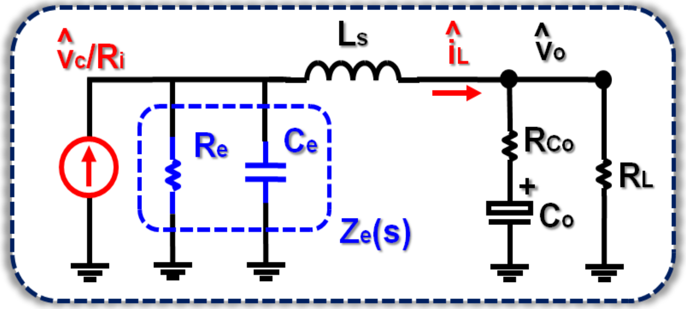
$$\omega_1=\frac{\pi}{T_{on}}\qquad Q_1=\frac{2}{\pi}$$



$$C_e=T_{on}^2/(L_s\pi^2)\qquad R_e=2L_s/T_{on}$$

R_e is always positive in constant on-time control

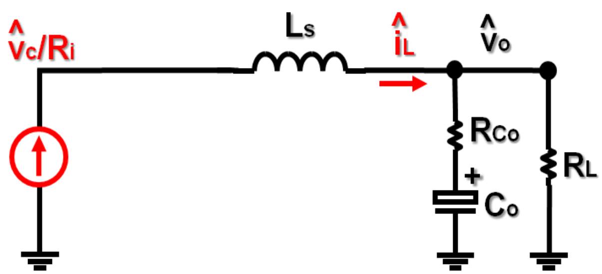
Equivalent Circuit Model
for Constant On-time Control



- R_e: *the current loop effect*

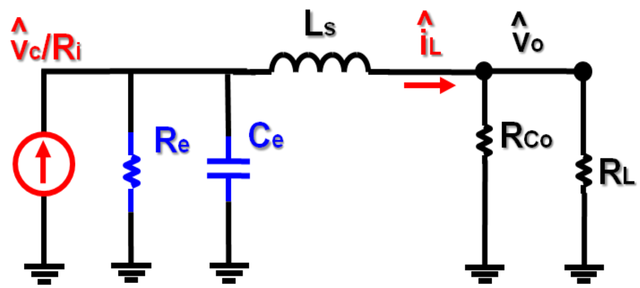
$$R_e = 2L_s / T_{on}$$

Re is large (no external ramp)



- C_e: *the high frequency response*

$$C_e = T_{on}^2 / (L_s \pi^2)$$



1. ***The resonant frequency between L_s and C_e is $1/(2T_{on})$***
2. ***R_e is always positive which results no instability issue***

Extension to Other Current-Mode Controls

Constant On-time Control

$$C_e = T_{on}^2 / (L_s \pi^2)$$

$$R_e = 2L_s / T_{on}$$

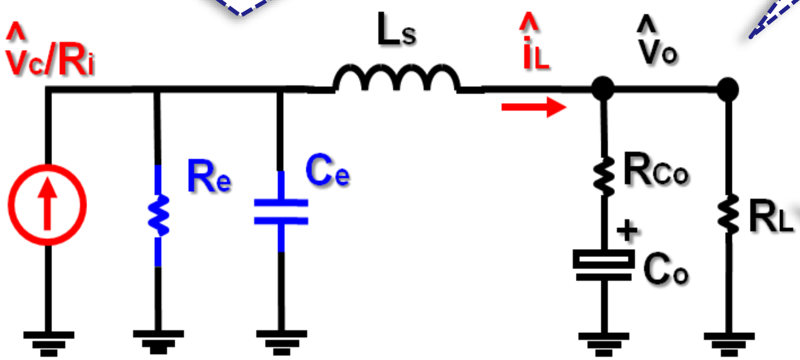
Re is always positive

Constant Off-time Control

$$C_e = T_{off}^2 / (L_s \pi^2)$$

$$R_e = 2L_s / T_{off}$$

Re is always positive



Charge Control

$$C_e = T_{sw}^2 / (L_s \pi^2)$$

$$R_e = L_s / [T_{sw}(\frac{L_s f_s}{R_L} - \frac{D}{2} + \frac{L_s C_T f_s}{V_o R_i} s_e)]$$

Re can be positive or negative

Peak Current-Mode Control

$$C_e = T_{sw}^2 / (L_s \pi^2)$$

$$R_e = L_s / [T_{sw}(\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

Re can be positive or negative

Valley Current-Mode Control

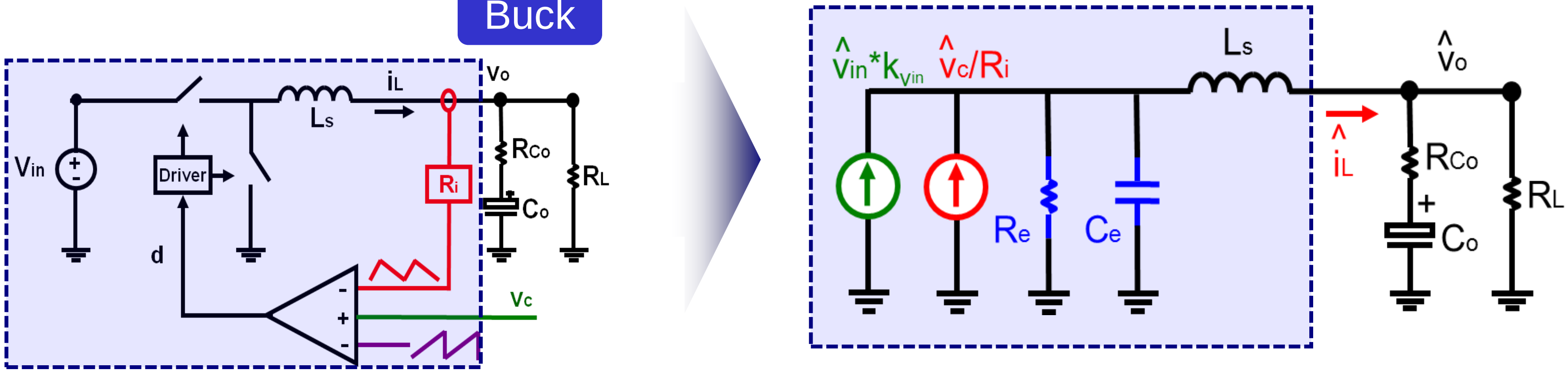
$$C_e = T_{sw}^2 / (L_s \pi^2)$$

$$R_e = L_s / [T_{sw}(\frac{s_f + s_e}{s_n + s_f} - 0.5)]$$

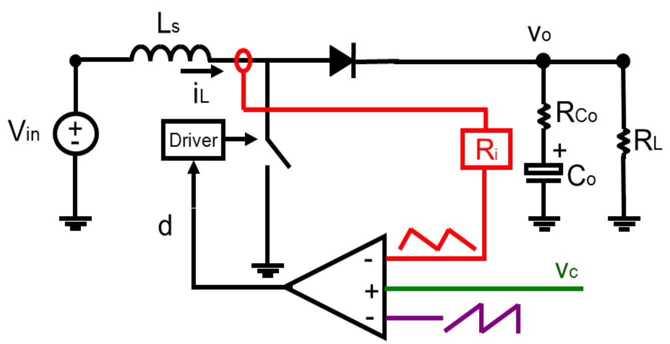
Re can be positive or negative

Extend to Other Converters ?

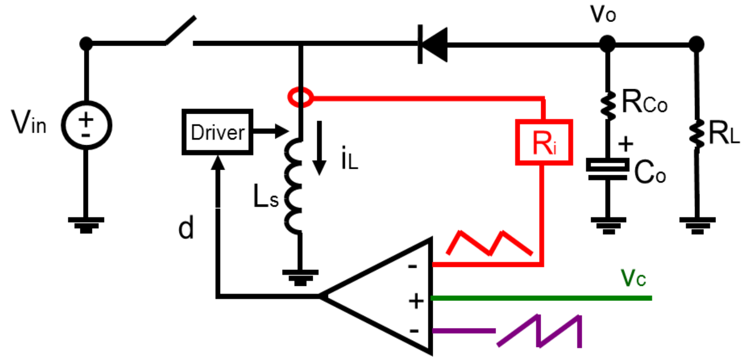
Buck



Boost



Buck-Boost

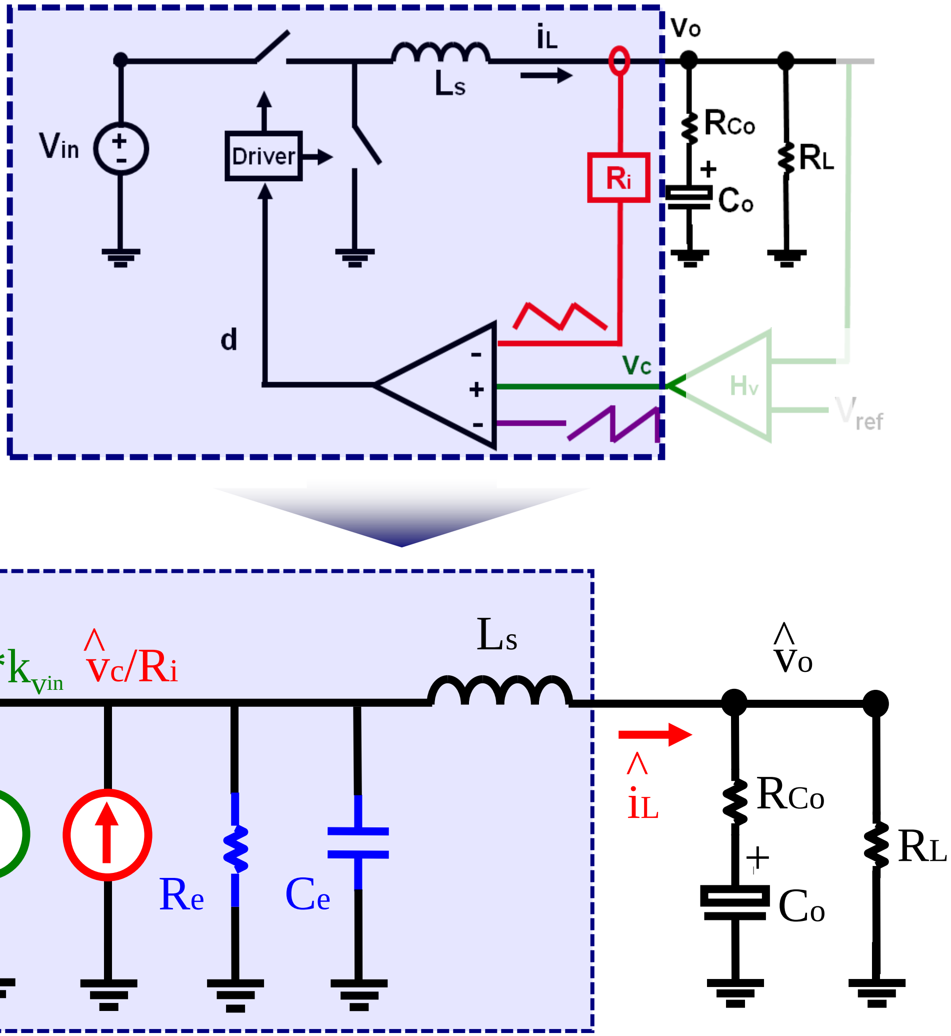


No equivalent circuit for other current mode control converters

Unified Three-Terminal Switch Model for Current Mode Controls

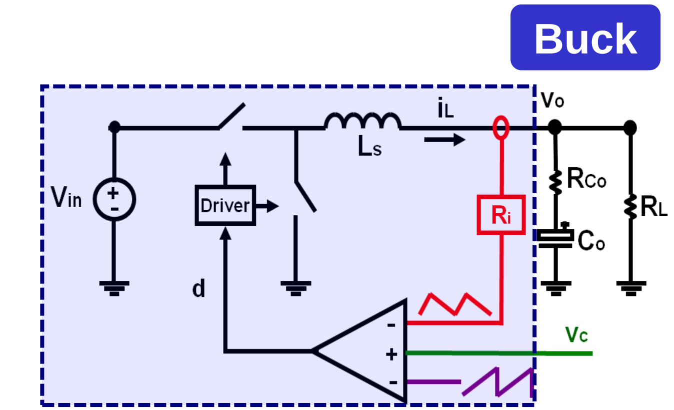
* Yingyi Yan, “Unified Three-terminal Switch Model for Current Mode Controls”, Master Thesis, VT 2010.

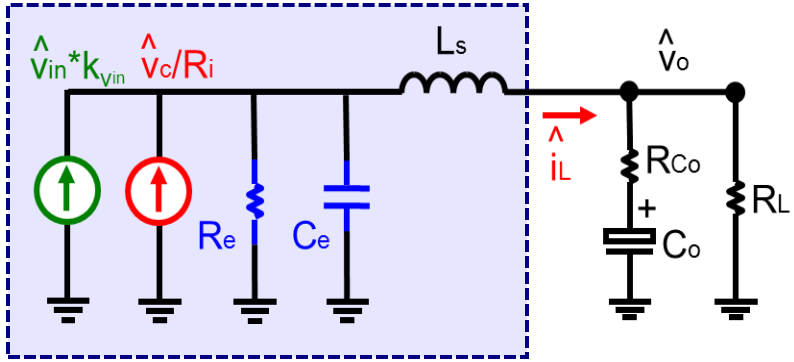
Motivation 1: Complete the Equivalent Circuit

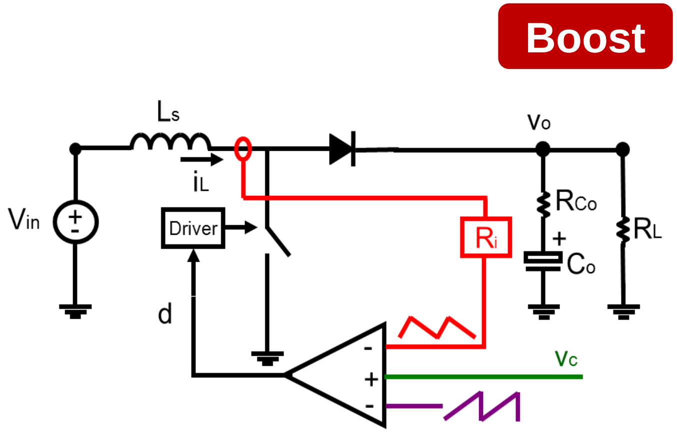


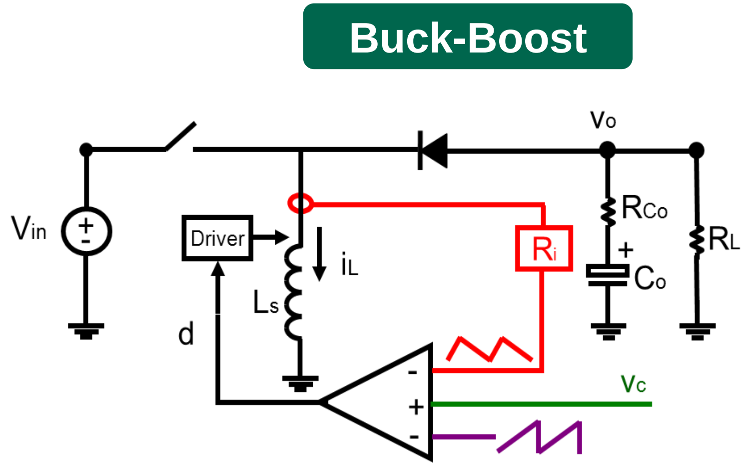
The model only shows relationship between v_{in}, v_c and v_o, but input current property is lost.

Motivation 2: Extend to Other Converters



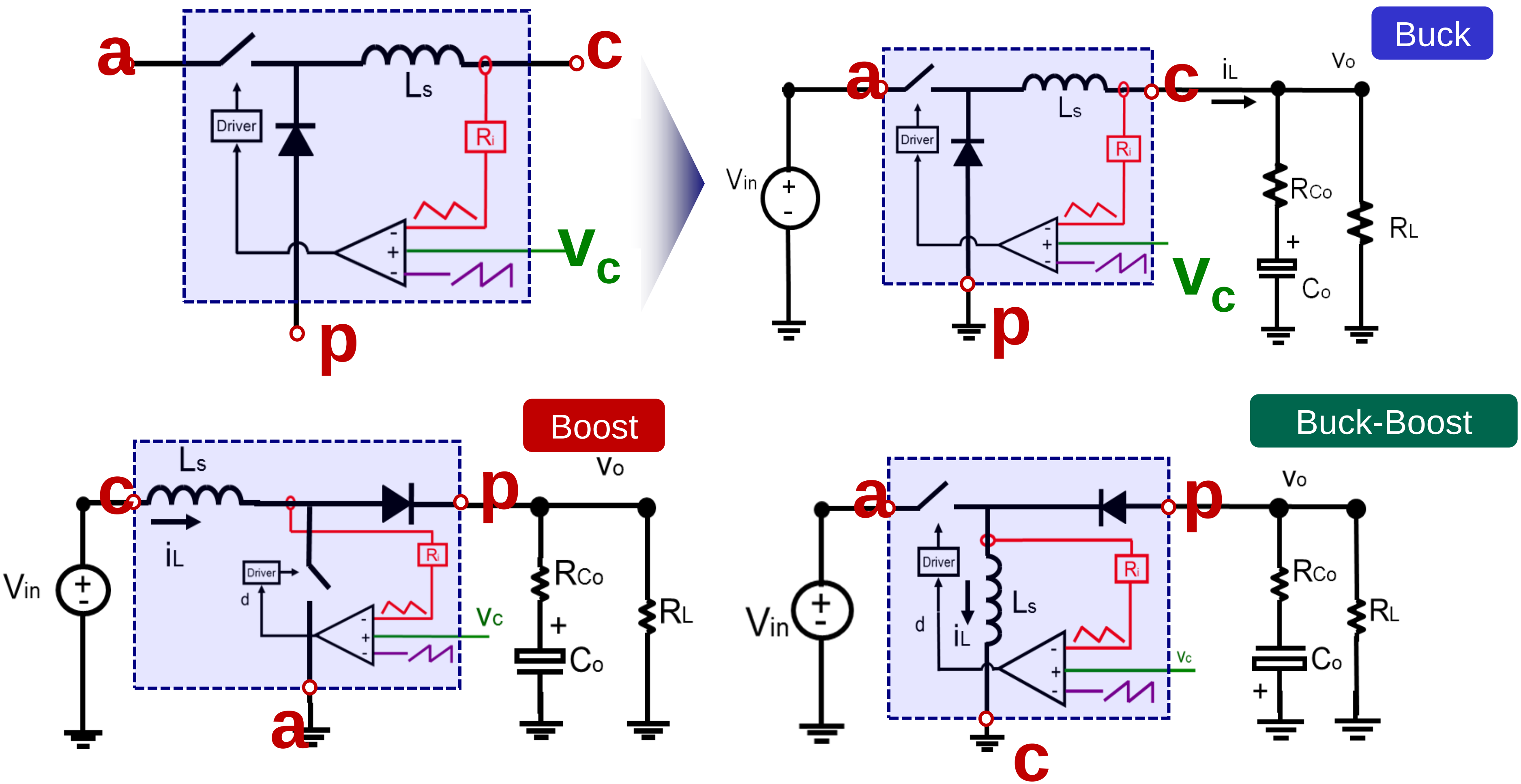






No equivalent circuit for other current mode control converters

Objective: Model the Common Structure

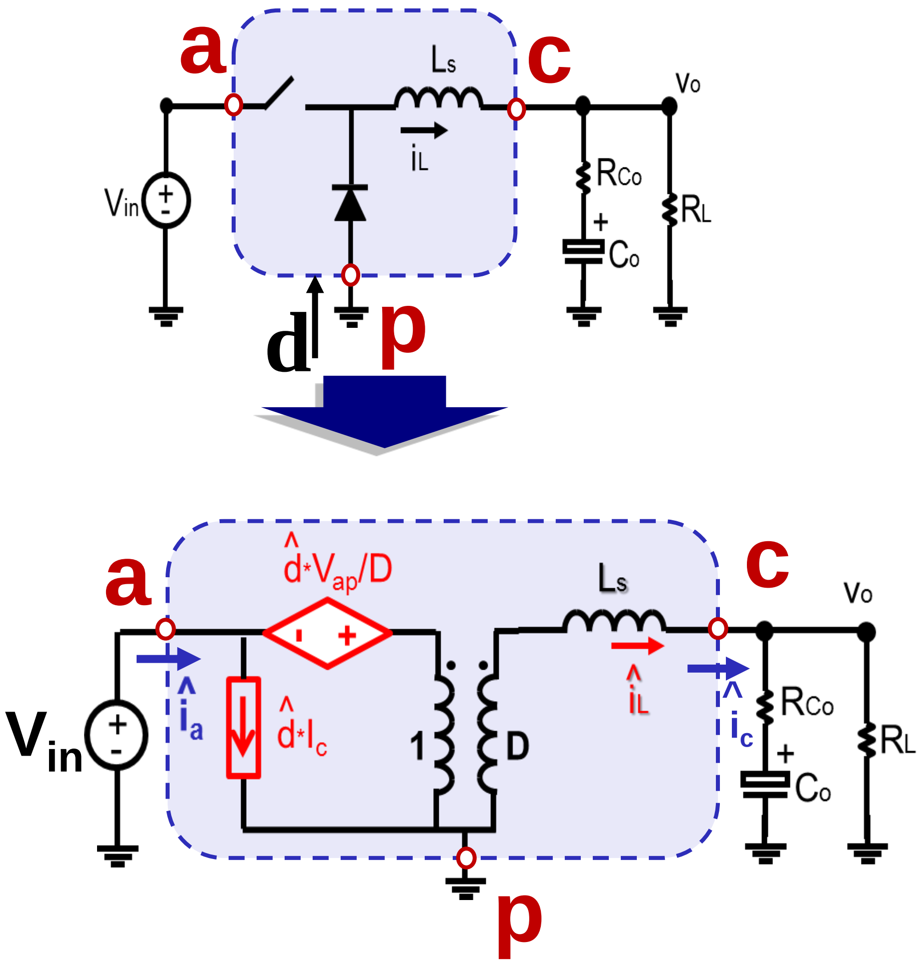


Objectives:

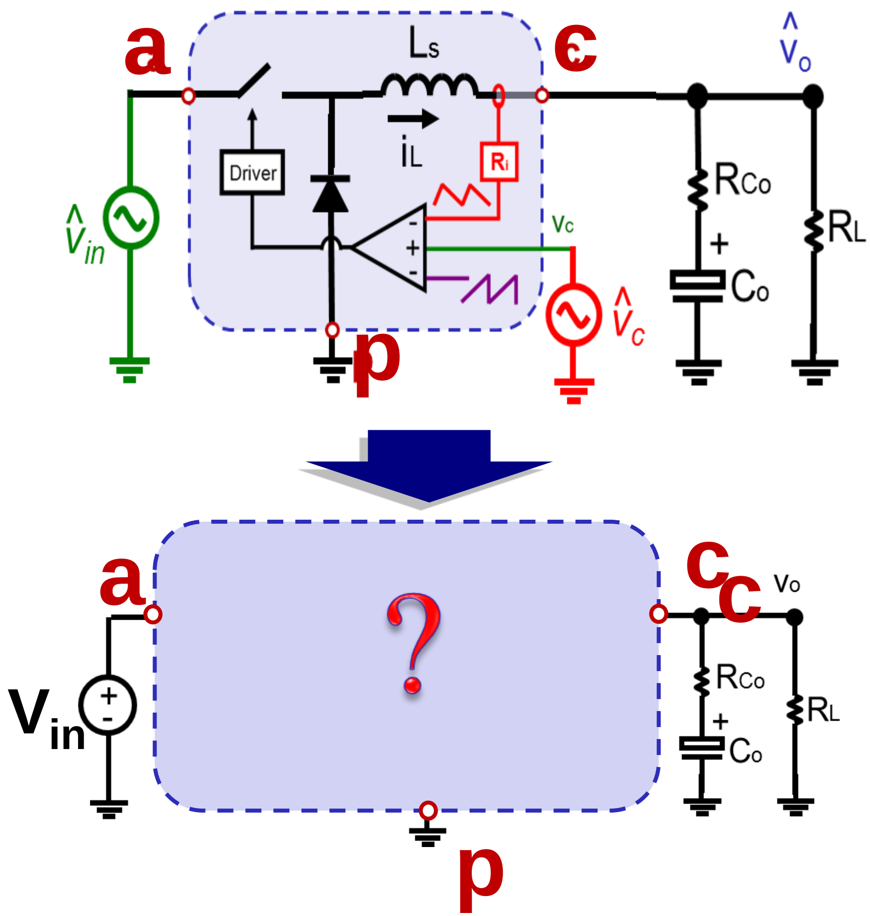
1. **To correctly represent input and output characteristic of Buck converter**
2. **To develop a unified equivalent circuit applicable to other topologies**

Three-terminal Switch Model Concept

- 3-terminal switch model



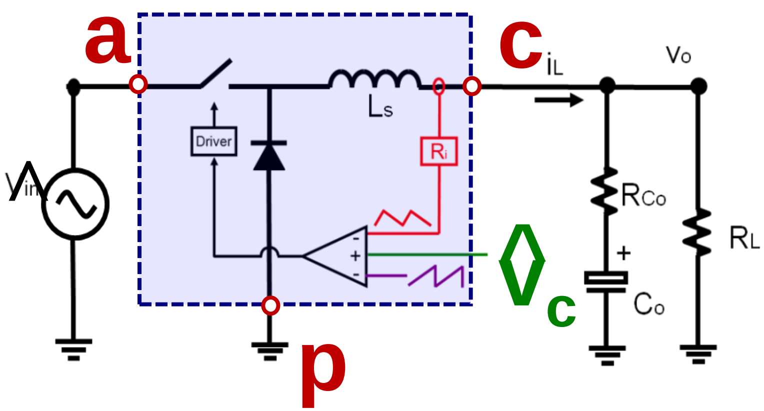
- 3-terminal switch model with current feedback

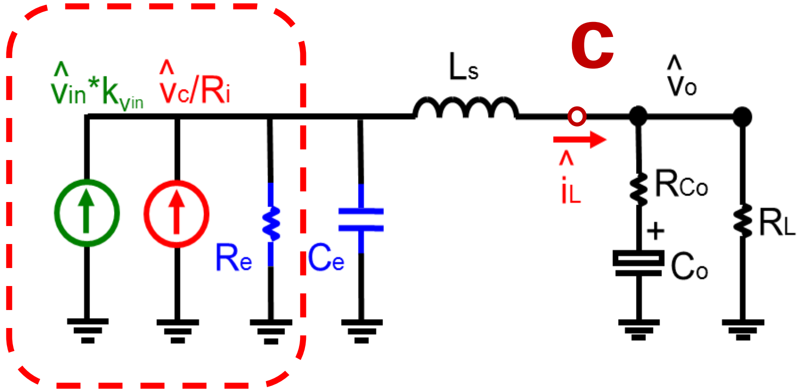


What is the equivalent circuit for current mode control PWM switch?

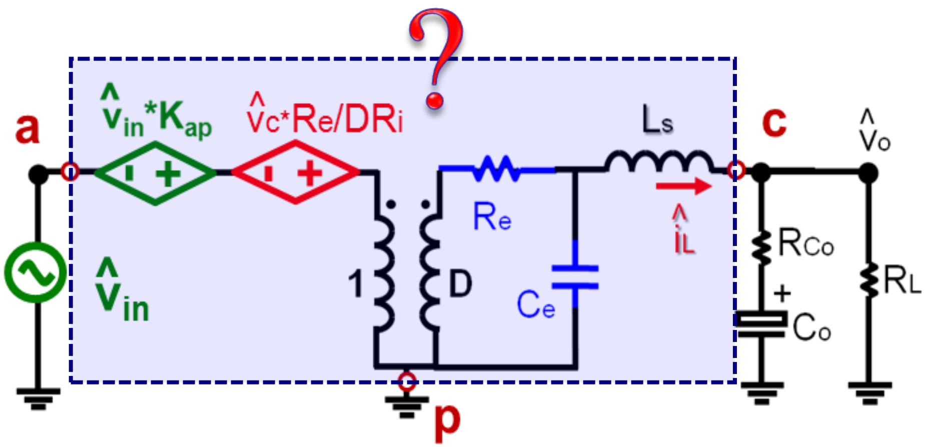
Start from

Equivalent Circuit for Buck Converter

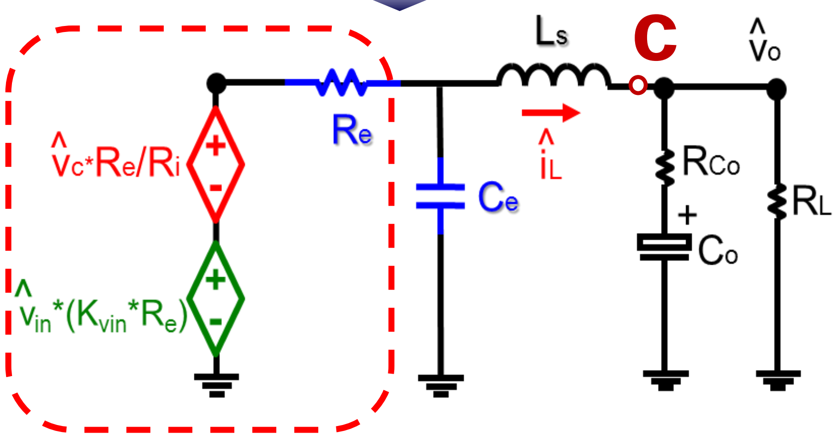




Small signal Equivalent circuit describing $\hat{v}_o/\hat{v}_c$



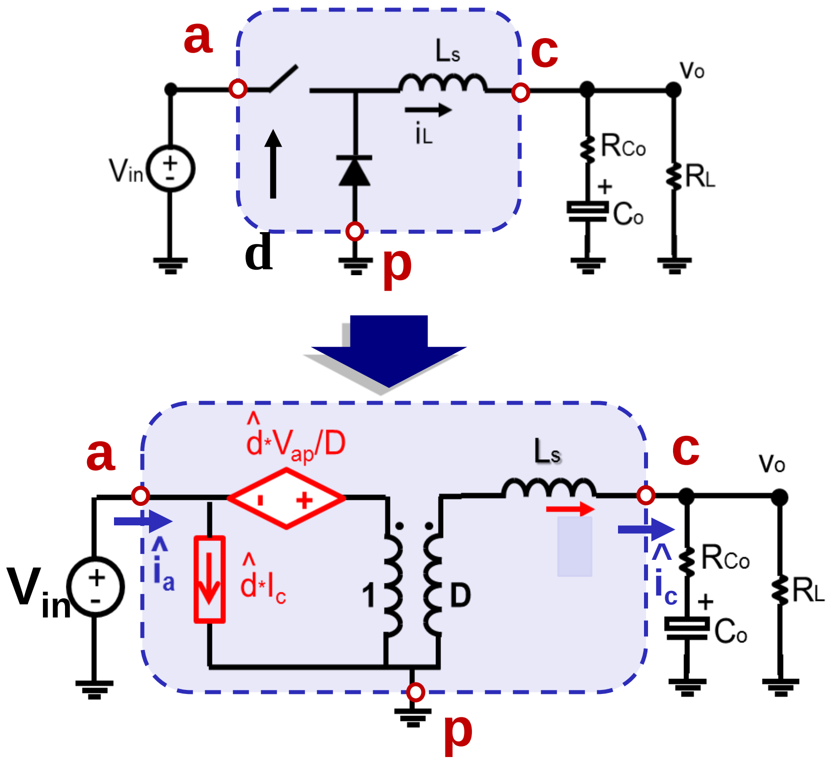
DC Transformer



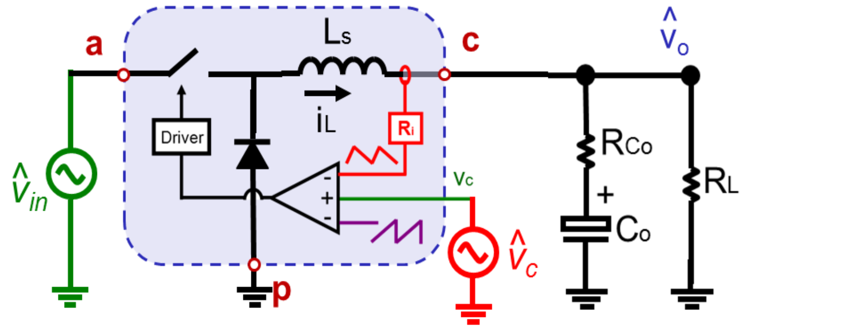
Thevenin Equivalent Circuit

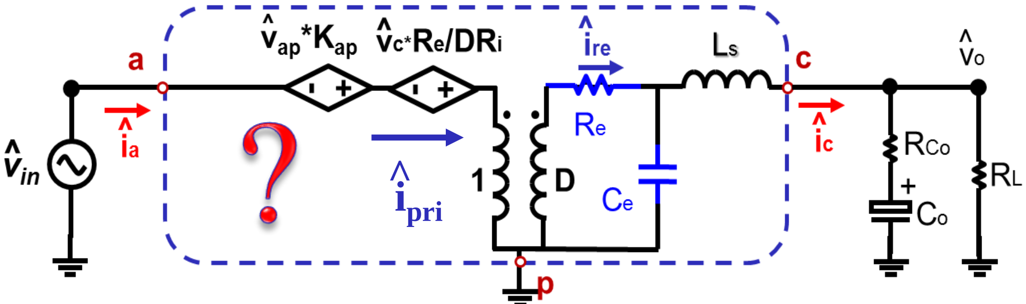
Universal Property of Switch Current

- **3-terminal switch model**



- **3-terminal switch model**





- ***Small Signal Relationship:***

$$\hat{i}_a = D \cdot \hat{i}_c + \hat{d} \cdot I_c$$

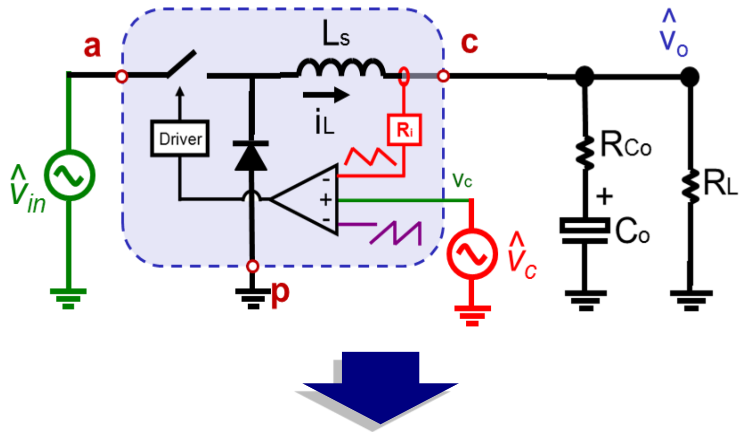
- ***Small Signal Relationship:***

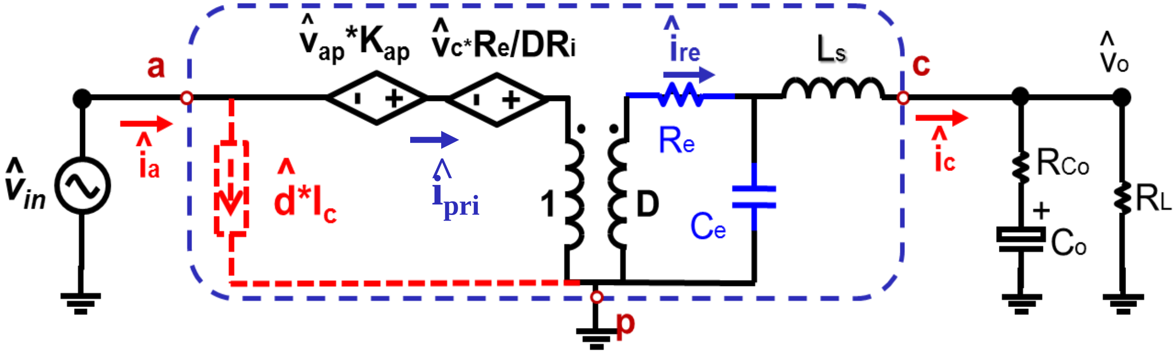
$$\hat{i}_a \approx D \cdot \hat{i}_c + \hat{d} \cdot I_c$$

The property of the switch current i_a should be preserved in the current mode control as well

Address the Missing Term $\hat{d}*I_c$

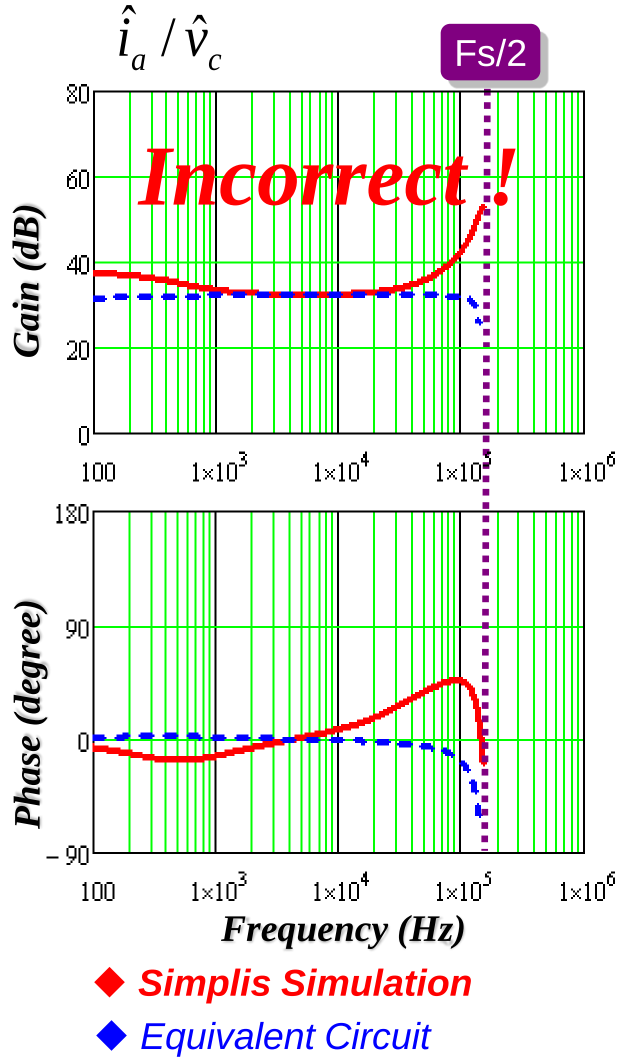
Equivalent Circuit for Buck with Current Mode Control





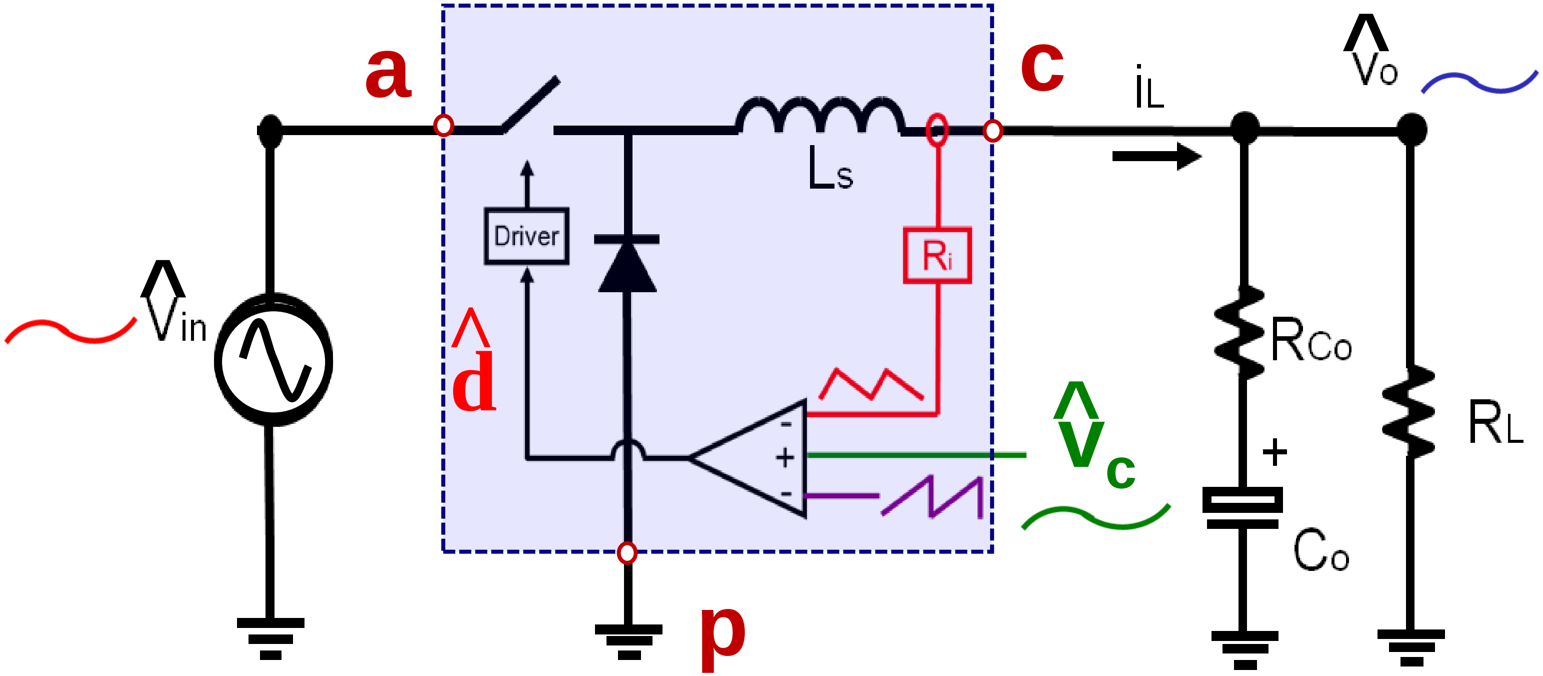
$$\hat{i}_{pri} = D \cdot \hat{i}_{\mathrm{Re}} \approx D \cdot \hat{i}_c$$

$$\hat{i}_a = \hat{d} \cdot I_c + \hat{i}_{pri} \approx \hat{d} \cdot I_c + D \cdot \hat{i}_c$$



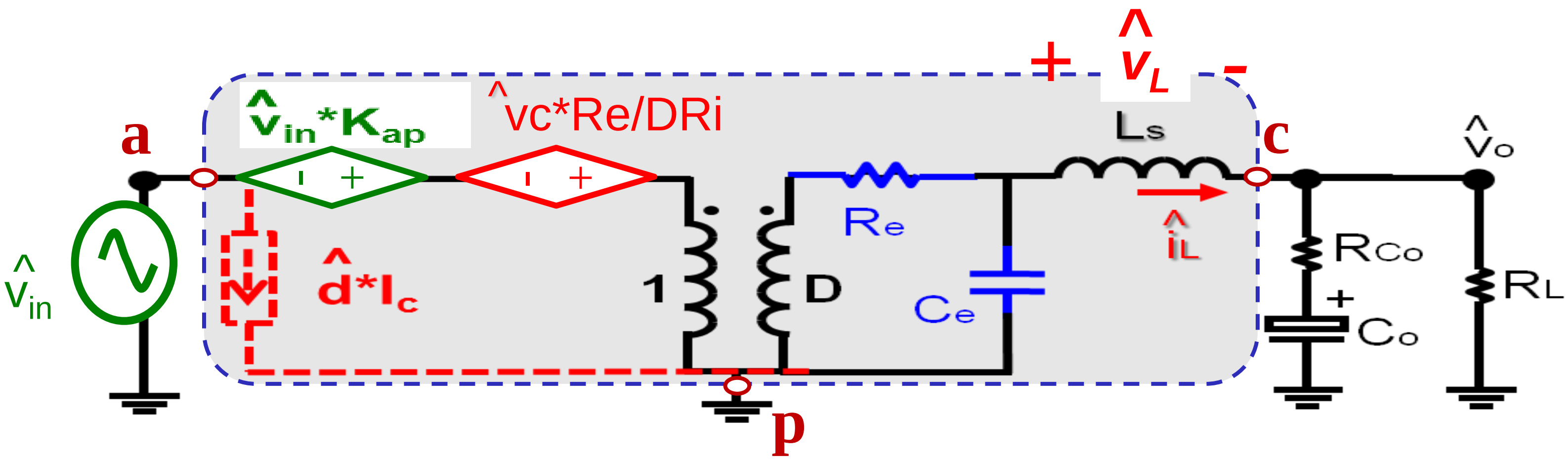
Complete the model by addressing $\hat{d}*I_c$ term

Duty Cycle Modulation



To correctly represent modulation effect, input and output voltage perturbation have to be considered.

$\hat{d} * I_c$ Term Representation in Equivalent Circuit



➢ **From Jian Li's describing function derivation:***

$$\hat{d} \approx \hat{v}_c \cdot A(s) + \hat{v}_{ap} \cdot B(s) + \hat{v}_{cp} \cdot C(s) \qquad \textbf{(1)}$$

➢ **From Equivalent Circuit:**

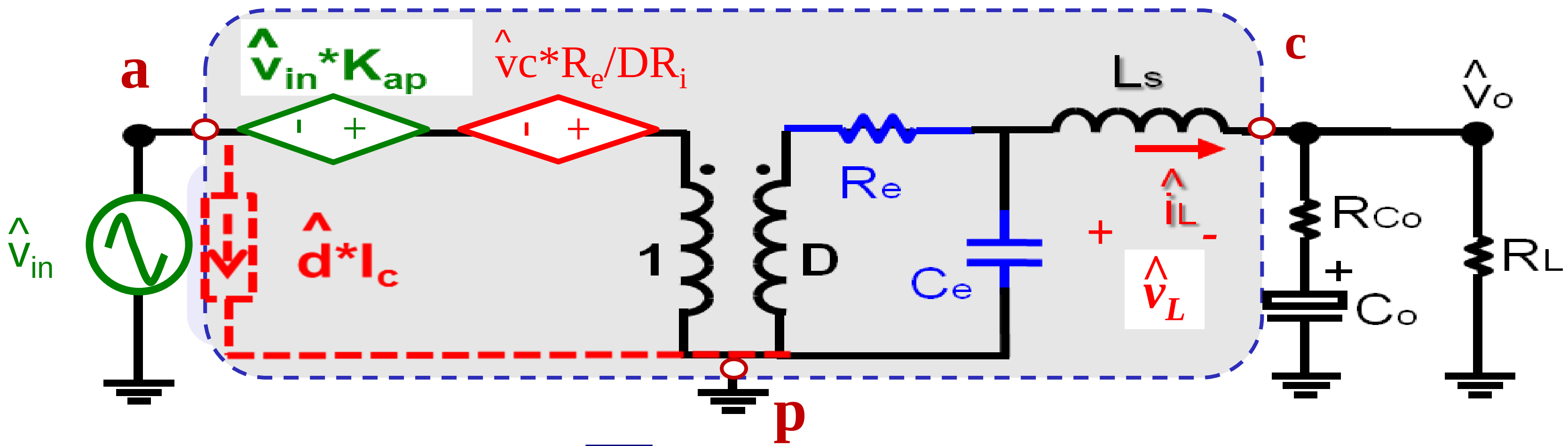
$$\hat{v}_c \cdot A(s) = \hat{v}_L / V_{ap} - \hat{v}_{ap} \cdot B(s) - \hat{v}_{cp} \cdot C(s) - \hat{v}_{ap} \cdot D / V_{ap} + \hat{v}_{cp} / V_{ap} \qquad \textbf{(2)}$$

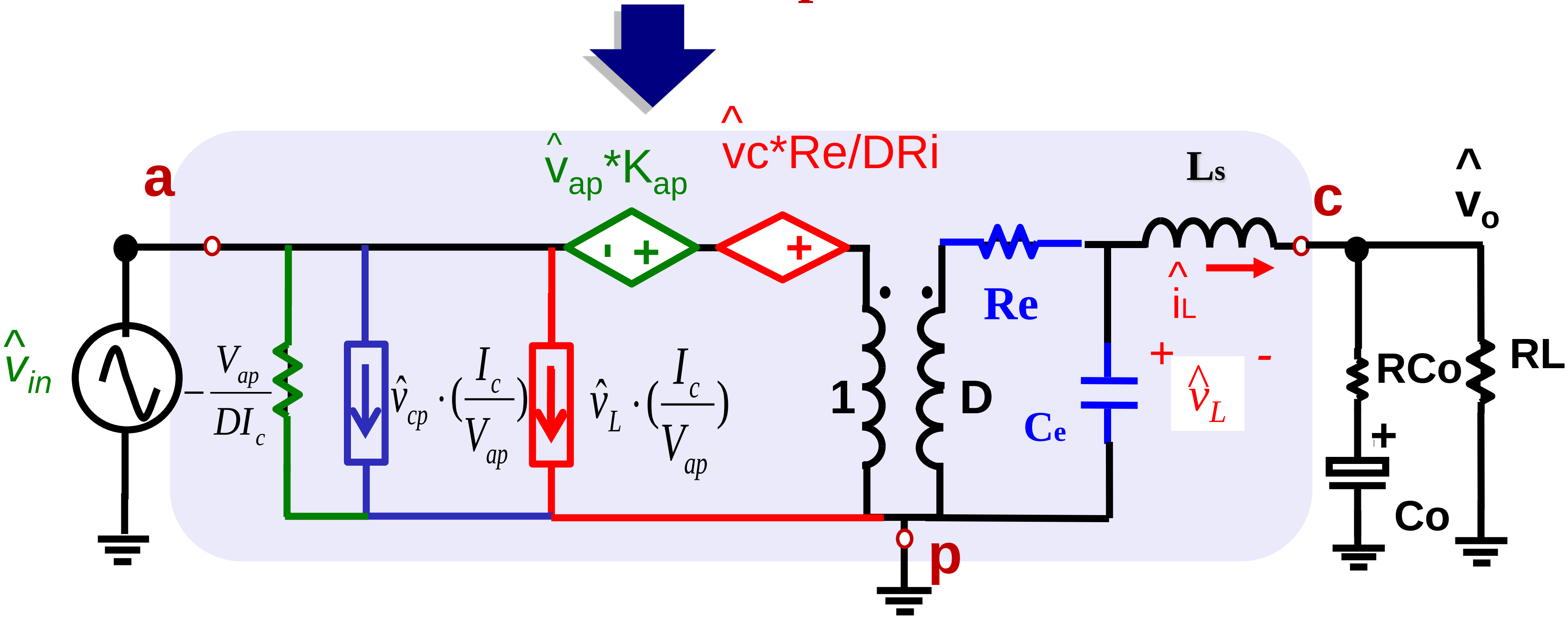
From equation 1 and 2

$$\hat{d} \cdot I_c = \hat{v}_L \cdot \left(\frac{I_c}{V_{ap}}\right) + \hat{v}_{cp} \cdot \left(\frac{I_c}{V_{ap}}\right) + \hat{v}_{ap} / \left(-\frac{V_{ap}}{DI_c}\right)$$

*** Jian Li, Current-Mode Control: Modeling and its Digital Application, Ph.D. Dissertation.**

Complete Equivalent Circuit Model

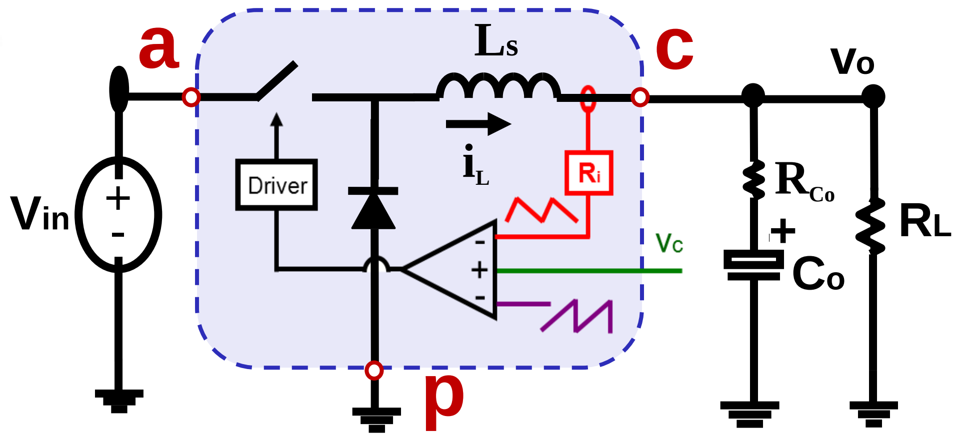




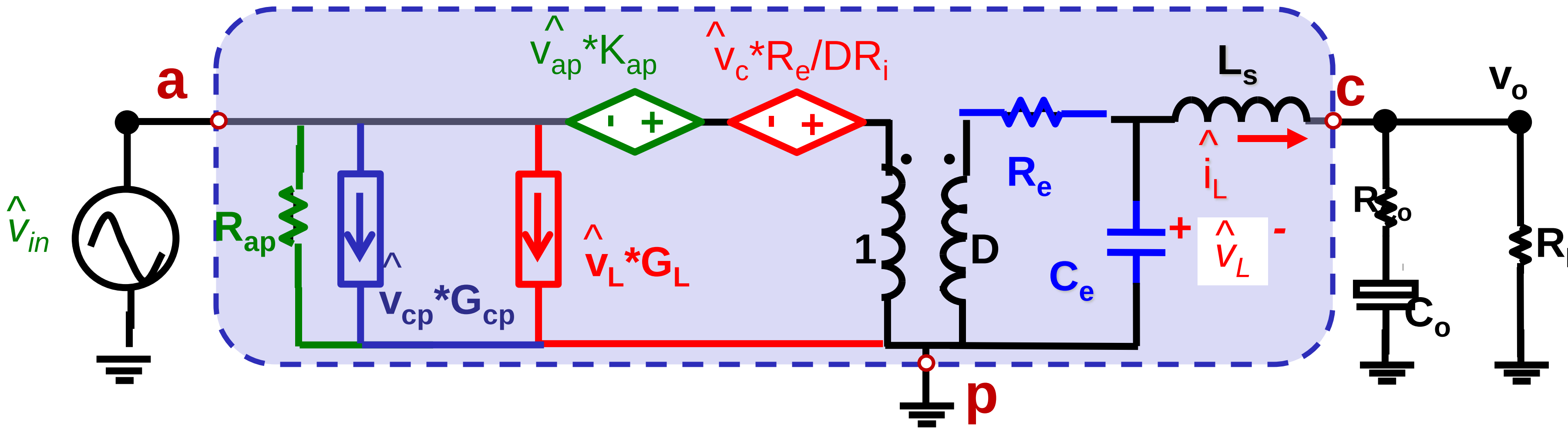
Three Terminal Switch Model

by Yingyi Yan

Buck Converter



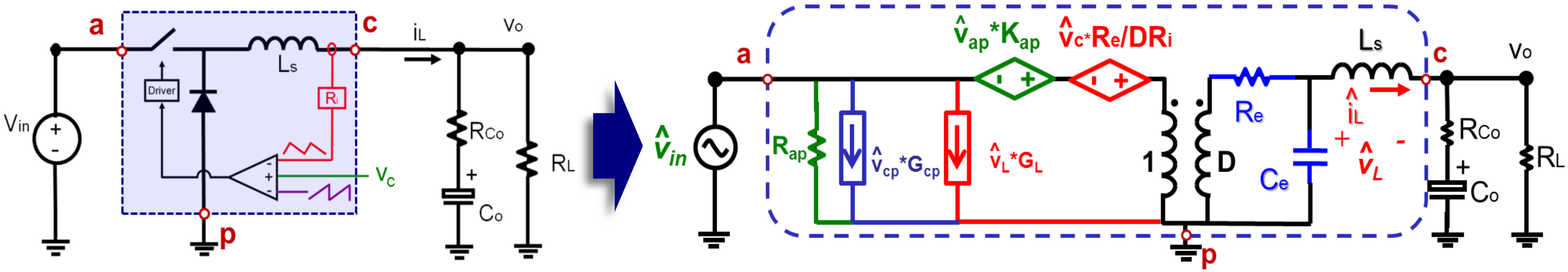
3 Terminal current-cell model

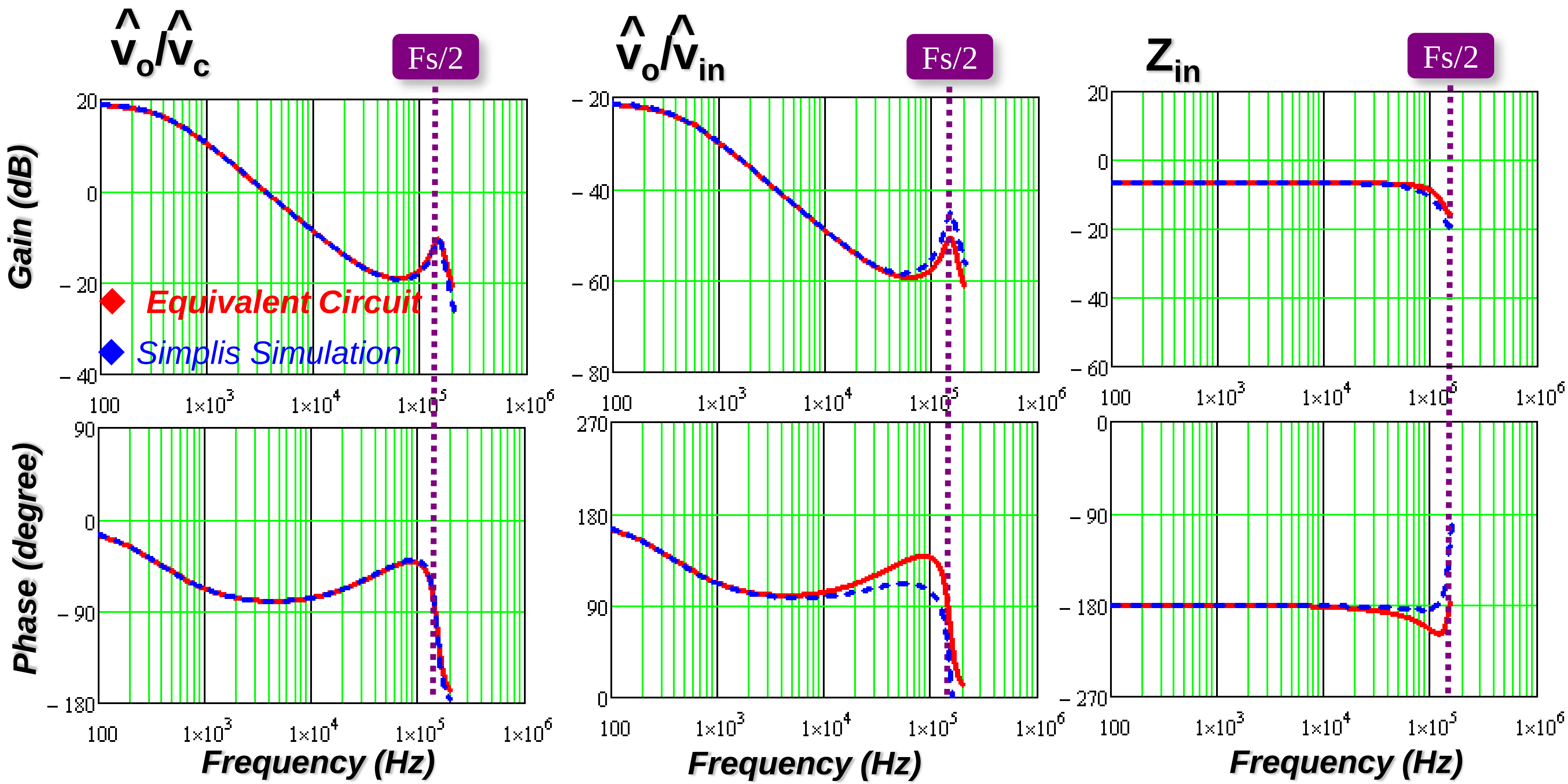


Three terminal equivalent circuit represents small signal characteristic of three terminal switch with closed current loop

Model Verification

Buck Converter with Peak current mode control





Model Extension to
Other Current-Mode Controls

Constant On-time Control

$$C_e = T_{on}^2/(L_s\pi^2) \qquad R_e = 2L_s/T_{on}$$

$$K_{ap} = T_{off}/T_{on}$$

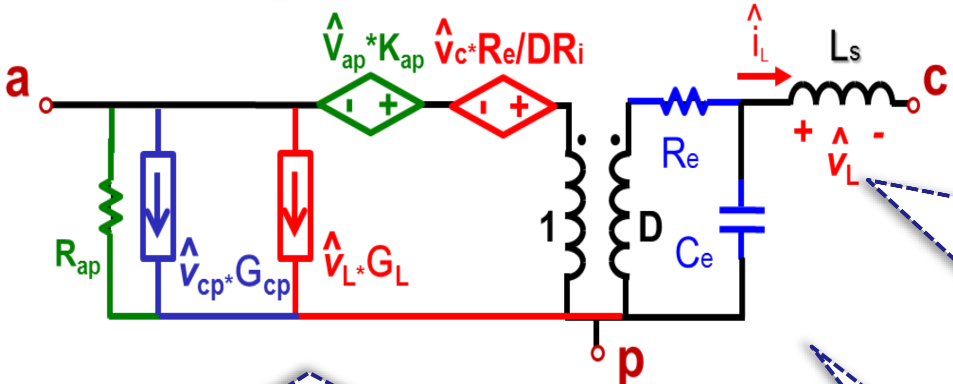
Re is always positive

Constant Off-time Control

$$C_e = T_{off}^2/(L_s\pi^2) \qquad R_e = 2L_s/T_{off}$$

$$K_{ap} = -\mathbf{1}$$

Re is always positive



Charge Control

$$C_e = T_{sw}^2/(L_s\pi^2) \quad R_e = L_s/[T_{sw}(\frac{L_s I_L}{V_{cp}T_{sw}} - \frac{D}{\mathbf{2}} + \frac{s_e}{s_f}\frac{C_T}{T_{sw}})]$$

$$K_{ap} = (T_{off}/\mathbf{2}L_s)\cdot R_e$$

Re can be positive or negative

Peak Current-Mode Control

$$C_e = T_{sw}^2/(L_s\pi^2) \quad R_e = L_s/[T_{sw}(\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

$$K_{ap} = -(T_{off}/\mathbf{2}L_s)\cdot R_e$$

Re can be positive or negative

Valley Current-Mode Control

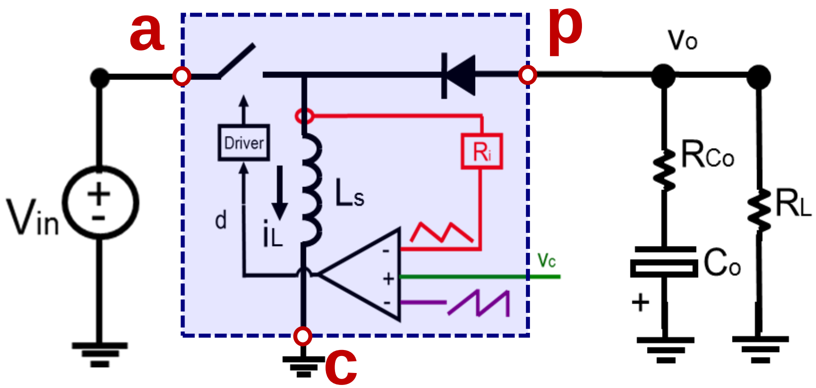
$$C_e = T_{sw}^2/(L_s\pi^2) \quad R_e = L_s/[T_{sw}(\frac{s_f + s_e}{s_n + s_f} - 0.5)]$$

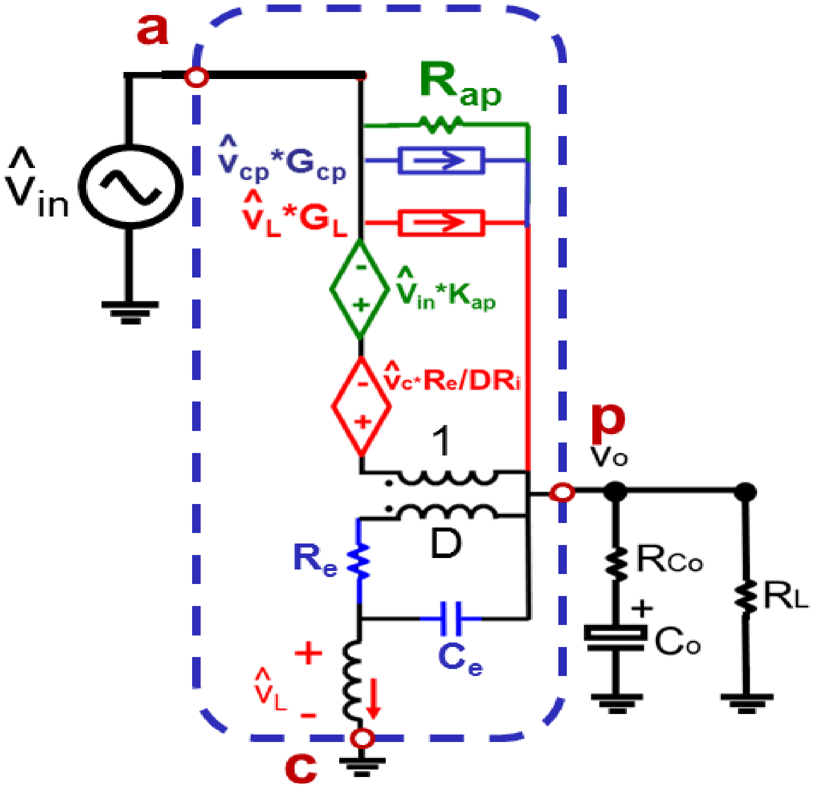
$$K_{ap} = (T_{on}/\mathbf{2}L_s)\cdot R_e$$

Re can be positive or negative

Control-to-v_o Transfer Function

Buck-boost with peak current mode control





V_{in}=500V, V_o=400V, R_L=150Ω,
f_{sw}=100kHz,L_s=300μH